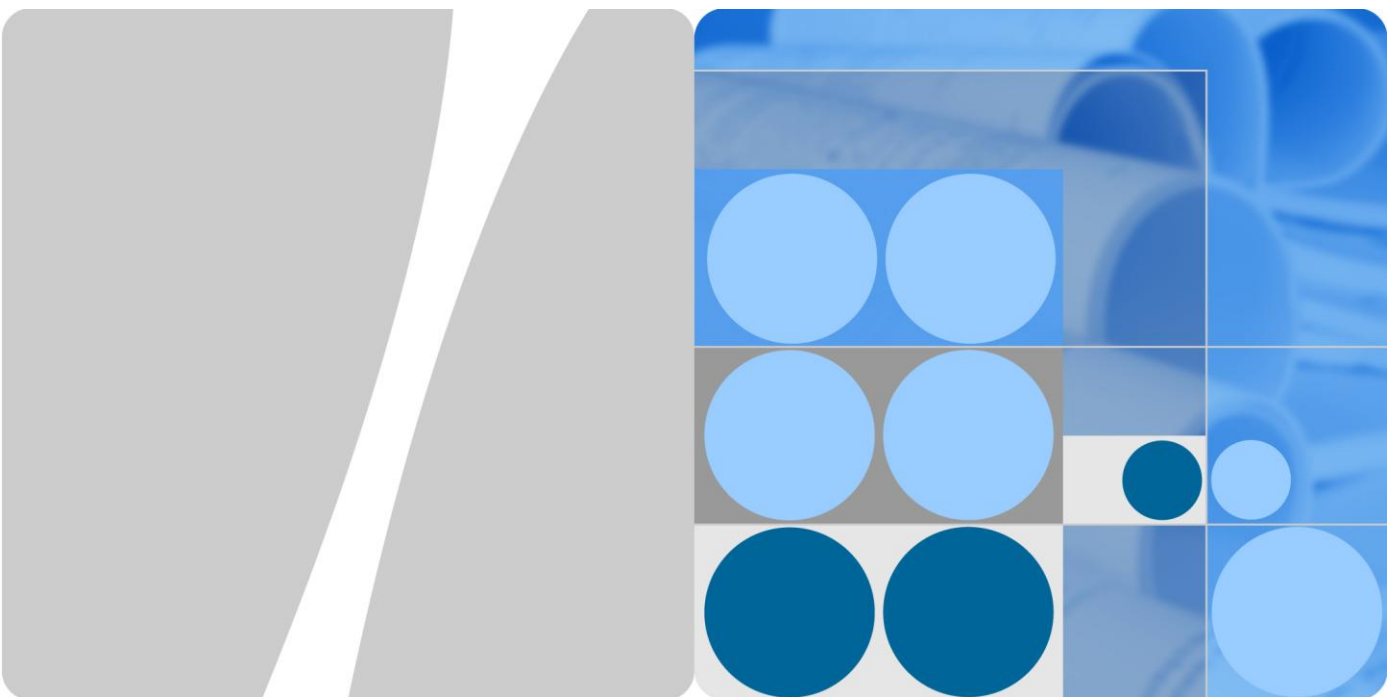




RAN Troubleshooting Guide

Issue 01
Date 2012-06-25

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Overview

Document Purpose

This document provides information on how to identify and repair common faults on RAN equipment that is working in a live network. Operation and maintenance (OM) personnel should use this guide in the following scenarios:

- User complaints are received
- Faults are detected in the routine maintenance processes
- Emergency faults are detected in the equipment
- Alarm analysis

Product Version

The following table lists the product versions related to this document.

Product Name	Product Model	Product Version
RNC	BSC6900	V900R013C00
NodeB	DBS3900/DBS3800/BTS3812E/BTS3900	V100R013C00/ V200R013C00

Intended Audience

This guide is intended for system engineers.

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Alerts you to a high risk hazard that could, if not avoided, result in serious injury or death.

Symbol	Description
 WARNING	Alerts you to a medium or low risk hazard that could, if not avoided, result in moderate or minor injury.
 CAUTION	Alerts you to a potentially hazardous situation that could, if not avoided, result in equipment damage, data loss, performance deterioration, or unanticipated results.
 TIP	Provides a tip that may help you solve a problem or save time.
 NOTE	Provides additional information to emphasize or supplement important points in the main text.

Change History

01 (2012-06-25)

This is the first release version.

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1 Troubleshooting Process and Methods

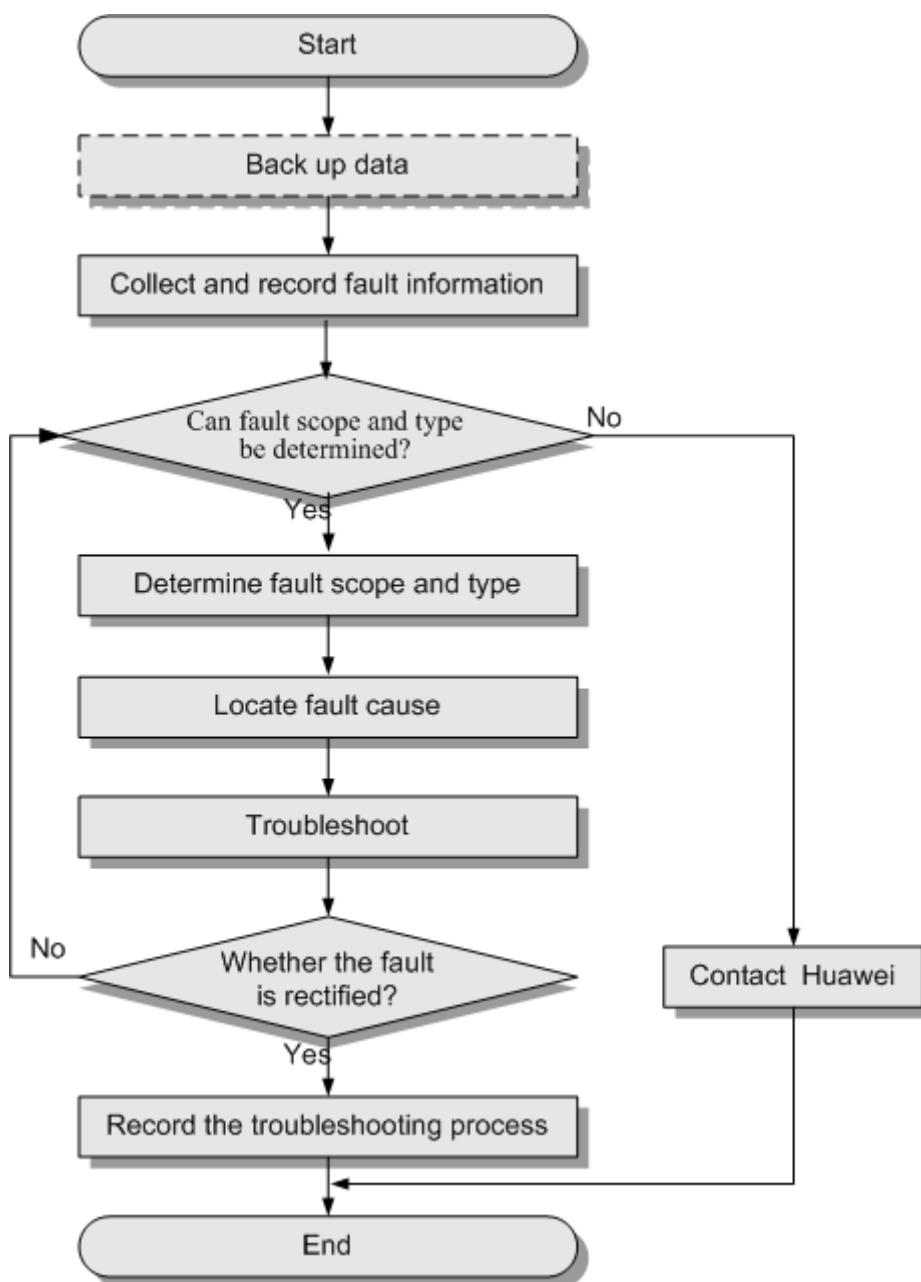
1.1 About this Chapter

This chapter describes the process for troubleshooting, common methods of fault location, and how to get technical support from Huawei.

1.2 Troubleshooting Process

1.2.1 Flowchart

Figure 1-1 shows the troubleshooting flowchart.

Figure 1-1 Troubleshooting flowchart

1.2.2 Collecting and Recording Fault Information

Fault Information to be Collected

When a fault occurs, OM personnel must start troubleshooting by obtaining fault information, which serves as a reference. OM personnel should collect as much fault information as possible. The following information must be collected before any operation:

- Symptoms, including details and basic data
- Time, location, and frequency of occurrence
- Scope and impact
- Equipment operating status before the fault occurred

- Operations performed on the equipment before the fault occurred, and the results of these operations
- Measures taken to deal with the fault, and the results
- Alarms resulting from the fault
- Status of board LEDs

Methods of Collecting Fault Information

To collect fault data, do as follows:

- Consult the personnel who reported the fault about symptoms, time, location, and frequency of the fault.
- Consult the OM personnel about the equipment operating status before the fault occurred, operations performed on the equipment before the fault occurred, fault symptoms, and measures taken to deal with the fault and the results.
- Observe board LEDs, the OM subsystem, and the alarm management system to obtain information about the status of system software and hardware.
- Estimate the impact of the fault by testing services, measuring performance, and tracing interface messages or signaling messages.

Strategies for Collecting Fault information

Strategies for collecting fault information are as follows:

- Do not handle a fault hastily. Collect as much information as possible before attempting to repair the fault.
- Maintain good communication with other departments and OM personnel of other sites. Ask them for technical support if necessary.

1.2.3 Determining Fault Scope and Type

After collecting all available fault information, analyze the fault symptoms, and determine the fault scope and type.

This document describes 11 types of faults, as listed in Table 1-1.

Table 1-1 Faults and scopes

No.	Category	Fault Type	Description
1	HSPA service	HSPA service setup failure	HSPA service setup failure, instead of a low rate of HSPA services
2		HSUPA rate fault	Fluctuating or low HSUPA rate
3		HSDPA rate fault	Fluctuating or low HSDPA rate
4	KPI	SLC fault	Cell access failure
5		RRC connection setup fault	Low RRC connection setup success rate
6		RAB connection setup fault	Low RAB access success rate
7		Call drop rate fault	High call drop rate

No.	Category	Fault Type	Description
8		Handover fault	Low handover success rate
9		Paging fault	Low paging success rate
10	Operation & Maintenance	Operation & Maintenance fault	Faults of OM on RAN devices
11	Transmission	ATM Transmission network fault	ATM transmission faults
12		IP Transmission network fault	IP transmission faults

1.2.4 Locating the Cause of the Fault

To locate the cause of the fault, first compare and analyze possible causes, and then eliminate causes that are unlikely or impossible.

Comparison and Interchange

- **Description**
OM personnel can compare the faulty components or symptoms with their normal equivalents to locate faults. Comparison is applied in scenarios where the scope of the fault is small.
If the fault scope and area cannot be determined after the replacement of some components with spare parts, then interchange a component that is suspected of being faulty with known good components that are being used in the system. For example, replace a board or optical cable that is suspected faulted with an equivalent item that is known to be good. Then compare the status before and after the operation to determine if the fault was repaired or to further determine the scope and area of the fault. Interchange is applied in scenarios where the scope of the fault is large.
- **Application Scenarios**
Comparison and interchange are used when faults occur after NE hardware, software or a new feature is introduced that may have caused a service outage.
- **Instructions**
Use this method to compare the performances and KPIs before and after hardware or software is changed, or a new feature is introduced.

Segment-by-Segment Location

- **Function**
A fault may occur at any node in an end-to-end network. Therefore, this method helps locating the fault quickly.
- **Application Scenario**
Transmission network fault or PS data transmission fault
- **Usage**
Locate the fault segment by segment.

Layer-by-Layer Location

- **Function**

As specified by the protocol, the upper layer can work properly only when its lower layers are working properly. When a fault occurs, all associated layers malfunction. In addition, the symptom of a fault may vary if different monitoring methods are used. Therefore, this method helps locating the layer where the fault is generated and facilitates the troubleshooting.

- Application Scenario
Transmission network fault or PS data transmission fault
- Usage
Locate the fault layer by layer.

1.2.5 Troubleshooting

To repair faults and restore the system, troubleshoot different faults using proper measures and procedures. Troubleshooting consists of checking cables, replacing boards, modifying data configuration, switching over boards, and resetting boards.

1.2.6 Ensuring that System Is Repaired

Test the system again after troubleshooting to ensure that the fault is completely repaired. Ensure the system works properly by observing the status of board LEDs and alarm information, and perform dial tests to ensure that all services are operational.

1.2.7 Recording the Troubleshooting Process

It is important to record the troubleshooting process in a way that OM personnel can use in the future. When the troubleshooting/repair action is complete, OM personnel should:

- Review the entire troubleshooting process
- Note key points of the process
- Summarize methods for improvement

of the system which could avoid recurrence of the faults of the same type.



NOTE

Ensure notes are recorded in a logbook or other method that OM personnel will have future access to.

1.2.8 Contacting Huawei for Technical Support

If faults are difficult to identify or solve, then prepare the following information, and contact Huawei for technical support.

Step 1 Collect general fault information.

The general information required is as follows:

- Full name of the office
- Contact name and number
- Time when the fault occurred
- Detailed symptoms of the fault
- Version of the host software
- Measures taken to deal with the fault, and the results
- Severity and expected repair time

Step 2 Collect fault location information.

Information to be collected is listed according to the related steps.

Step 3 Use the following methods to contact Huawei for technical support:

- E-mail: support@huawei.com
- Website: <http://support.huawei.com>

**NOTE**

<http://support.huawei.com> contains contact information for the local office in your region.

2 Common Maintenance Functions

2.1 About This Chapter

This chapter describes common maintenance functions and how to perform the functions during troubleshooting.

2.2 Transmission Maintenance Function

This section describes the common maintenance function during the diagnosis of transmission faults.

2.2.1 Checking for Faults in Crossed Pair Connections

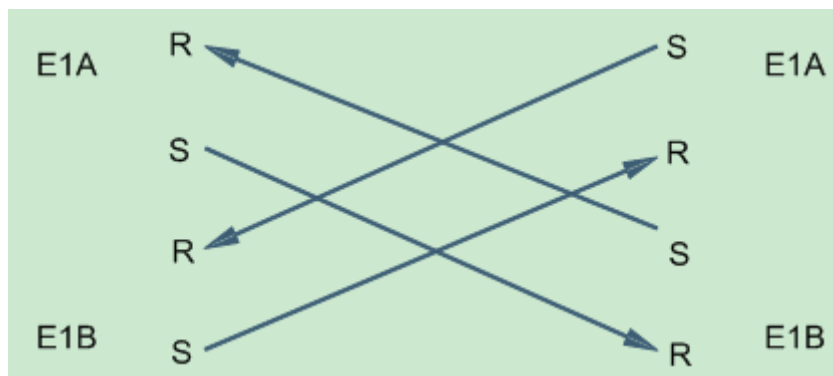
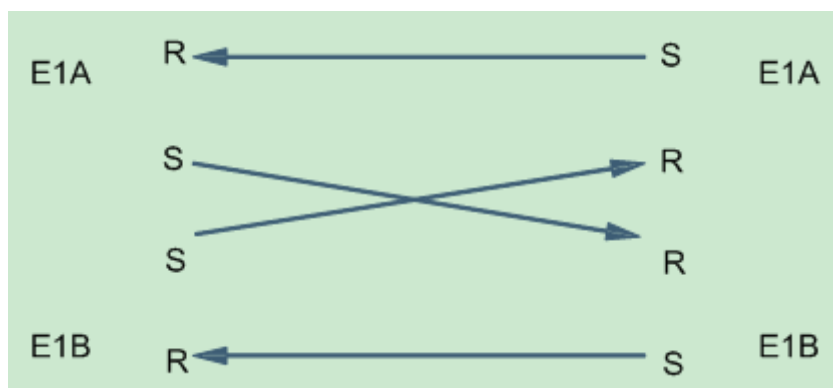
Function Description

This function allows users to detect faults caused by crossed pair connections at E1 ports when equipment at two ends interconnects. Crossed pair connections cause the communication of signals at the physical layer, upper link failure, and service setup failure.

There are two crossed pair connections, which are described as follows:

Crossed pair connection 1: The TX ends of two pairs of E1 lines are connected to the RX ends, as shown in Figure 2-1.

Crossed pair connection 2: The TX end of an E1 line is connected to the RX end of the other E1 line, as shown in Figure 2-2.

Figure 2-1 Crossed pair connection 1**Figure 2-2** Crossed pair connection 2

Prerequisites

No alarms are generated on the E1 port to be detected.

Operation Procedure

- Step 1** Perform a remote loopback detection on the local RNC.
- Step 2** Run **SET E1T1LOP** on the RNC, and set **LOPT** to **REMOTE_LOOP**.



CAUTION

Ongoing services will be affected. Therefore, do not perform this operation without permission. Exercise caution while performing the operation, if required.

- Step 3** Check for loopback alarms on the peer NodeB.

----End

Operation Results

Check whether the ALM-25807 E1/T1 alarm is generated on the NodeB, with the cause value of physical loopback.

If the alarm is generated, crossed pair connections fail.

If no alarm is generated, crossed pair connections are correct.

2.2.2 Performing a Bit Error Monitoring on the E1/T1 Port

Function Description

This function enables users to monitor statistical information about bit errors on the E1/T1 port and learn the transmission quality on links of the port in real time.

This function is applicable to the AEUa/PEUa/EIUa/OIUa/POUc board.

Operation Procedure

Step 1 Log in to the RNC LMT.

Step 2 On the LMT, click **Monitor**. The **Monitor** tab page is displayed.

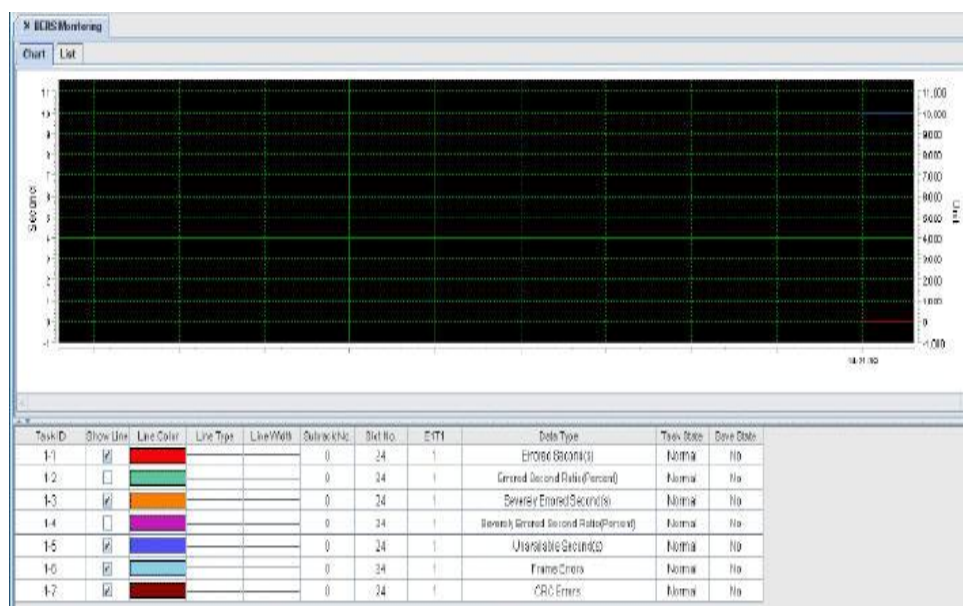
Step 3 In the monitor navigation tree, choose **Monitor** > **Common Monitoring**, and then double-click **Bit Error Monitoring**.

Step 4 In the displayed **Bit Error Monitoring** dialog box, set parameters, and then click **OK** to start monitoring.

----**End**

Operation Results

After the bit error monitoring starts, a monitoring window is displayed. The toolbar shows the task name and related parameters and real-time monitoring results are displayed in the list or chart mode, as shown in Figure 2-3.

Figure 2-3 Bit error monitoring results

2.2.3 Using VCLCC to Check for Link Faults

Function Description

This function enables users to check for faults on the SAAL link, IPoA PVC, and AAL2 path. This function is applicable to the AEUa/AOUa/AOUc/UOIa (ATM) /UOIc board.



NOTE

Before you perform this operation, the peer end (MGW/MSC/SGSN) complies with the ATM F5 protocol and the virtual channel link continuity check (VCLCC) function has been activated. The NodeB only responds to the detection function.

The function is activated only when upper-layer applications (NCP/CCP/ADJNODE/MTP3LNK) are configured on the SAAL link.

Operation Procedure

- Step 1** Determine the links to be monitored according to alarms and performance counters.
- Step 2** Start a monitoring task of a specified link. Run **ACT VCLCC** on the RNC and set **Activation Mode** to **CC**.
- Step 3** Run **DSP VCLCC** on the RNC to query monitoring results.
- Step 4** Run **DEA VCLCC** on the RNC to stop the monitoring task.

----End

Operation Results

VCLCC has been activated if no ALM-21324 VCL CC alarms are generated on the RNC.

Check whether the following alarms are generated:

1. ALM-21321 VCL CC Detection Failure
2. ALM-21322 VCL Alarm Indication Signal
3. ALM-21323 VCL Remote Alarm Indication

If one of the alarms is generated, the links fails or packets are discarded. If no alarm is generated, the link is normal.

2.2.4 Using VCLCC to Check for Link Delays

Function Description

This function enables users to detect whether the SAAL link, IPoA PVC and AAL2 path is delayed. The local end transmits detected signals to the peer end and the peer end directly transmits the received signals back to the local end. Then, the local end calculates the difference from the time when signals are transmitted to the time when signals are received, which is called link delay.

This function is applicable to the AEUa/AOUa/AOUc/UOIa (ATM)/UOIc board.



NOTE

Before you perform this operation, the peer end (MGW/MSC/SGSN) complies with the ATM F5 protocol and the virtual channel link continuity check (VCLCC) function has been activated. The NodeB only responds to the detection function.

The function is activated only when upper-layer applications (NCP/CCP/ADJNODE/MTP3LNK) are configured on the SAAL link.

Operation Procedure

- Step 1** Determine the links to be monitored according to alarms and performance counters.
- Step 2** Start a monitoring task of a specified link. Run **ACT VCLCC** on the RNC and set **Activation Mode** to **LOOKBACK**.
- Step 3** Run **DSP VCLCC** on the RNC to query monitoring results.
- Step 4** Run **DEA VCLCC** on the RNC to stop the monitoring task.

----End

Operation Results

Loopback detection succeeds if no ALM-21326 VCL LB alarms are generated on the RNC.

Analyze the **DSP VCLCC** command execution result. If LB **Test Result** is **Succeeded**, you can obtain the link delay. Run the command for multiple times to check a change in the link delay.

```
+++ WCDMA-RNC 2010-09-21 11:53:22
O&M #7112
%%DSP VCLCC: LNKT=AAL2PATH, ANI=150, PATHID=4;%%
RETCODE = 0 Execution succeeded.
```

Continuous check result

```
Adjacent node of AAL2 path = 150
AAL2 path ID = 4
```

```

        SINK activated state = CC_DOWN
        SOURCE activated state = CC_DOWN
        LB Test result = Succeeded
        LOC alarm = Normal
        AIS alarm = Normal
        RDI alarm = Normal
    CC activated failure alarm = Normal
        LB failure alarm = Normal
    Average Time Delay[ms] = 8
        Max Time Delay[ms] = 8
        Min Time Delay[ms] = 8
    (Number of results = 1)

---    END

```

2.2.5 Using VCLPM to Check for Abnormal Links

Prerequisites

The VCLCC function has been activated at local and peer ends and remains activated during VCLPM.

Function Description

This function enables user to check the number of discarded cells and the number of misinsertion cells on the VCL of multiple SAAL links, AAL2 paths, and IPOA PVC links at the same time.



NOTE

This function is applicable to the AOUc/UOIC board on the RNC and not applicable to NodeB V1.

If the version of the backplane subrack that houses the boards is VER.A or VER.B. (the version is queried by running **DSP BRDVER**), the MSP 1+1 single-end mode (in the SET MSP command execution, **MODE** is set to **MODE2**) does not support the VCL PM function. If the version is VER.C or a later version, the MSP 1+1 single-end mode supports the VCL PM function.

Operation Procedure

- Step 1** Determine the links to be monitored according to alarms and performance counters.
 - Step 2** Run **ACT VCLPM** on the RNC or NodeB to activate the PM function of the specified PVC.
 - Step 3** Run **DSP VCLPM** on the RNC or NodeB to query and record the results.
 - Step 4** Run the command for five consecutive times at an interval of three minutes.
- Note: If you run the preceding command once, only the accumulated values of the counters can be queried. Generally, you can obtain the link quality in a certain period by running the command for multiple times and calculating the difference of the counter values.
- Step 5** Run **DEA VCLPM** on the RNC to stop the monitoring task.

----End

Operation Results

Analyze the **DSP VCLPM** command execution result. If one of the following parameter value increases, the link fails:

- Number of Discard Cells by Send
- Number of Wrong Inserted Cells by Send
- Number of Discard Cells by Receive
- Number of Wrong Inserted Cells by Receive
- Wrong Cells calculated by BIP16 in SOURCE
- Wrong Cells calculated by BIP16 in SINK

Otherwise, the link is normal.

2.2.6 Performing VCL Link Performance Query

Function Description

This function enables users to query the number of transmitted/received cells, packets, bytes, and error bytes of the SAAL link, AAL2 path and IPOA PVC.

Operation Procedure

Step 1 Determine the links to be monitored according to alarms and performance counters.

Step 2 Run **DSP AALVCCPFM** on the RNC to query and record the results.

Step 3 Run the command for five consecutive times at an interval of three minutes.



NOTE

Note: If you run the preceding command once, only the accumulated values of the counters can be queried. Generally, you can obtain the link quality in a certain period by running the command for multiple times and calculating the difference of the counter values.

----End

Operation Results

Analyze the **DSP AALVCCPFM** command execution result. If one of the following parameter value increases, the link fails or is of poor transmission quality:

- Number of Sent/Received Cells
- Number of Sent/Received Packets
- Number of Sent/Received Bytes
- Number of Sent/Received Error Bytes

Otherwise, the link is normal or of poor quality.

2.2.7 Performing the IP over ATM OMCH Continuity Check

Function Description

This function enables users to check IP over ATM OMCH connectivity on the RNC.

Operation Procedure

Step 1 Check RNC scripts and locate the local IP address of the OMCH based on the NodeB ID.

```
ADD UNODEBIP:NODEBID=10009, NBTRANTP=ATMTRANS_IP, ATMSRN=3,  
ATMSN=24, NBATMOAMIP="10.136.76.36".
```

Step 2 Locate the peer IP address of the OMCH based on the NodeB IP address.

```
ADD IPOAPVC:IPADDR="10.136.76.1", PEERIPADDR="10.136.76.36",  
CARRYT=NCOPT, CARRYNCOPTN=1, CARRYVPI=1, CARRYVCI=33, TXTRFX=240,  
RXTRFX=240, PEERT=IUB;
```

Step 3 Run **PING IP** on the RNC from the local IP address to the peer IP address of the OMCH.

```
PING IP: SRN=3, SN=24, SIPADDR="10.136.76.1", DESTIP="10.136.76.36",  
CONTPING=NO, PKTSIZE=56;
```

Step 4 Perform the continuity check using different ping packets.

1. Set the **PKTSIZE** parameter in the **PING IP** command to adjust packet sizes. Use 64, 500, 1000, and 1500 bytes packets to verify whether all packets fail to be transmitted or whether only large packets fail to be transmitted.
2. Set the **TIMES** parameter in the **PING IP** command to adjust detection times. Set this parameter to a large value, for example, 1000, to ensure the accuracy of the detection result under different conditions.

----End

Operation Results

For details, see "Operation Results" in 2.2.10 "Using the Ping Operation to Perform the IP Continuity Check."

2.2.8 Using LOP VCL to Check for Link Faults or Link Delays

Function Description

This function enables users to check for faults or delays of the SAAL link, IPoA PVC and AAL2 path.



NOTE

Before you perform this operation, the peer end (MGW/MSC/SGSN) complies with the ATM F5 protocol. The NodeB only responds to the detection function and NodeB V1 only supports the function of detecting the AAL2 path link.

Operation Procedure

Run **LOP VCL** on the RNC to start the detection.

----End

Operation Results

In the command execution result, if **Loopback result** is **Succeeded**, the local end receives IEs from the peer end and the PVC link is normal. In this case, the round trip time (RTT) of the detected IE is displayed.

If **Loopback result** is **Failed**, the local end fails to receive IEs from the peer end and the PVC link fails.



NOTE

You are advised to run **LOP VCL** for multiple times to ensure that the detected VCL link quality is accurate.

```
O&M      #73423
%%LOP VCL: LNKT=AAL2PATH, ANI=14, PATHID=5;%%
RETCODE = 0  Execution succeeded.

Loopback result
-----
Loopback result = Succeeded
Time Delay[ms] = 9
(Number of results = 1)

---      END

+++      HWBSC6810      2010-11-17 10:14:05
O&M      #73555
%%LOP VCL: LNKT=IPOAPVC, IPADDR="192.168.1.250", PEERIPADDR="192.168.1.251";%%
RETCODE = 0  Execution succeeded.

Loopback result
-----
Loopback result = Failed
Time Delay[ms] = <NULL>
(Number of results = 1)

---      END
```

2.2.9 Checking the Operating Status of the Ethernet Port

Function Description

This function enables users to query the operating status and traffic volume on the Ethernet port. The traffic volume is accumulative and you can analyze the data change by running the command for multiple times.

This function is applicable to the FG2a/GOUa/FG2c/GOUc board.

Operation Procedure

Run **DSP ETHPORT** on the RNC or NodeB.

Operation Results

In the command execution result, if **Link Availability Status** is **Unavailable**, IP transmission fails.

You can run the commands for multiple times and calculate the value differences to check whether the number of received and transmitted correct bytes increases. If the number of correct bytes increases, the transmission and reception of the port are abnormal.

If the number of incorrect bytes increases, the link of the port encounters error packets.

If the value of **Number of received Multicast frame** or **Number of received broadcast frame** increases, broadcast or multicast packet shocks occur.

2.2.10 Using the Ping Operation to Perform the IP Continuity Check

Function Description

This function can be used to check the connectivity of the IP layer between the local end and the destination end. It also enables users to check the connectivity, delay, jitter, packet loss, and transient interruption on the network. You can perform ping operations segment by segment to locate the area where the fault occurs.

Use 20, 500, and 1500 bytes packets to verify whether all packets fail to be transmitted or whether only large packets fail to be transmitted.

Use different DSCP values configured on multiple paths to verify whether each DSCP packet is transmitted properly.

Set this parameter to a large value, for example, 1000, to ensure the accuracy of the detection result under different conditions.

Operation Procedure

- Step 1** Determine the local IP address, subrack of the local IP address, slot of the local IP address, and peer IP address before performing the ping operation.
- Step 2** Run **PING IP** on the RNC or **PING** on the NodeB.
- Step 3** Perform IP continuity check using different ping packets.
1. Set the **PKTSIZE** parameter in the **PING IP** command on the RNC or the **PING** command on the NodeB to adjust the packet size. Use 20, 500, and 1500 bytes packets to verify whether all packets fail to be transmitted or whether only large packets fail to be transmitted.
 2. Set the **DSCP** parameter in the **PING IP** command on the RNC or the **PING** command on the NodeB to adjust the DSCP value. Use DSCP values on other links to verify whether each DSCP packet is transmitted properly.
 3. Set the **TIMES** parameter in the **PING IP** command on the RNC or set the **NUM** parameter in the **PING** command on the NodeB to adjust detection times. Set this parameter to a large value, for example, 1000, to ensure the accuracy of the detection result under different conditions.

----End

Operation Results

Adjust the packet size and DSCP value to verify whether each packet is transmitted properly.

Check for the transmission delay or jitter according to the **time** value and the change in the **time** value.

Check for transient interruption according to the number of times **Request time out** is displayed.

Check for the packet loss rate according to the **packet lost** value.

The following is an example of the command execution result:

Example 1: After you perform the ping operation on the RNC, the transmission network is normal. The command execution result is as follows:

```
Reply from 18.30.1.10: bytes=56 Sequence=1 ttl=252 time=10 ms
Reply from 18.30.1.10: bytes=56 Sequence=2 ttl=252 time=10 ms
Reply from 18.30.1.10: bytes=56 Sequence=3 ttl=252 time=10 ms
Reply from 18.30.1.10: bytes=56 Sequence=4 ttl=252 time=11 ms

--- 18.30.1.10 Ping statistics ---
  4 packet(s) transmitted
  4 packet(s) received
Percent 0.00 packet lost
round-trip min/avg/max = 10/10/11 ms

+++   MBSC15           2010-12-03 16:27:42
O&M   #3837
%%PING IP: SRN=0, SN=24, SIPADDR="15.1.26.10", DESTIP="18.30.1.10", CONTPING=NO,
TXINT=2000;%%
RETCODE = 0 Execution succeeded.

10 reports in total
(Number of results = 1)

---   END
```

Example 2: After you perform the ping operation on the RNC, the command execution results are all **Request time out**, which indicate that the transmission network is abnormal.

```
PING 18.30.1.10: 56 data bytes

Request time out
Request time out
Request time out
Request time out

--- 18.30.1.10 Ping statistics ---
  4 packet(s) transmitted
  0 packet(s) received
Percent 100.00 packet lost
```

Example 3: After you perform the ping operation on the RNC, **Request time out** is displayed occasionally in the command execution results, which indicate that packet loss occurs on the transmission network and the packet loss rate is displayed.

```
PING 18.30.1.10: 56 data bytes

Request time out
Reply from 18.30.1.10: bytes=56 Sequence=1 ttl=252 time=10 ms
```

```
Reply from 18.30.1.10: bytes=56 Sequence=1 ttl=252 time=10 ms
Request time out
```

```
--- 18.30.1.10 Ping statistics ---
    4 packet(s) transmitted
    2 packet(s) received
    Percent 50.00 packet lost
```

2.2.11 Using the Trace Operation to Check for Abnormal Transmission Nodes

Function Description

When the network is disconnected, this function detects the connectivity of each hop from the local end to the destination end, obtain the IP address along the path, and locate the hop where faults occur.

Operation Procedure

- Step 1** Determine the local IP address, subrack of the local IP address, slot of the local IP address, and peer IP address before performing the trace detection.
 - Step 2** Run **TRC IPADDR** on the RNC or **TRACERT** on the NodeB.
 - Step 3** Estimate a possible **MAX TTL** value, and continue the detection by increasing the estimated **MAX TTL** value. If an IP address that is not displayed exists in the output, the estimated **MAX TTL** value is larger than the actual number of hops.
 1. It is the maximum TTL value of the transmitted TRACERT packets if you run **TRC IPADDR** on the RNC.
 2. It is the maximum TTL value if you run **TRACERT** on the NodeB.
- End

Operation Results

The network is normal if the output shows the IP address of the last hop matches the destination IP address.

The network is abnormal if the output shows the IP address of the last hop does not match the destination IP address and some IP addresses fail to be displayed. Locate the hop where the faults occur and check for the faulty device.

Example 1: After you run **TRC IPADDR** on the RNC, the network is normal. The command execution result is as follows:

```
%%TRC IPADDR: SRN=0, SN=24, DESTIP="18.30.1.10", MAXTTL=4, %%
RETCODE = 0 Execution succeeded.
  traceroute to 18.30.1.10(18.30.1.10) 4 hops max, 40 bytes packet
  1 15.1.26.1 3 ms 4 ms 4 ms
  2 40.3.2.3 2 ms 3 ms 3 ms
  3 40.3.1.1 9 ms 8 ms 7 ms
  4 18.30.1.10 3 ms 3 ms 3 ms
(Number of results = 1)
--- END
```

From the result, you can obtain each next-hop address on the path designated for the destination address 18.30.1.10.

Example 2: After you run **TRC IPADDR** on the RNC, the network is abnormal. The command execution result is as follows:

```
%%TRC IPADDR: SRN=0, SN=24, DESTIP="18.30.1.10", MAXTTL=4, %%
RETCODE = 0 Execution succeeded.
  traceroute to 18.30.1.10(18.30.1.10) 4 hops max,40 bytes packet
  1 15.1.26.1 3 ms 4 ms 4 ms
  2 * * *
  3 * * *
  4 * * *
(Number of results = 1)
--- END
```

From the result, the last IP address is not the destination IP address and some IP addresses fail to be displayed, indicating that the transmission over the port with its IP address of 15.1.26.1 fails.

2.2.12 Using the Ping Operation to Check the IP Path Status

Function Description

The path ping function checks the IP path connectivity and link status.

In the path ping process, the RNC sends ICMP packets continuously to the destination IP address and receives response packets along the IP path where this function is activated. You can learn about the transmission status of the IP path according to the statistics of response packets.

Operation Procedure

Run **ADD IPPATH** on the RNC or run **MOD IPPATH** on the NodeB. Set **PATHCHK** to **ENABLED** to enable the IP path check function.

Operation Results

Check for the ALM-21581 Path Fault alarms. If such alarms are generated due to IP path ping failures, the IP path is faulty.

Check for the delay, jitter, packet loss, and congestion of an IP path from the performance measurement counters listed below.

Performance Measurement Data
VS.IPPATH.PING.MeanDELAY
VS.IPPATH.PING.MaxDELAY
VS.IPPATH.PING.MeanJITTER
VS.IPPATH.PING.MaxJITTER
VS.IPPATH.PING.MeanLOST
VS.IPPATH.PING.MaxLOST

Performance Measurement Data
VS.IPPATH.Fwd.Cong
VS.IPPATH.Fwd.Cong.Dur
VS.IPPATH.Bwd.Cong
VS.IPPATH.Bwd.Cong.Dur

2.2.13 Performing IP Loopback Detection to Check for Abnormal Transmission Nodes

Function Description

This function checks for faults in the RNC, the Iub interface or the Uu interface. Perform a local loopback in the RNC to check whether faults occur in the RNC, and perform a remote loopback between the RNC and the NodeB to check whether Iub transmission faults occur.

If the IP loopback result shows no packet loss and the delay is less than 15 ms, the fault occurs in the Iu interface or the Uu interface.

This function is applicable to the IP networking over the Iub interface.



CAUTION

Do not perform this operation without permission, because ongoing services will be interrupted.

Operation Procedure

Step 1 Determine the local and peer IP address, subrack and slot of the board.

Step 2 Run **STR IPLOPTST** on the RNC.

If **Loop type** is set to **LOCAL_LOOP**, detect the connectivity between the DSP and the interface board.

If **Loop type** is set to **REMOTE_LOOP**, run **SET UDPLOOP** on the NodeB to start the IP remote loopback according to the configured IP and the port number.



NOTE

The detection time on the RNC must be shorter than the loopback time on the NodeB to ensure that the NodeB is performing remote loopback.

Step 3 Run **DSP IPLOPTST** on the RNC.

Step 4 Stop the loopback on the RNC and on the NodeB.

Run **SET UDPLOOP: LM=NOLOOP** on the NodeB.

Run **STP IPLOPTST** on the RNC.

----End

Operation Results

In the command execution result, check whether the number of transmitted packets is the same with that of received packets. If not, packet loss occurs.

```
%%DSP IPLOPTST: SRN=0, DPUSN=8, DSPNO=0;%%  
RETCODE = 0 Execution succeeded.
```

```
Result of IP loopback test  
-----  
Subrack No. = 0  
DPU slot No. = 8  
DSP No. = 0  
INT Subrack No. = 2  
INT slot No. = 24  
Local IP = 15.0.24.10  
Local port No. = 65040  
Peer IP = 115.7.0.2  
Peer port No. = 65040  
Number of sent packets = 161  
Number of received packets = 160  
Average Time Delay[ms] = 1  
(Number of results = 1)  
--- END
```

2.2.14 Performing IP PM Detection to Check IP Path Performance on the Iub Interface

Function Description

This function detects delay, variation (that is, jitter), and packet loss rate of the IP path on the Iub interface.

If packet loss occurs, IP PM activated on the RNC detects the downlink packet loss, and IP PM activated on the NodeB detects the uplink packet loss.

Operation Procedure

Step 1 Determine the IP path to be detected.

Step 2 Run **ACT IPPM** on the RNC or **ADD IPPMSESSION** on the NodeB.

----End

Operation Results

Check for the following alarms on the RNC or NodeB:

1. NodeB ALM-25900 IP PM Activation Failure
2. RNC ALM-21341 IP PM Activation Failure

If one alarm is generated, the IP PM function is unavailable.

If no alarm is generated, check the following performance counters to obtain the TX rate, packet loss rate, jitter, and delay of the IP path.

TX rate	VS.IPPM.Bits.MeansTx VS.IPPM.Peak.Bits.RateTx VS.IPPM.Pkts.MeansTx VS.IPPM.Peak.Pkts.RateTx
Packet loss rate	VS.IPPM.Forward.DropMeans VS.IPPM.Forward.Peak.DropRates
Jitter	VS.IPPM.Forward.JitterStandardDeviation VS.IPPM.Back.JitterStandardDeviation
Delay	VS.IPPM.Rtt.Means IPPM VS.IPPM.MaxRttDelay IPPM

2.2.15 Performing Node Synchronization Detection to Check for Transmission Delay and Jitter on the User Plane

Function Description

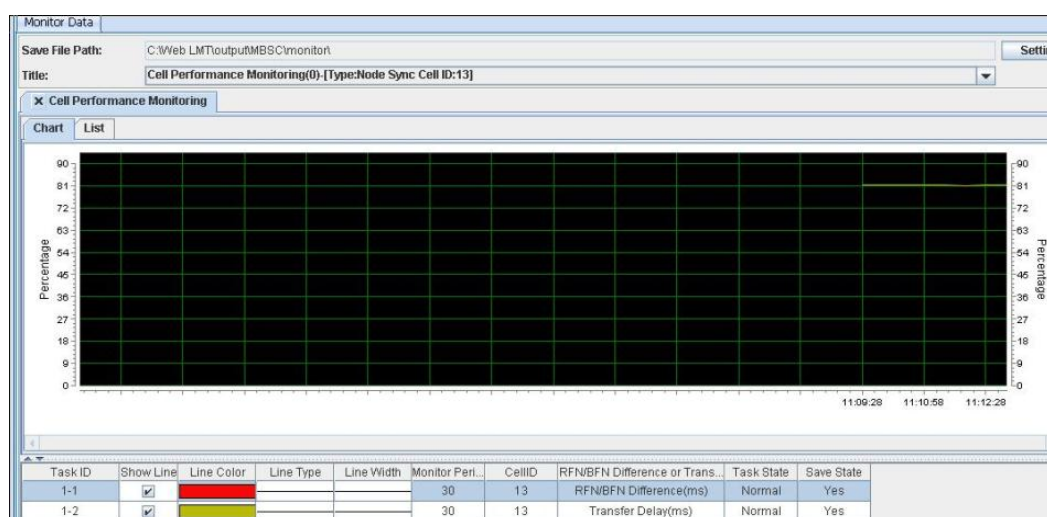
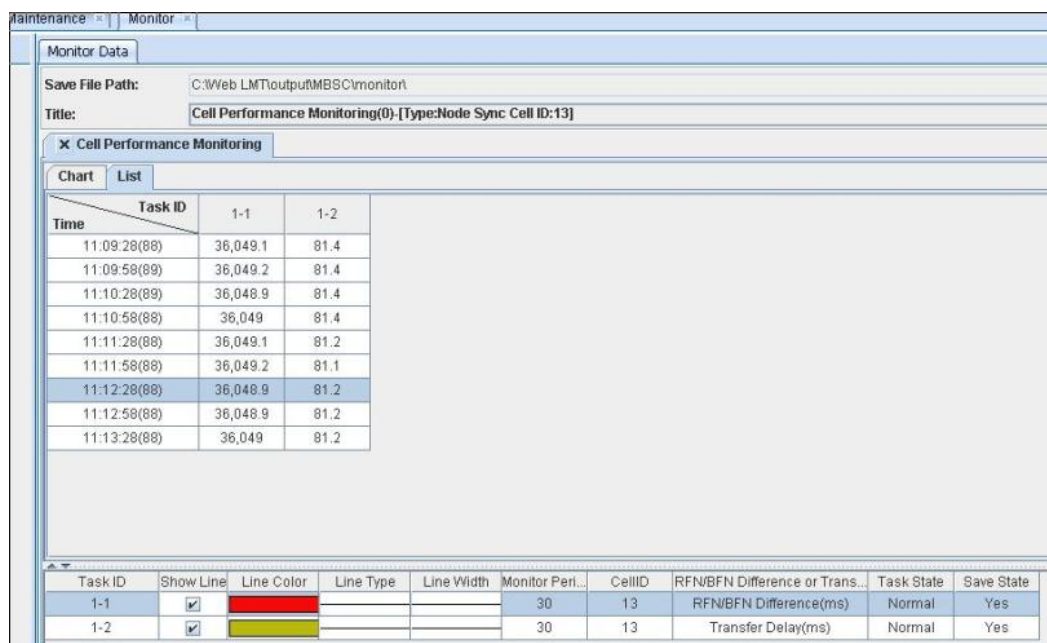
This function enables users to check the delay and jitter of the Iub interface on the user plane.

Operation Procedure

- Step 1** In the LMT window, click **Monitor** to display the **Monitor** tab page.
- Step 2** In the monitor navigation tree, choose **Monitor** > **UMTS Monitoring** > **Cell Performance Monitoring**.
- The **Cell Performance Monitoring** dialog box is displayed.
- Step 3** In the displayed **Cell Performance Monitoring** dialog box, set **Monitor Item** to **Node Synchronization**. Then click **Submit** to start performance monitoring.
- End**

Operation Results

Two types of monitoring data, RFN/BFN difference and transmission delay are displayed in table and chart mode.



2.3 Clock Maintenance Function

This section describes the common maintenance function during the diagnosis of clock faults.

2.3.1 Querying the Status of the BSC Reference Clock

This function enables users to query the status of the BSC reference clock.

Function Description

On the M2000 or LMT client, query the status of the clock used by the current system and the clock switching mode of the current clock phase-locked loop (PLL) according to the clock

status of the GCGa/GCUa board. If the status of the clock source stratum is **Unavailable** or the current state of phase-lock loop is **Unknown**, the clock is lost. Contact associated engineers to rectify the fault until the status of the clock source stratum is **Available** or the current state of phase-lock loop is **Traceable**.

Operation Procedure

1. Menu Mode

In the LMT window, click the **Device Maintenance** tab.

The **Device Maintenance** tab page is displayed.

On the device panel, right-click the GCUa/GCGa board and choose **BSC Board Clock Status Query** from the shortcut menu.

In the **Query BSC Board Clock Status** dialog box, click **Query** to check the clock status of the board, as shown in Figure 2-4.

Figure 2-4 Querying the status of the BSC reference clock

Attribute Name	Attribute Value
Slot No.	12
Board type	GCUa
Phase-locked loop state of cur	Free-running
LevelH state of clock source	Initialization
LevelB state of clock source	Initialization
LevelD state of clock source	Initialization
LevelI state of clock source	Initialization
LevelO state of clock source	Available
System Clock working mode	Auto Handover
Clock source level currently us	0
LevelH clock source	<NULL>
LevelB clock source	<NULL>

2. Using MML commands

Run **DSP CLK** on the RNC to query the status of the clock boards in the MPS. In this step, enter the subrack number and slot number. GCUa and GCGa boards are fixedly configured in slots 12 and 13 in the MPS.

2.3.2 Querying the Status of the BSC Board Clock

This function enables users to query the status of the BSC board clock.

Function Description

This function enables users to query the working status of each board clock according to the clock status of the BSC board and to query the status of the clock used by the current system and the clock switching mode of the current clock phase-locked loop (PLL) according to the clock status of the GCUa board.



CAUTION

The function is not applicable to the FG2a, GOUa, FG2c, or GOUc board.

Operation Procedure

1. Menu Mode

In the LMT window, click the Device Maintenance tab. The Device Maintenance tab page is displayed.

On the device panel, choose a board in position, right-click and choose **BSC Board Clock Status Query** from the shortcut menu to display the **Query BSC Board Clock Status** dialog box.

In the **Query BSC Board Clock Status** dialog box, set parameters and click **Query** to check the clock status of the board.

2. Using MML commands

Run **DSP CLK** on the RNC to query the status of the BSC board clock.

3 Troubleshooting HSPA Service Setup Failures

3.1 About This Chapter

This document describes how to troubleshoot the HSPA service setup failure in the PS domain.

3.2 Definition of HSPA Service Setup Failures

The R99 service in the PS domain is normal and only HSPA services cannot be performed.

NOTE

The cell HSPA function is properly activated. That is, the ALM-22217 UMTS Cell HSDPA Function Fault and ALM-22218 UMTS Cell HSUPA Function Fault are not generated.

3.3 Related Information

The RNC determines whether HSPA services are set up on the HS-DSCH or E-DCH based on the MBR assigned by the CN and the HSPA bearer rate threshold set by the RNC. If the DL MBR assigned by the CN exceeds the HSDPA bearer rate threshold set by the RNC, the HSDPA service is set up on the HS-DSCH. If the UL MBR assigned by the CN exceeds the HSUPA bearer rate threshold set by the RNC, the HSUPA service is set up on the E-DCH. Otherwise, the HSPA services will be set up on the DCH.

Admission of HSUPA and HSDPA user quantity is performed on NodeB level and cell level respectively. If admission fails on either level, the corresponding service will be rejected.

Maximum number of HSUPA users supported by cells = MIN (Maximum number of HSUPA users in a single cell limited by the RNC license, Maximum number of HSUPA users supported by the NodeB)

Maximum number of HSDPA users supported by cells = MIN (Maximum number of HSDPA users in a single cell limited by the RNC license, Maximum number of HSDPA users supported by the NodeB)

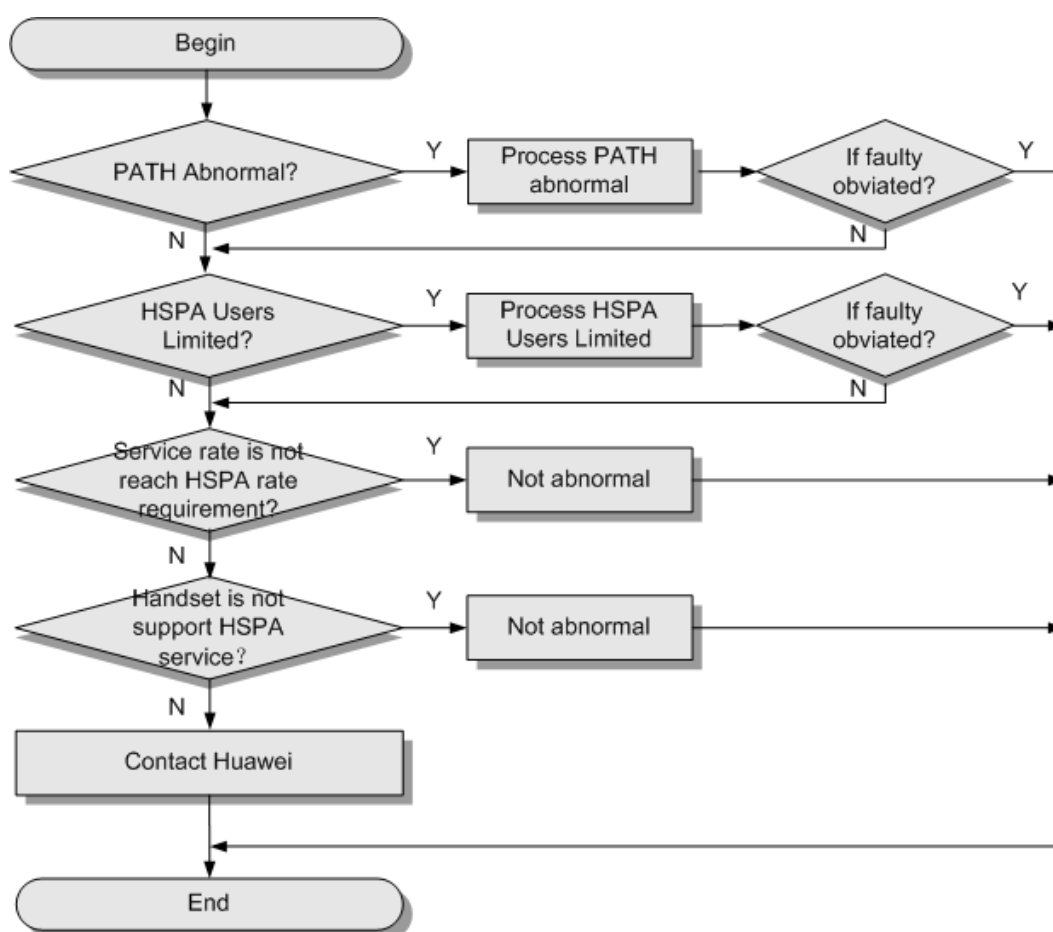
3.4 Possible Causes

- The AAL2PATH or IPPATH is abnormal.
- Failure to admit HSUPA and HSDPA user quantity occurs.
- The service rate does not meet the threshold of HSPA services.
- The terminal does not support HSPA services.

3.5 Troubleshooting Flowchart

Figure 3-1 shows the troubleshooting flowchart.

Figure 3-1 Troubleshooting flowchart



3.5.1 Troubleshooting Abnormal AAL2PATH or IPPATH



NOTE

The MML commands involved in this section are all executed on the RNC. Troubleshooting methods for the HSUPA and HSDPA service are the same in different scenarios. So make the HSUPA service as an example.

- Step 1** Check whether the VS.HSUPA.RAB.FailEstab.ULIUBBAnd.Cong of faulty cells increases obviously.

If yes, go to Step 2; if no, exit.

Step 2 Run **LST UCELL** to find the corresponding NodeB Name (**NodeBName**) based on Cell ID (**CellId**).

Step 3 Run **LST ADJNODE** to find the corresponding Adjacent Node ID based on Adjacent Node Name (**NodeBName**) in Step 2.

Step 4 Run **LST ADJMAP** to find Gold user TRMMAP index, Silver user TRMMAP index, and Bronze user TRMMAP index based on Adjacent Node ID (**ANI**) in Step 3.

Step 5 Run the **LST TRMMAP** to find the corresponding **path type** set up for the service based on TRMMAP index in Step 4.

Step 6 Check whether the path exists based on the transmission mode of the Iub interface.

If...	Then...
Interface type is Iub interface and Transport type includes ATM	Go to Step 7.
Interface type is Iub interface and Transport type includes IP	Go to Step 11.

Step 7 Run **LST ATMTRF** to check whether there are the ATM traffic records of the Service type upon the **path type** in Step 5.

If yes, record Traffic index and go to Step 8.

If no, **path type** corresponding to this site does not exist. Go to Step 9.

Step 8 Run **LST AAL2PATH**. Check whether the path whose AAL2 Path Type matches path type in Step 5 and TX traffic record index, RX traffic record index value matches Traffic index in Step 7 exists.

If yes, record the AAL2 path ID and go to Step 10.

If no, go to Step 9.

Step 9 Run **MOD TRMMAP** to change the path of corresponding services to the corresponding service category or run **ADD AAL2PATH** to initially configure a link. Check whether the fault is rectified. If yes, no further action is required. If no, go to Step 14.

Step 10 Check whether the AAL2PATH link is normal.

Run **DSP AAL2PATH** or check for the ALM-21581 Path Fault to determine whether link status is normal.

If yes, exit.

If no, see section 13.4 "Troubleshooting AAL2 Path Faults."

Step 11 Run **LST IPPATH** to determine whether the path in Step 5 exists based on IP path type value

If yes, go to Step 12.

If no, go to Step 13.

Step 12 Check whether the IPPATH is available.

Analyze whether the ALM-21581 Path Fault is generated based on alarms.

If yes, see section 14.4 "Troubleshooting IP Path Faults."

If no, go to Step 13.

Step 13 Run **MOD TRMMAP** to change corresponding **path** of the service to the existing link type or run **ADD IPPATH** to initially configure a link. Check whether the fault is rectified. If yes, no further action is required. If no, contact Huawei technical support.

Step 14 Collect fault information and the following information and provide the information for Huawei technical support.

- MML scripts of RNC configuration data
- RNC Iub interface tracing
- RNC UE tracing
- Results of running **DSP UCELLCHK** on the RNC
- RNC alarm logs

3.5.2 Troubleshooting Failures to Admit HSUPA User Number and HSDPA User Number



NOTE

The MML commands involved in this document are all executed on the RNC. Troubleshooting methods for HSUPA and HSDPA service are the same in different scenarios. So make HSUPA service as an example.

Step 1 Run **DSP UCELLCHK** to query the number of current cell HSUPA and HSDPA users.

```

Cell Uu uplink congested state = Not congested
Cell Uu downlink congested state = Not congested
Cell uplink load[%] = 0
Cell downlink load[%] = 0
Current cell DCH user number = 0
Current cell FACH user number = 0
Current cell HSDPA user number = 0
Current cell HSUPA user number = 0
Count of cell current 64QAM users = 0
Count of cell current MIMO users = 0
Count of cell current EFach users = 0
Count of cell current DtxDrx users = 0
Count of cell current HsScchLessOperation users = 0
Current cell uplink equivalent user number[*100] = 0
Current cell downlink equivalent user number[*100] = 0
Current user uplink switch algorithm = algorithm two

```

Step 2 Run **LST LICENSE** to query related switch items with the maximum number of HSUPA users and HDPA users in functional items.

For example, if the query results are as follows, the maximum number of HSUPA users supported by the cell is 128.

60 HSUPA users per cell = ON

96 HSUPA users per cell = ON

128 HSUPA users supported by a single cell = ON

Step 3 Run **LST UCELLCAC** to query the maximum number of HSUPA users and HSDPA users based on the cell admission algorithm.

```

HSDPA best effort PBR threshold = 30
Maximum HSDPA user number = 64
Max UL TX power of conversational service = 24
Max UL TX power of streaming service = 24
Max UL TX power of interactive service = 24
Max UL TX power of background service = 24
Background noise = 61
Default PCPICH Ec/No = -13
Auto-adaptive background noise update switch = ON
Background noise update continuance Time = 120
Equivalent user number threshold for background noise = 0
Maximum HSUPA user number = 20
DL HSUPA reserved factor = 0

```

Step 4 Run **LST UNODEBALGOPARA** to check for the maximum number of HSUPA and HSDPA users supported by the NodeB.

```

%%LST UNODEBALGOPARA: LstFormat=VERTICAL;%%
RETCODE = 0 Execution succeeded.

List NodeB Algorithm Parameters
-----
NodeB Name = "RNC182-BBU1"
NodeB LDC algorithm switch = IUB LDR Algorithm::OFF
                          = NodeB Credit LDR Algorithm::OFF
                          = LCG Credit LDR Algorithm::OFF
                          = IUB OLC Algorithm::OFF
NodeB Max HSDPA User Number = 3840
NodeB Max HSUPA User Number = 3840
HSUPA Credit Consume Type = MBR
Reserved parameter 1 = Reserved Switch 1::OFF
                    = Reserved Switch 2::OFF

```

Step 5 Determine the relationship between current users and maximum number of users supported.

If the Number of Current Users is close to the Maximum Number of Users Supported, the number of HSUPA users is insufficient. Increase the number of supported HSUPA users.

- If the fault is rectified, no further action is required.
- If no, go to Step 6.

Number of Current Users	Maximum Number of Users Supported
Number of current HSUPA users of cells in Step 1	MIN (Maximum number of users in a single cell limited by the RNC license in Step 2, Maximum number of HSUPA users set in the cell admission algorithm in Step 3, Maximum number of HSUPA users supported by the NodeB in Step 4)
Total number of current HSUPA users of cells in Step 1	Maximum number of HSUPA users supported by the NodeB in Step 4

Step 6 Collect fault information and the following information and provide the information to Huawei technical support.

- MML scripts of RNC configuration data
- RNC Iub interface tracing
- RNC UE tracing
- Results of running **DSP UCELLCHK** on the RNC
- RNC alarm logs

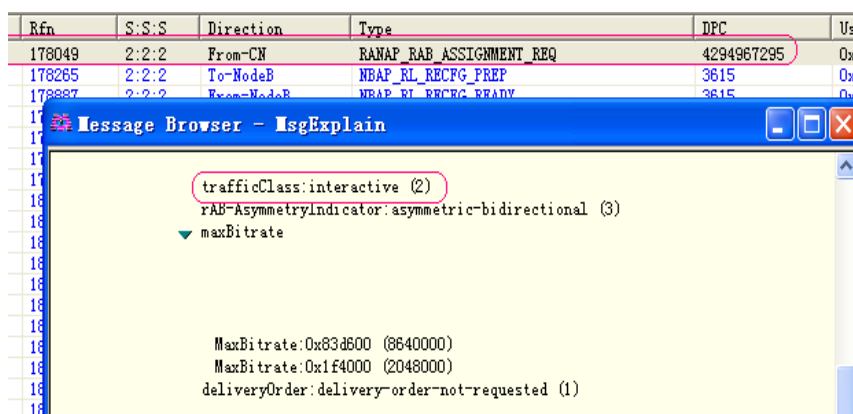
3.5.3 Determining Whether the Service Rate Mismatch the Threshold of HSPA Services



NOTE

The MML commands involved in this section are all executed on the RNC.

Step 1 Check service categories. Query the value of the trafficClass information element in the RANAP_RAB_ASSIGNMENT_REQ message delivered by the CN.



Step 2 Query the HSPA rate threshold related to the traffic in Step 1. Run **LST UFRCHLTYPEPARA**.

```

+++   bsc6900           2010-03-15 20:38:39
O&M   #148653
%%LST UFRCHTYPEPARA:;%
RETcode = 0   Execution succeeded.

List Channel Type Parameters
-----
                CS voice channel type = UL_DCH, DL_DCH
                VOIP channel type = UL_DCH, DL_DCH
                IMS channel type = UL_DCH, DL_DCH
                SRB channel type = UL_DCH, DL_DCH
                SRB channel type RRC effect flag = FALSE
                UL BE traffic DCH decision threshold[kbit/s] = D8
                DL BE traffic DCH decision threshold[kbit/s] = D8
                DL streaming traffic threshold on HSDPA[kbit/s] = D64
                DL BE traffic threshold on HSDPA[kbit/s] = D128
                UL streaming traffic threshold on HSUPA[kbit/s] = D256
                UL BE traffic threshold on HSUPA[kbit/s] = D608
(Number of results = 1)

```

Step 3 Determine the relationship between the actual rate and the HSPA rate threshold in Step 2.

If the actual rate is less than the HSPA rate threshold, the actual rate does not meet the HSPA rate requirements and no further action is required. Otherwise, go to Step 4.

Step 4 Collect fault information and the following information and provide the information to Huawei technical support.

- MML scripts of RNC configuration data
- RNC Iub interface tracing
- RNC UE tracing
- Results of running **DSP UCELLCHK** on the RNC
- RNC alarm logs

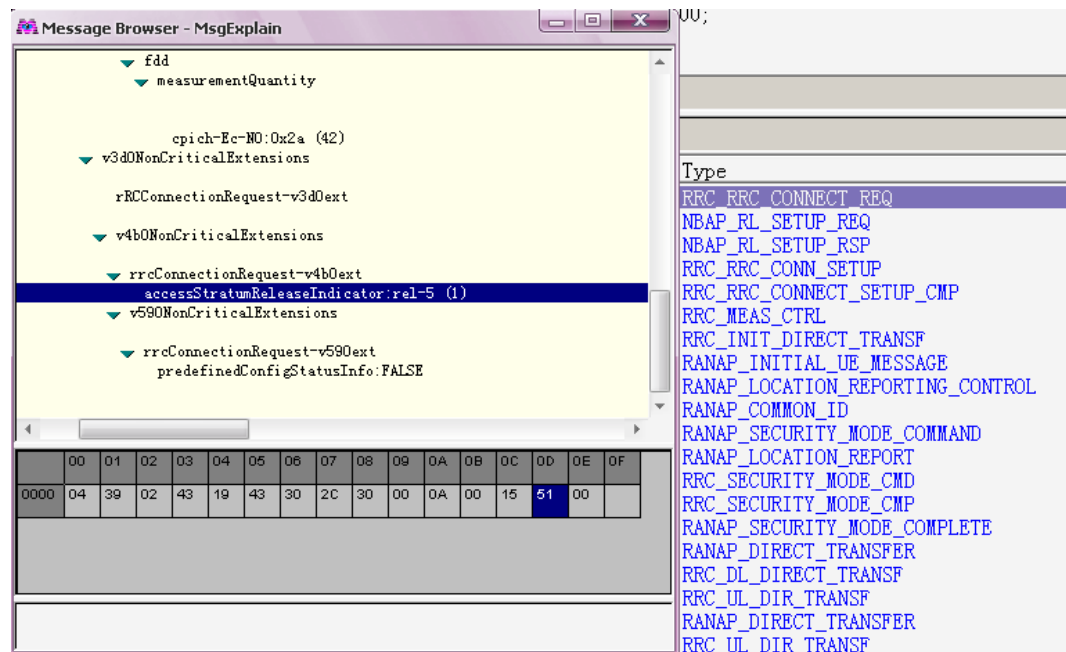
3.5.4 Determining Whether the Terminal Supports the HSPA Services

Step 1 (Optional) Determine whether the terminal supports the HSDPA service.

Query the accessStratumReleaseIndicator information element of the RRC CONNECTION SETUP REQ message.

If rel-5 and later versions are displayed, go to Step 2.

Otherwise, the terminal does not support the HSDPA service and no further action is required.



Step 2 (Optional) Determine whether the terminal supports the HSUPA service.

Query the **accessStratumReleaseIndicator** information element of the **RRC CONNECTION SETUP REQ** message.

If rel-6 and later version are displayed and the **ueCapabilityIndication** information element is displayed as the **hsdch-edch** information element, go to step 3.

Otherwise, the terminal does not support the HSUPA service and no further action is required.

Step 3 Collect fault information and the following information and provide the information to Huawei technical support.

- MML scripts of RNC configuration data
- RNC Iub interface tracing
- RNC UE tracing
- Results of running **DSP UCELLCHK** on the RNC
- RNC alarm logs

3.6 Typical Cases

Fault Description

The RNC HSUPA RAB success rate becomes small and the HSUPA RAB success rate of several cells is 0.

Fault Handling

Step 1 Analyze one site whose HSUPA RAB success rate is 0. The Iub interface is in ATM transmission mode and the ANI is 287. The VS.HSUPA.RAB.FailEstab.ULIUBBand.Cong increases significantly.

Step 2 Run **LST ADJMAP** and get the value of TMI (24) based on the ANI (287).

- Step 3** Run **LST TRMMAP**. Find that the HUSRBPRIPATH is the RT_VBR based on the TMI (24).
- Step 4** Run **LST AAL2PATH**. There is one link with AAL2PATH equals HSPA. It's TXTRFX and RXTRFX is 158.
- Step 5** Run **LST ATMTRF**. Find that the ST is UBR based on the TRFX (158). So The HSPA AAL2PATH of the RT_VBR does not exist.
- Step 6** Get the Conclusion:
- The RNC is not configured with the PATH for the HSUPA signaling bearer. This results in failure to set up the HSUPA service.

Fault Rectification

The PATH for the HSUPA signaling bearer is added.

4 Troubleshooting HSUPA Data Transmission Faults

4.1 About This Chapter

This chapter describes the types of HSUPA data transmission faults, the handling procedure.

4.2 Definition of HSUPA Data Transmission Faults

The uploading rate of the PS HSUPA service is low or fluctuates.

4.3 Related Information

4.3.1 Requisites for Reaching HSUPA Maximum Rate

- **Capabilities of UEs during HSUPA service**

According to 3GPP TS25.306 specifications, there are six categories of HSUPA supporting categories for UEs. As show in Figure 4-1, these UEs support a rate ranging from 711 kbit/s to 5.74 Mbit/s at the MAC layer. Only UEs in Category 6 support a rate up to 5.74 Mbit/s at the MAC layer.

Figure 4-1 HSUPA supporting capabilities of UEs

E-DCH Category	Max. Capability Combination	E-DCH TTI	Max. Data Rate (Mbit/s)		
			MAC Layer 10 ms TTI	MAC Layer 2 ms TTI	Air Interface
Category 1	1 x SF4	10 ms only	0.71	-	0.96
Category 2	2 x SF4	10 ms and 2 ms	1.45	1.40	1.92
Category 3	2 x SF4	10 ms only	1.45	-	1.92
Category 4	2 x SF2	10 ms and 2 ms	2.0	2.89	3.84
Category 5	2 x SF2	10 ms only	2.0	-	3.84
Category 6	2 x SF4 + 2 x SF2	10 ms and 2 ms	2.0	5.74	5.76

- **Channelization code available in E-DPDCH during HSUPA service**

According to the 3GPP TS25.213 specifications, a UE can be assigned four EDPDCHs to reach a maximum channelization code of $2 \times \text{SF4} + 2 \times \text{SF2}$ only when the SRB is set up on the HSUPA (that is, no DPDCH channels exist). A UE can be assigned two EDPDCHs to reach a maximum channelization code of $2 \times \text{SF2}$ when the SRB is set up on the DCH (that is, one DPDCH exists) as shown in Figure 4-2.

Figure 4-2 E-DPDCH channelization code

$N_{\text{max-dpdch}}$	E-DPDCH _k	Channelisation code $C_{\text{ed},k}$
0	E-DPDCH ₁	$C_{\text{ch},\text{SF},\text{SF}/4}$ if $\text{SF} \geq 4$ $C_{\text{ch},2,1}$ if $\text{SF} = 2$
	E-DPDCH ₂	$C_{\text{ch},4,1}$ if $\text{SF} = 4$ $C_{\text{ch},2,1}$ if $\text{SF} = 2$
	E-DPDCH ₃ E-DPDCH ₄	$C_{\text{ch},4,1}$ 0
1	E-DPDCH ₁	$C_{\text{ch},\text{SF},\text{SF}/2}$
	E-DPDCH ₂	$C_{\text{ch},4,2}$ if $\text{SF} = 4$ $C_{\text{ch},2,1}$ if $\text{SF} = 2$

4.4 Troubleshooting Low or Fluctuating HSUPA Rate

4.4.1 Fault Description

The uploading rate of PS HSUPA services is low or fluctuates.

4.4.2 Possible Causes

- The path where the SRB is located does not support HSUPA.
- The SIM card has a low data rate upon subscription.
- UEs have poor support for HSUPA.
- CE resources are insufficient.
- The uplink signal in the cell is of poor quality.
- Some cells do not support the corresponding data rate.

4.4.3 Fault Handling Procedure

Step 1 (Optional) According to section 4.3.1 "Requisites for Reaching HSUPA Maximum Rate," check whether the path for SRB over HSUPA is available when the target data rate is 5.74 Mbit/s.

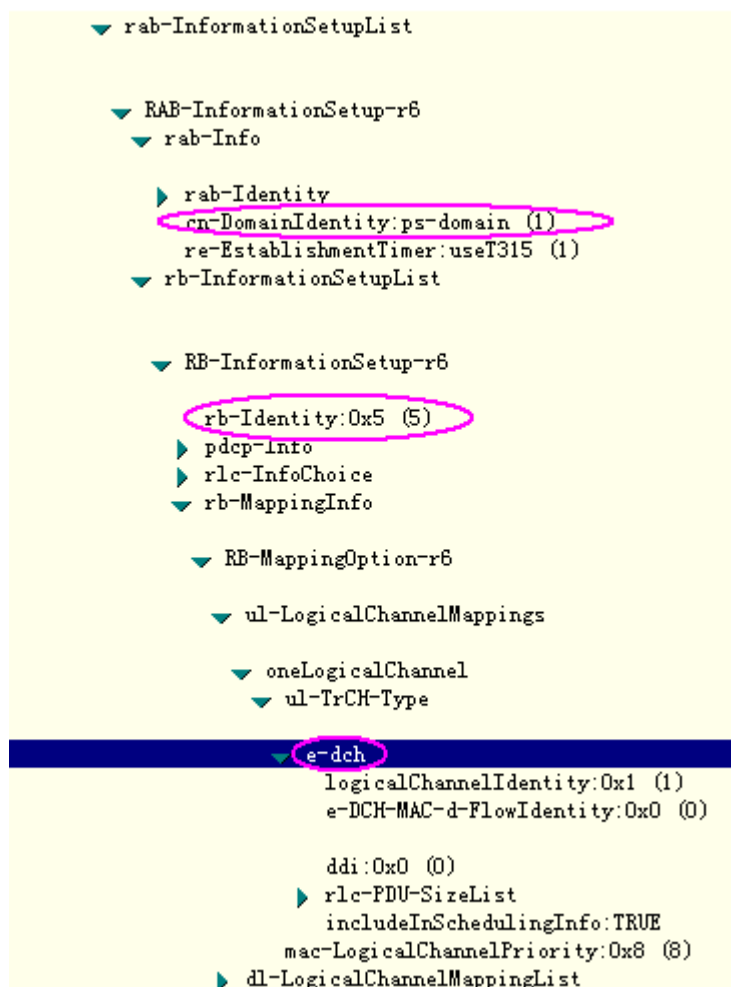
Checking path according to section 3.5.1 "Troubleshooting Abnormal AAL2PATH or IPPATH."

- If yes, go to Step 2.
- Otherwise, if the problem is solved, troubleshooting ends; if not, go to Step 2.

Step 2 Check whether the service is set up on the EDCH.

Check the **cn-DomainIdentity**, **rb-Identity**, and **ul-LogicalChannelMappings** information elements (IE) in the **RRC_RB** message:

- If the value of **cn-DomainIdentity** is **ps-domain** and the value of **ul-TrCH-Type** under this **rb** is **edch** when the value of **rb-Identity** is greater than 4, as shown in the Figure 4-3, the PS service is set up on the EDCH. Go to Step 3.
- Otherwise, go to chapter 3 "Troubleshooting HSPA Service Setup Failures."

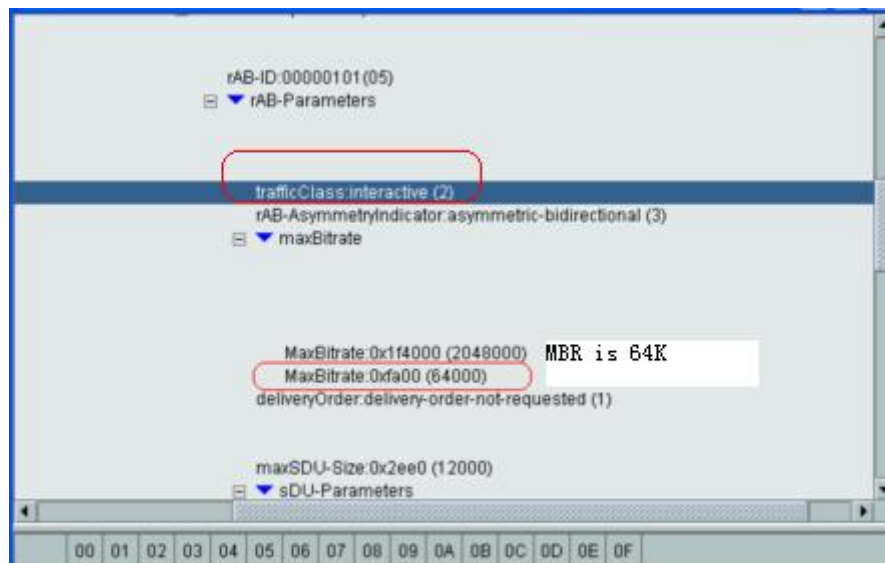
Figure 4-3 IEs of the RRC RB SETUP message

Step 3 Check whether the assigned maximum rate is greater than the required rate.

Check the MaxBitRate IE in the RANAP_RAB_ASSIGNMENT_REQ message to determine whether the maximum uplink bit rate assigned by the CN is greater than the required rate.

- If yes, go to Step 4.
- If no, ask the CN side to solve the problem by checking USIM card subscription information.

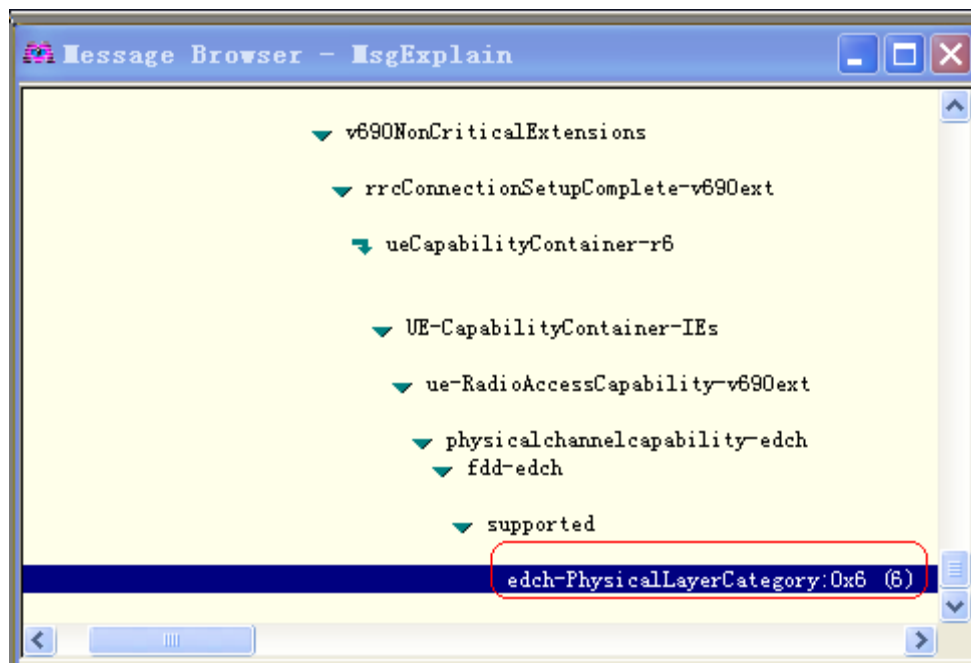
Figure 4-4 Service type and maximum bit rate information in RANAP_RAB_ASSIGNMENT_REQ message



Step 4 Check whether the UE supports the required rate.

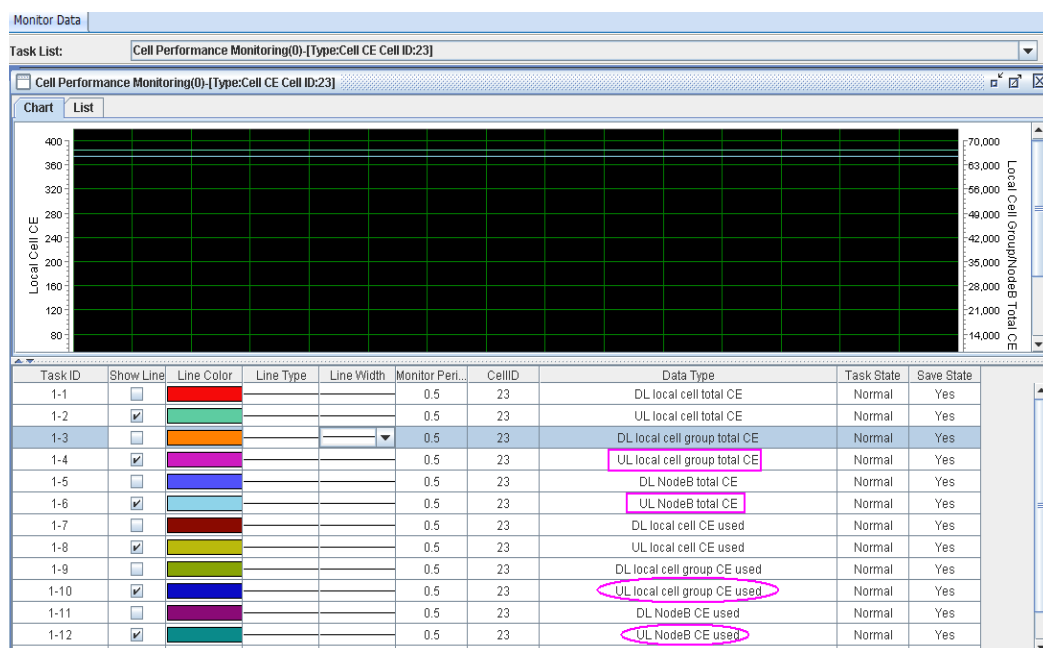
View the **edch-PhysicalLayerCategory** IE in the **RRC_CONNECT_SETUP_CMP** message as shown in Figure 4-5 and then determine whether the value of **Max.Data Rate** corresponding to the UE capability based on Figure 4-1 HSUPA supporting capabilities of UEs is greater than the required rate.

- If yes, go to Step 5.
- Otherwise, the UE does not support the rate. Change another UE. If the problem is solved,, the troubleshooting ends; if not, go to Step 8.

Figure 4-5 The UE Capacity information in RRC_CONNECT_SETUP_CMP message**Step 5** Check whether uplink CE resources are insufficient.

Start **Cell Performance Monitoring**, set **Monitor Item** to **Cell CE**, and check whether the value of **UL Local Cell Group Total CE Used** or **UL NodeB Total CE Used** is close to that of **UL Local Cell Group Total CE** or **UL NodeB Total Cell** as shown in Figure 4-6.

- If yes, insufficient CE resources can be determined as the problem. The troubleshooting ends.
- If no, go to step 6.

Figure 4-6 Checking cell CE on the RNC

Step 6 Check whether the UE transmit power is limited.

Start **Connection Performance Monitoring**, and set **Monitor Item** to **UE Tx Power**.

- If the tracking result shows that the UE transmit power often reaches 20 dBm, the air interface is of poor uplink quality, and the UE transmit power is close to the maximum value (typically 24 dBm), resulting in a low HSUPA rate. It is recommended that you observe the transmit power in areas with good coverage (RSCP > -90 dBm). The troubleshooting ends.
- If the transmit power hardly reaches 20 dBm, go to Step 7.

Step 7 Check for changes in the uplink bandwidth assigned by the RNC.

Start **Connection Performance Monitoring**, set **Monitor Item** to **UL Throughput Bandwidth**.

- If the uplink bandwidth assigned by the RNC decreases, view the signaling to check whether the UE is handed over to a cell with a different HSUPA supporting capability (for example, the UE is handed over from a cell that supports 5.76 Mbit/s to a cell that only supports 1.44 Mbit/s). If yes, modify the neighboring cells and check again.
- If no, go to Step 8.

Step 8 Contact Huawei.

4.4.4 Typical Cases

Fault Description

In office L in country C, the HSUPA rate stays around 600 kbit/s at some sites and reaches a maximum of 608 kbit/s, unable to reach the bandwidth of 5.4 Mbit/s.

Possible Causes

As the path for SRB over HSUPA has not been set, the service cannot be set up at the 5.4 Mbit/s rate.

Fault Handling

Check whether the configuration meets the following requirements:

1. Typical services at the uplink rate of 5.44 Mbit/s have been configured.

```
ADD UTYPRABASIC:RABINDEX=54, APPLIEDDIRECT=APPLIED_ON_DOWNLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=1800000,
ADD UTYPRABASIC:RABINDEX=55, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=2048000, SSD=
ADD UTYPRABASIC:RABINDEX=56, APPLIEDDIRECT=APPLIED_ON_UPLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=2720000, SS=
ADD UTYPRABASIC:RABINDEX=57, APPLIEDDIRECT=APPLIED_ON_DOWNLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=3600000, SS=
ADD UTYPRABASIC:RABINDEX=58, APPLIEDDIRECT=APPLIED_ON_UPLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=5440000, SS=
ADD UTYPRABASIC:RABINDEX=59, APPLIEDDIRECT=APPLIED_ON_DOWNLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=7200000,
ADD UTYPRABASIC:RABINDEX=60, APPLIEDDIRECT=APPLIED_ON_DOWNLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=10100000,
ADD UTYPRABASIC:RABINDEX=61, APPLIEDDIRECT=APPLIED_ON_DOWNLINK, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=INTERACTIVE, MAXBITRATE=13900000,
```

2. The SRB over HSPA function is enabled on the RNC.

In the SET UFRCHLTYPEPARA command, SRBCHLTYPE is set to HSPA.

3. For the HSUPA rate, 64 kbit/s, 384 kbit/s, 608 kbit/s and 5.44 Mbit/s are used.

In SET EDCHRATEADJUSTSET, RATE_64KBPS, RATE_384KBPS, RATE_608KBPS, and 5.44 Mbit/s are selected.

4. The site uses the transmission mapping table of 66. In the transmission mapping table, the AAL2 path of RT_VBR is set to carry SRB over HSUPA data.

```
ADD TRMAP:TMI=66, ITFT=IUB, TRANST=ATM_IP, CCHPRIPATH=RT_VBR, CCHSECPATH=EF, SRBPRIPATH=RT_VBR, SRBSECPATH=EF,
VOICEPRIPATH=RT_VBR, VOICESECPATH=EF, CSConvPRIPATH=RT_VBR, CSConvSECPATH=AF43, CSSTRMPRIPath=RT_VBR, CSSTRMSECPATH=AF43,
PSConvPRIPATH=RT_VBR, PSConvSECPATH=AF43, PSSTRMPRIPath=RT_VBR, PSSTRMSECPATH=AF43, PSINTGHPRIPATH=NRT_VBR,
PSINTGHSECPATH=AF23, PSINTMDPRIPATH=NRT_VBR, PSINTMDSECPATH=AF23, PSINTLOWPRIPATH=NRT_VBR, PSINTLOWSECPATH=AF23,
PSBKGPRIPATH=NRT_VBR, PSBKGSECPATH=AF23, HDSRBPRIPATH=RT_VBR, HDSRBSECPATH=EF, HDConvPRIPATH=RT_VBR, HDConvSECPATH=AF43,
HDSRMPRIPath=RT_VBR, HDSRMPSECPATH=AF43, HDINTGHPRIPATH=AF13, HDINTGHSECPATH=UBR, HDINTMDPRIPATH=AF13, HDINTMDSECPATH=UBR,
HDINTLOWPRIPATH=AF13, HDINTLOWSECPATH=UBR, HDBKGPRIPATH=AF13, HDBKGSECPATH=UBR, HUSREPRIPATH=RT_VBR, HUSRBSECPATH=EF,
HUConvPRIPATH=RT_VBR, HUConvSECPATH=AF43, HUSRMPRIPath=RT_VBR, HUSRMPSECPATH=AF43, HUINTGHPRIPATH=AF13, HUINTGHSECPATH=UBR,
HUINTMDPRIPATH=AF13, HUINTMDSECPATH=UBR, HUINTLOWPRIPATH=AF13, HUINTLOWSECPATH=UBR, HUBKGPRIPATH=AF13, HUBKGSECPATH=UBR,
REMARK="ATM_IP", SIPPRIPATH=RT_VBR, SIPSECPATH=EF, HDVOICEPRIPATH=RT_VBR, HDVOICESECPATH=AF43, HUVOICEPRIPATH=RT_VBR,
HUVOICESECPATH=AF43, HDSIPPRIPATH=RT_VBR, HDSIPSECPATH=EF, HUSIPPRIPATH=RT_VBR, HUSIPSECPATH=EF;
```

5. Check whether the TRFX of RTVBR is 140.

```
ADD ATMTRF:TRFX=100, ST=CBR, UT=KBIT/S, PCR=512, CDVT=1024
ADD ATMTRF:TRFX=101, ST=CBR, UT=KBIT/S, PCR=128, CDVT=1024
ADD ATMTRF:TRFX=102, ST=UBR, PLUS, UT=KBIT/S, MCR=64, CDVT=
ADD ATMTRF:TRFX=120, MBS=1000, ST=RTVBR, UT=KBIT/S, PCR=38
ADD ATMTRF:TRFX=121, MBS=1000, ST=NRTVBR, UT=KBIT/S, PCR=3
ADD ATMTRF:TRFX=130, MBS=1000, ST=RTVBR, UT=KBIT/S, PCR=57
ADD ATMTRF:TRFX=131, MBS=1000, ST=NRTVBR, UT=KBIT/S, PCR=5
ADD ATMTRF:TRFX=132, ST=UBR, UT=KBIT/S, PCR=5712, CDVT=102
ADD ATMTRF:TRFX=140, MBS=1000, ST=RTVBR, UT=KBIT/S, PCR=76
ADD ATMTRF:TRFX=141, MBS=1000, ST=NRTVBR, UT=KBIT/S, PCR=7
ADD ATMTRF:TRFX=142, ST=UBR, UT=KBIT/S, PCR=7616, CDVT=102
ADD ATMTRF:TRFX=180, MBS=1000, ST=RTVBR, UT=KBIT/S, PCR=15
ADD ATMTRF:TRFX=181, MBS=1000, ST=NRTVBR, UT=KBIT/S, PCR=1
ADD ATMTRF:TRFX=182, ST=UBR, UT=KBIT/S, PCR=15232, CDVT=1C
```

6. Check whether the AAL2 path type is R99 when TRFX is 140. If yes, HSPA services cannot be carried.

```
ANI=23,PATHID=6,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=55,TXTRFX=142,RXTRFX=142,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=HSPA;  
ANI=23,PATHID=1,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=50,TXTRFX=140,RXTRFX=140,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=R99;  
ANI=23,PATHID=2,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=51,TXTRFX=140,RXTRFX=140,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=R99;  
ANI=23,PATHID=3,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=52,TXTRFX=140,RXTRFX=140,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=R99;  
ANI=23,PATHID=4,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=53,TXTRFX=141,RXTRFX=141,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=R99;  
ANI=23,PATHID=5,CARRYF=0,CARRYT=IMA,CARRYSN=16,CARRYIMAGRPN=8,VP1=1,VCI=54,TXTRFX=142,RXTRFX=142,RSCGRPFLAG=NO,OWNERSHIP=LOCAL,TIMERCU=10,TRMLOADTHINDEX=0,AAL2PATHT=HSPA;
```

Location Result

As the path for SRBoverHSUPA is not set, the service cannot be set up at 5.44 Mbit/s.

Solution

Modify path attributes to allow the path for SRBoverHSUPA to carry HSPA services. The problem is solved.

```
MOD AAL2PATH: ANI=23, PATHID=1, AAL2PATHT=SHARE;  
MOD AAL2PATH: ANI=23, PATHID=2, AAL2PATHT=SHARE;  
MOD AAL2PATH: ANI=23, PATHID=3, AAL2PATHT=SHARE;  
MOD AAL2PATH: ANI=23, PATHID=2, AAL2PATHT=SHARE;  
MOD AAL2PATH: ANI=23, PATHID=3, AAL2PATHT=SHARE;
```

5 Troubleshooting HSDPA Service Rate Faults

5.1 About This Chapter

This chapter describes how to locate abnormalities in the rate of the HSDPA service in the PS domain.

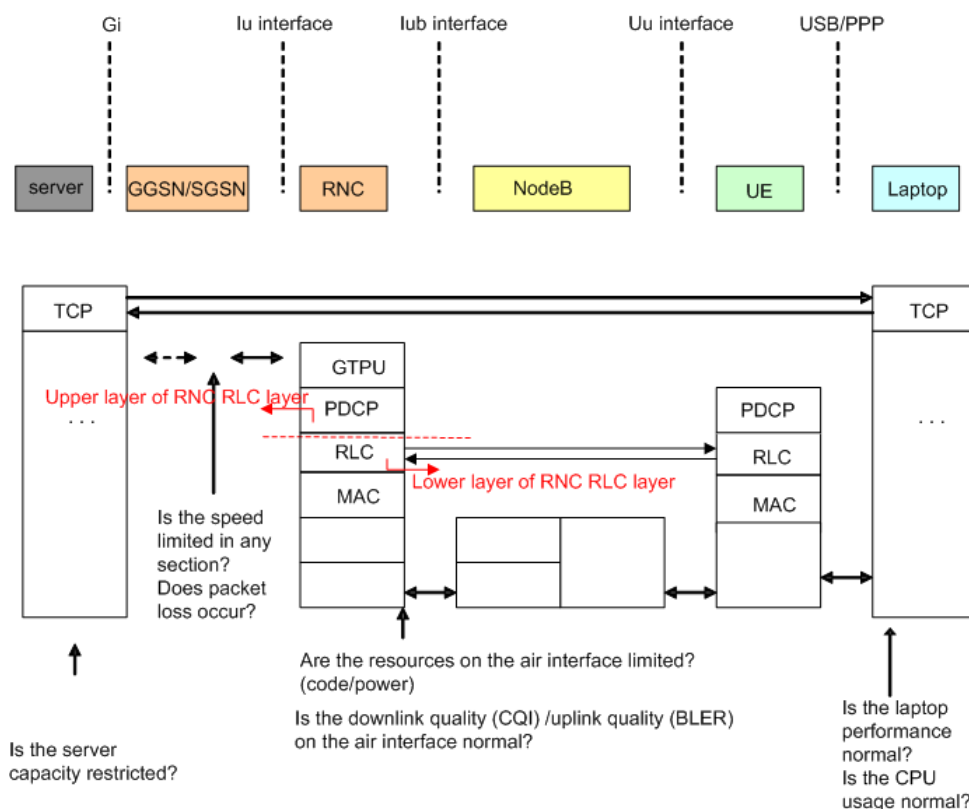
5.2 Definition of HSDPA Service Rate Faults

The PS service is set up on the HSDSCH, and the downlink rate is low or fluctuates.

5.3 Related Information

E2E Handling Process

The HSDPA service rate indicates end-to-end system performance. Abnormalities in any part of the system affect data transmission. Figure 5-1 shows the network elements (NEs) and important processes involved.

Figure 5-1 NEs involved in HSDPA data transmission

Main-layer Handling Process

At the TCP layer, feedback is used for acknowledgement. The data packets in the transmission window are cleared only after receipt of acknowledgement to release space for subsequent data packets. The transmission end caches all data that has been sent but not confirmed, to make sure they can be resent upon negative acknowledgement or the timer is out. If the transmission end still fails to receive acknowledgement, the data packets transmission fails.

At the GTPU and PDCP layers, data packets are transmitted transparently and no problems are incurred.

When the HSDPA service rate is normal, the TCP layer on the server side continuously transmits data to the RNC RLC layer through the Iu interface, and the RNC RLC layer steadily transmits data to the UE through the Iub and Uu interfaces. At this time, the RLC buffer keeps transmitting data and receiving new data.

Methods to Narrow Fault Range

Upon troubleshooting, the segment where the problem occurs can be determined by transmitting emulated packets to the RNC RLC layer.

- If the rate is normal, the abnormality exists above the RNC RLC layer.
- If the rate is abnormal, check for abnormalities below the RNC RLC layer, and recheck whether any abnormality exists above the RNC RLC layer after the problem is solved.

Supporting CQI with Maximum Physical Rate

Table 5-1 lists the mapping between the theoretical rates of HSDPA terminals and the minimum CQI requirements.

Table 5-1 Mapping between the theoretical rates of HSDPA terminals and the minimum CQI requirements

HSDPA handset type	Support Physical Rate	HS-PDSCH code num	The least CQI for Peak Rate
Cat12	1.8Mbit/s	5	18
Cat6	3.6Mbit/s	5	22
Cat8	7.2Mbit/s	10	25
Cat10	14.4Mbit/s	15	26
Cat14	21.56Mbit/s	15	30
Cat18	28.8Mbit/s	15	14

5.4 Troubleshooting Low or Fluctuating HSDPA Service Rate

5.4.1 Fault Description

After the service is set up on the HSDPA channel, the rate does not reach the anticipated level. The following symptoms may appear.

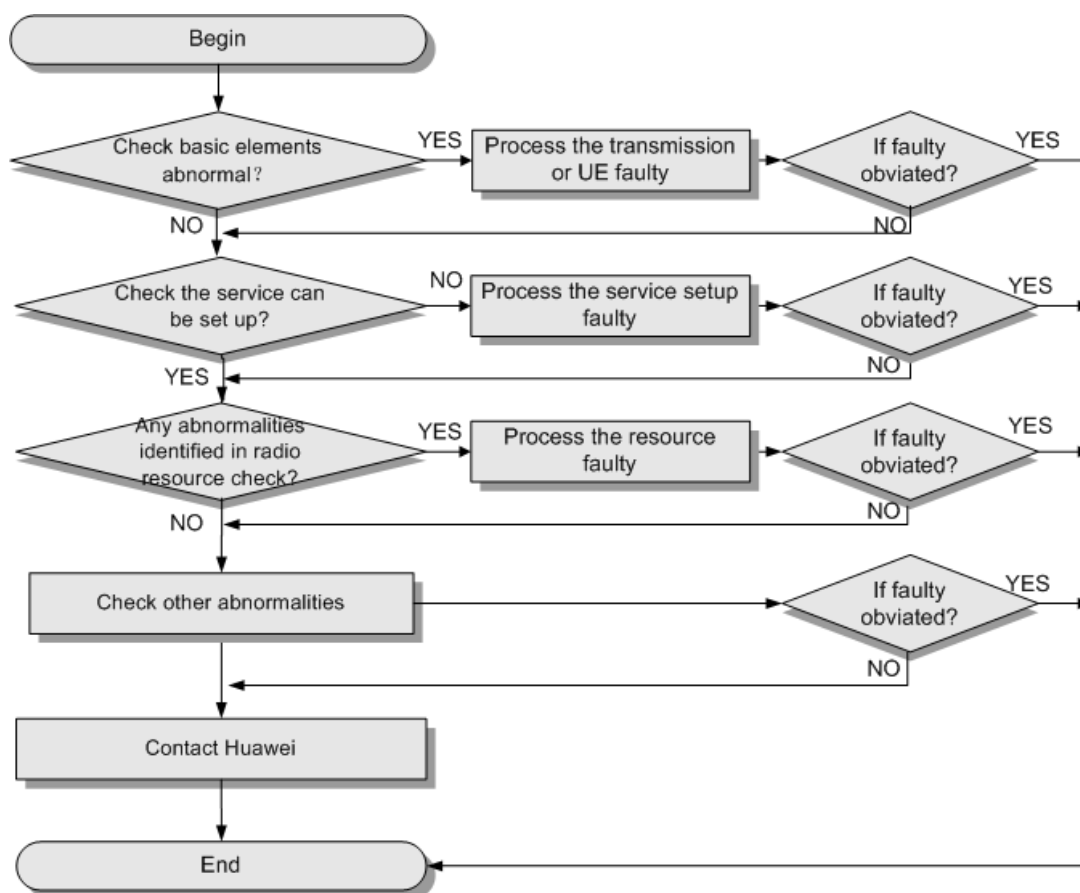
Symptom 1: The downloading rate is low and steady.

Symptom 2: The downloading rate fluctuates regularly, either ascending or descending in steps, or fluctuating in a square wave. During fluctuation, the throughput occasionally reaches the theoretical value.

Symptom 3: The downloading rate fluctuates irregularly, and occasionally reaches the theoretical value but fluctuates dramatically.

5.4.2 Fault Handling Flowchart

Figure 5-2 shows the fault handling flowchart for the low or fluctuating HSDPA service rate.

Figure 5-2 Fault handling flowchart for the low or fluctuating HSDPA service rate

5.4.3 Checking Basic Elements

Step 1 Troubleshoot alarms at the Iub interface link in the homing cell and troubleshoot alarms at the Iu link of the homing RNC.

- If the fault is rectified, no further action is required.
- If the fault persists, go to Step 2.

Step 2 Determine whether the problem lies in a specific terminal by eliminating the following abnormalities.

1. Whether a rate limit is set on the portable computer.

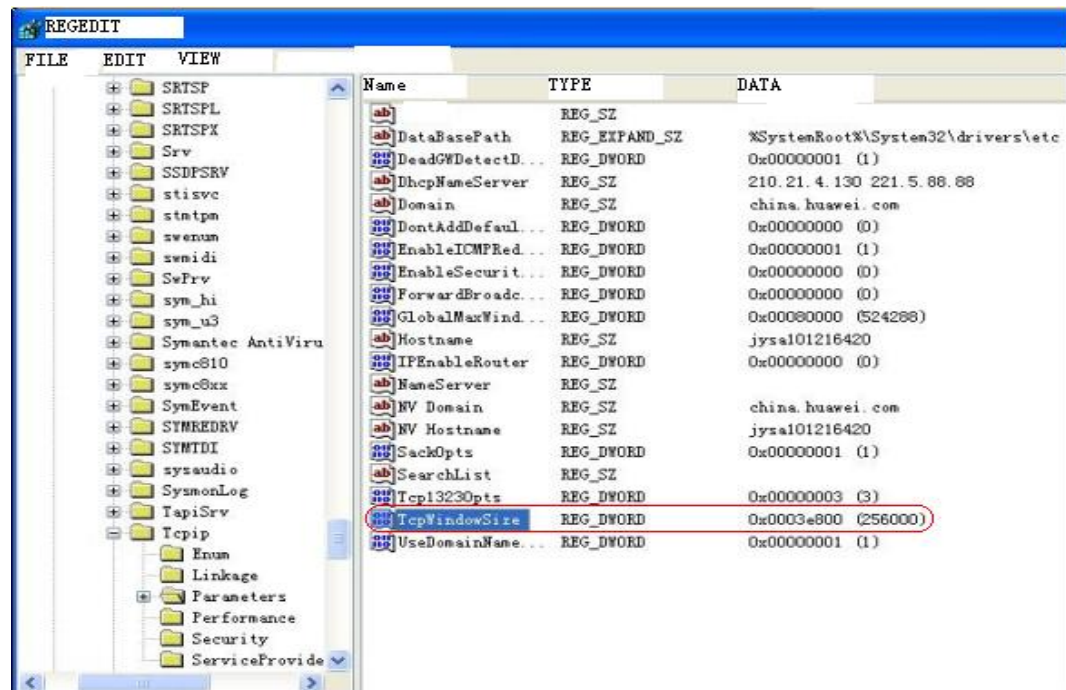
In Windows, choose **Computer Management** -> **MODEM**, and select the relevant terminal. Double-click **Advanced**, and see if the following setting appears in the window.

- If yes, remove the AT command line. If the fault is rectified, no further action is required. If the fault persists, go to Step 3.
- If no, no AT limit is set, go to 2.

For example: in the setting format at +cgeqreq = 1,2,2048,7200, 2 indicates the service type (interactive), and 2048 and 7200 indicate the uplink rate (2 Mbit/s) and the downlink rate (7.2 Mbit/s), respectively.

2. Whether CPU usage of the portable computer is greater than 95%.
 - If yes, change the portable computer.

- If no, go to 3.
- Whether the TCP window on the UE (handset) meet the required rate.
View the TCP window size in the following location of the registry:
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\Tcpip\Parameters\TcpWindowSize.



Check whether the TCP window meet the required rate according to the following table.

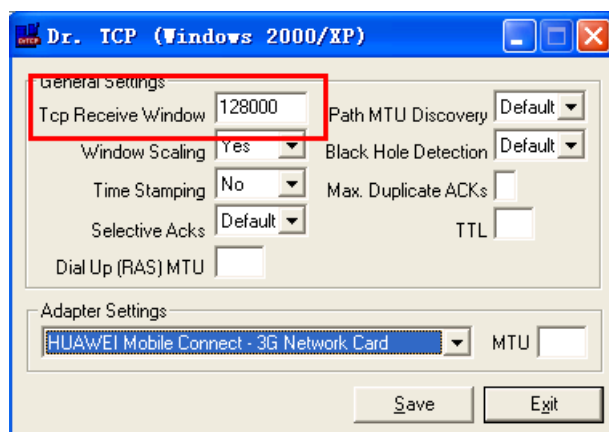
Table 5-2 DL bandwidth VS the minimum values of receive and send window sizes

DL Bandwidth	Scenario	Minimum Value of Receive Window Size
2048 Kbit/s	Only Download	64 Kbytes
3648 Kbit/s	Only Download	128 Kbytes
7200 Kbit/s	Only Download	256 Kbytes

If yes, go to 4.

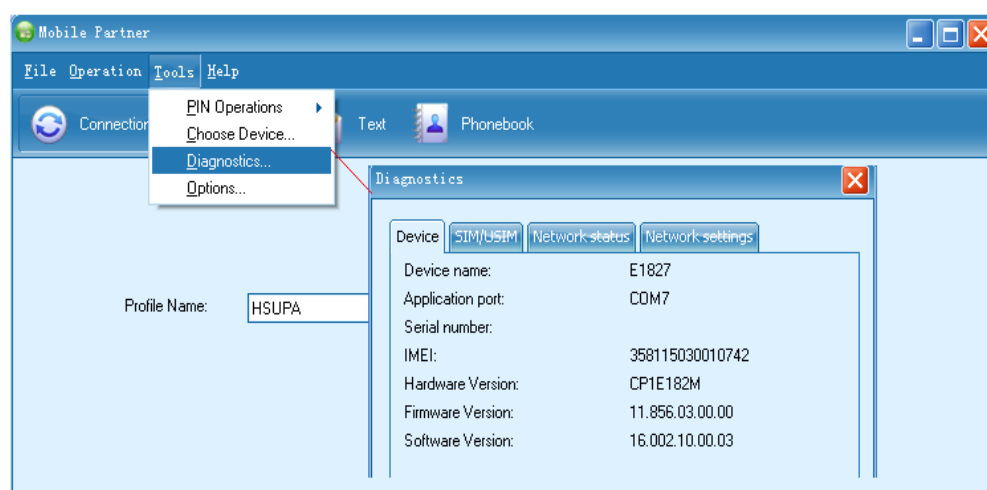
If no, modify the Tcp Receive Window.

Example: Complete setting on the DRTCP software, and restart the RNC after the setting is complete.



4. Make sure the correct terminal driver is used, and otherwise the rate fluctuates or stays low. If a definite result cannot be determined, follow the example below to query the device information. Then, return the device information to the terminal manufacturer for confirmation.

Device information query



- If the correct terminal driver is used, change the portable computer.
- If the correct terminal driver is not used, go to Step 3.

Step 3 Contact Huawei.

5.4.4 Determining Whether the Service Can Be Set Up

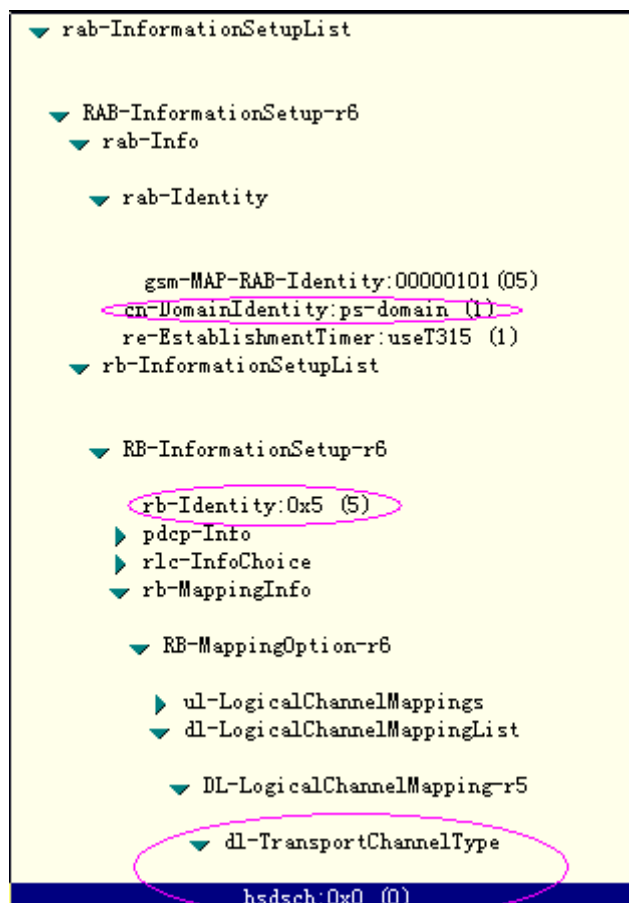
Step 1 Determine whether the service is set up on an HSDSCH.

Check the **cn-DomainIdentity**, **rb-Identity** and **dl-TransportChannelType** IEs in the RRC_RB SETUP messages shown in Figure 5-3.

- If the value of **cn-DomainIdentity** is **ps-domain**, and the value of **dl-TransportChannelType** is **hdsch** when the value of **rb-Identity** is greater than 4, as shown in the figure, the PS service is set up on the HSDSCH. Go to Step 2.

- If the PS service is not set up on the HSDSCH, go to chapter 3 "Troubleshooting HSPA Service Setup Failures."

Figure 5-3 RRC_RB SETUP message



Step 2 Determine whether the enabled license item supports the required rate.

- Run the RNC MML command **LST LICENSE: FN= "license file name"** to check the relevant license item.

Examples of RNC-related license items:

High Speed Downlink Packet Access=ON

High Speed Downlink Packet Access Function 3.6M=ON

High Speed Downlink Packet Access Function 7.2M=ON

High Speed Downlink Packet Access Function 13.976Mbps=ON

HSPA + Downlink 28 Mbit/s Per User=ON

HSPA + Downlink 21 Mbit/s Per User=ON

HSPA+ Downlink 42 Mbit/s per User=OFF

HSPA+ Downlink 84 Mbit/s per User=OFF

- Run the NodeB MML command **DSP LICENSE** to check the relevant license item.

Examples of HSPA related license items:

Operator Index	Operator Name	License Identifier	License Item
65535	Shared	LQW9ULCE01	UL CE Num
65535	Shared	LQW9DLCE01	DL CE Num
65535	Shared	LQW9LOCELLO1	Local Cell Num
65535	Shared	LQW9HSDPA01	HSDPA Function(per NodeB)
65535	Shared	LQW9HUSER01	HSDPA User Num
65535	Shared	LQW9HDPCK01	HSDPA RRM Package 1(per NodeB)
65535	Shared	LQW9HDCODE01	HSDPA Code Number

Examples of HSPA + (64QAM, MIMO, DC) feature related license items:

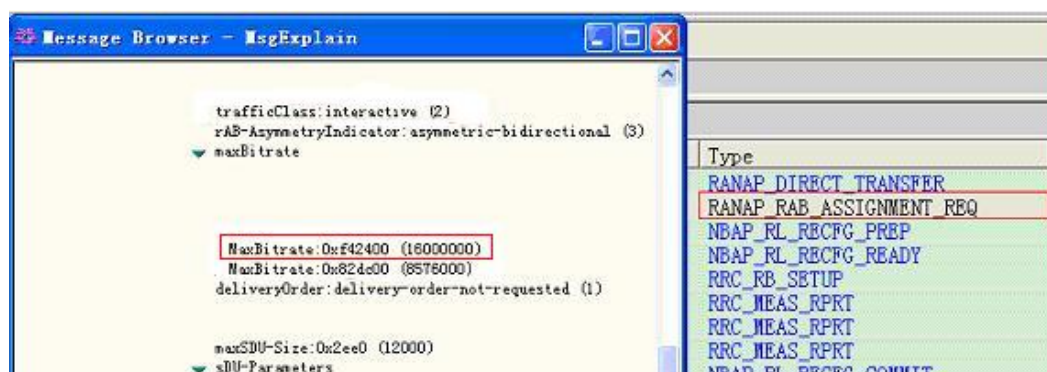
LQW964QAM01	DL 64QAM Function(per Cell)
LQW9MIMO01	MIMO Function(per Cell)
LQW9DDCO1	DC-HSDPA Function(per Cell)

Step 3 Determine whether the assigned maximum rate is greater than the required rate.

Check the **MaxBitRate** IE in the **RANAP_RAB_ASSIGNMENT_REQ** message to determine whether the maximum downlink bit rate assigned by the CN is greater than the required rate as shown in the Figure 5-4.

- If yes, go to Step 4.
- If no, ask the CN side to solve the problem by checking USIM card subscription information.

Figure 5-4 Service type assigned in the RAB assignment message and maximum uplink/downlink bit rate



Step 4 Determine whether the terminal supports the required rate.

Check the **hdsch - physical - layer - category** IE in the **RRC_CONNECT_SETUP_CMP** message as shown in Figure 5-5.

Determine whether the value of "the total number of soft channel bits" corresponding to the **hdsch - physical - layer - category** value of HS-DSCH category is greater than the required rate in the Table 5-3 below.

Table 5-3 HSDPA terminal capacity table

HS-DSCH category [ⓐ]	Maximum number of HS-DSCH codes received [ⓐ]	Minimum inter-TTI interval [ⓐ]	Maximum number of bits of an HS-DSCH transport block received within an HS-DSCH TTI [ⓐ] NOTE 1 [ⓐ]	Total number of soft channel bits [ⓐ]	Supported modulations without MIMO operation [ⓐ]	Supported modulations simultaneous with MIMO operation [ⓐ]
Category 1 [ⓐ]	5 [ⓐ]	3 [ⓐ]	7298 [ⓐ]	19200 [ⓐ]	QPSK, 16QAM [ⓐ]	Not applicable (MIMO not supported) [ⓐ]
Category 2 [ⓐ]	5 [ⓐ]	3 [ⓐ]	7298 [ⓐ]	28800 [ⓐ]		
Category 3 [ⓐ]	5 [ⓐ]	2 [ⓐ]	7298 [ⓐ]	28800 [ⓐ]		
Category 4 [ⓐ]	5 [ⓐ]	2 [ⓐ]	7298 [ⓐ]	38400 [ⓐ]		
Category 5 [ⓐ]	5 [ⓐ]	1 [ⓐ]	7298 [ⓐ]	57600 [ⓐ]		
Category 6 [ⓐ]	5 [ⓐ]	1 [ⓐ]	7298 [ⓐ]	67200 [ⓐ]		
Category 7 [ⓐ]	10 [ⓐ]	1 [ⓐ]	14411 [ⓐ]	115200 [ⓐ]		
Category 8 [ⓐ]	10 [ⓐ]	1 [ⓐ]	14411 [ⓐ]	134400 [ⓐ]		
Category 9 [ⓐ]	15 [ⓐ]	1 [ⓐ]	20251 [ⓐ]	172800 [ⓐ]		
Category 10 [ⓐ]	15 [ⓐ]	1 [ⓐ]	27952 [ⓐ]	172800 [ⓐ]		
Category 11 [ⓐ]	5 [ⓐ]	2 [ⓐ]	3630 [ⓐ]	14400 [ⓐ]	QPSK [ⓐ]	
Category 12 [ⓐ]	5 [ⓐ]	1 [ⓐ]	3630 [ⓐ]	28800 [ⓐ]		
Category 13 [ⓐ]	15 [ⓐ]	1 [ⓐ]	35280 [ⓐ]	259200 [ⓐ]	QPSK, 16QAM, 64QAM [ⓐ]	
Category 14 [ⓐ]	15 [ⓐ]	1 [ⓐ]	42192 [ⓐ]	259200 [ⓐ]		
Category 15 [ⓐ]	15 [ⓐ]	1 [ⓐ]	23370 [ⓐ]	345600 [ⓐ]	QPSK, 16QAM [ⓐ]	
Category 16 [ⓐ]	15 [ⓐ]	1 [ⓐ]	27952 [ⓐ]	345600 [ⓐ]		
Category 17 NOTE 2 [ⓐ]	15 [ⓐ]	1 [ⓐ]	35280 [ⓐ]	259200 [ⓐ]	QPSK, 16QAM, 64QAM [ⓐ]	— [ⓐ]
			23370 [ⓐ]	345600 [ⓐ]	— [ⓐ]	QPSK, 16QAM [ⓐ]
Category 18 NOTE 3 [ⓐ]	15 [ⓐ]	1 [ⓐ]	42192 [ⓐ]	259200 [ⓐ]	QPSK, 16QAM, 64QAM [ⓐ]	— [ⓐ]
			27952 [ⓐ]	345600 [ⓐ]	— [ⓐ]	QPSK, 16QAM [ⓐ]
Category 19 [ⓐ]	For future use; supports the capabilities of category 17 in this version of the protocol [ⓐ]					
Category 20 [ⓐ]	For future use; supports the capabilities of category 18 in this version of the protocol [ⓐ]					

- If the **hdsch-physical-layer-category** reported by the UE meets the theoretical rate requirement, go to Step 5.
- If no, terminal capacity does not support the rate. Replace the terminal and observe again. If the alarm is cleared, the troubleshooting ends. If no, go to Step 5.

Example: hsdSCH – physical – layer – category:0xe indicates the UE is an HSDPA category 14 terminal and supports a throughput of 21 Mbit/s at the physical layer.

Figure 5-5 Capacity information reported by the UE in the RRC_CONNECT_SETUP_CMP message

[illegible]

Step 5 Contact Huawei.

5.4.5 Determining Whether Radio Resources Are Limited

Step 1 Determine whether the quality of the downlink signal meets any of the following conditions.

- Determine whether the CQI measured from the UE stays greater than the minimum CQI needed by the required rate.

Check the CQI value reported by the UE during the service in the **HSDPA Link Statistics** item of drive test software (such as QXDM, Probe).

According to the Table 5-1 Mapping between the theoretical rates of HSDPA terminals and the minimum CQI requirements, check **The least CQI for Peak Rate** value when the **Support Physical Rate** is greater than the required rate.

Determine whether the CQI value reported by the UE stays greater than **The least CQI for Peak Rate** value.

- Determine whether the RSCP reported by the UE is greater than -80 dBm and whether EcN0 is greater than -3 dB (no users exist in the cell) or -11 dB (during HSPA service).

Enable the **Connection Performance Monitoring** function, and set **Monitoring Item** to **Cell SNR and Reception Signal Code Power**.

If yes, go to Step 2.

If no, poor air interface quality can be identified as the problem. Check air interface quality and observe again. If the problem is solved, the troubleshooting ends; if not, go to Step 4.

Step 2 Determine whether code resources are used up.



NOTE

C(016, number):0 indicates the status of the SF16 code whose code number value equals number, and 0 indicates that the code status is idle.

C(016, number):5 indicates the status of the SF16 code whose code number value equals number, and 5 indicates that the code status is the HSPDSCH channel is occupied.

1. Open the **Cell Performance Monitoring** dialog box of each cell under the local NodeB, set **Monitoring Item** to **Cell Code Tree Usage** and save the file.
 - Observe the status of the SF16 code on the LMT interface, which applies to the real-time monitoring scenarios.
 - Analyze the usage of C(016, number) codes in the saved result file, which applies to scenarios of analyzing the whole process.
2. Determine whether the cell contains any SF16 code under the code free status.
 - If yes, go to Step 4.
 - If no, go to 3.
3. Run the NodeB MML command **DSP license** to query the value of the license item HSDPA Code Number.
4. Determine whether the total number of SF16 codes under the Code Assigned to HSPDSCH status in 1 of all cells under NodeB is close to the number specified by the license item HSDPA Code Number in 3.
 - If yes, insufficient code resources can be determined as the problem.
 - Check if the rate is normal with sufficient code resources under the idle status.
 - If yes, increase code resources.
 - If no, contact Huawei.

Step 3 Determine whether power resources are used up.

1. Run the RNC MML command **LST UCELLHSDPA** to check whether The Offset of HSPA Total Power in the cell is the baseline value of 0.
If yes, go to 2.
If no, run the RNC MML command **MODUCELLHSDPA** to set the The Offset of HSPA Total Power (HspaPower) to 0.
2. Run the NodeB MML command **LST MACHSPARA**. Check whether the power margin is 5, and whether the Max Power Per Hs-user is 100.
If yes, go to 3.
If no, run the NodeB MML command **SET MACHSPARA** to set the values.
3. Open the **Cell Performance Monitoring** dialog box, and set Monitoring Item to Cell Downlink Carrier Tx Power.
4. Determine whether 95% is reached during data transmission.
 - If yes, the transmission power is limited. Check if the rate is normal with sufficient transmission power resources under the idle status. If yes, expand the NodeB. If no, contact Huawei.
 - If no, contact Huawei.

Step 4 Contact Huawei.

5.4.6 Check Faults Segment by Segment

Step 1 Simulate downlink data transmission by using the **Auto Ping** function as shown in Figure 5-6.

Determine whether the target rate is reached.

- If yes, no abnormalities exist below the RNC, and abnormalities above the Iu interface result in insufficient data sources. Go to Step 2.
- If no, check for abnormalities below the RNC. Go to Step 3.



NOTE

set appropriate **Ping Interval** and **Packet Length** values based on the target rate required.

If **Ping Interval** = 10 x 0.1 ms = 1 ms and **Packet Length** = 1000 bytes = 8000 bits, the source rate of packet transmission is 8000 bits/1 ms = 8 Mbit/s.

Figure 5-6 Auto Ping

The screenshot shows the 'UE Trace' configuration window. On the left is a tree view with options like Packet Capture, UMTS Services, and various interface traces. The 'UE Trace' option is selected. The main area contains several tabs: UE, IUR, IUB, IU, IUPC, UU, Monitor, RNTI, Other, Other1, and Bind. The 'UE' tab is active, showing fields for IMSI, MSISDN, IMEI(HEX), TMSI(HEX), LAC(HEX), MCC, and MNC. Below these are checkboxes for 'Debug Mode' and 'Save File'. The 'Trace Mode' is set to 'Report And Show'. The 'Trace Time(OMU Time)' is set from '2012/03/12 10:33:16' to '2012/03/12 10:33:16'. The 'File Name' is 'C:\Web LMT\MBSC\output\trace\tmfFile\UE_2012-3-12-10-35-48_IMSI_tmf'.

The screenshot shows the 'UE Trace' configuration window with the 'Other' tab selected. It contains several sections for configuring reports. The 'Layer 2 data transfer Periodic Report' is checked with a 'Report Period' of '200ms'. The 'Transport Data Periodic Report' is also checked. The 'AL Collect Period(s)' is set to '1'. The 'L2 Data Report' section has 'L2 SRB Data Report' checked with a 'Report time(s)' of '0'. The 'L2 TRB Data Report(IUB interface)' is checked with a 'L2 TRB Data Length(Byte)' of '2000'. The 'L2 TRB Data Report(IU interface)' is also checked. The 'Auto Ping' checkbox is checked, with a 'Ping Interval(0.1ms)' of '10' and a 'Packet Length(bytes)' of '1000'.

Step 2 Check Iu interface abnormalities and CN abnormalities.

Contact the CN engineer. Troubleshoot Iu interface transmission, CN packet loss and FTPServer transmission capability.

Step 3 Determine whether bottlenecks exist at the Iub interface.

1. Determine whether the path exists based on the transmission mode of the Iub interface.

IF	Then
ATM transmission is applied over the Iub interface.	Go to 2.

IP transmission is applied over the Iub interface.	Go to 5.
--	----------

2. Run the RNC MML command **DSPE1T1**, check the number of available E1s at the NodeB, estimate physically available bandwidth (a pair of E1s can provide a rate of about 0.75-0.8 Mbit/s), and determine whether the physical bandwidth is greater than the required rate. If yes, go to step 3; If no, increase E1.
3. Run the RNC MML command **LST AAL2PATH** (if data is carried by the optical port) or the **LST IMAGRP** (if data is carried in the form of IMA Group) to check the traffic record index used by NodeB; then, run the RNC MML command **LST ATMTRF** to check the sustainable cell rate (SCR) and determine whether SCR is greater than the required rate.
If yes, go to Step 4.
If no, modify and make SCR greater than the required rate.
4. Run the NodeB MML command **LST AAL2PATH** to query the reception cell rate (RCR) and determine whether RCR is smaller than or equal to the SCR in step 2.
If yes, go to Step 4.
If no, modify and make RCR smaller than or equal to SCR.
5. Run the RNC MML command **LST IPPATH** to check the transmission rate and determine whether the transmission rate is greater than the required rate. If yes, go to Step 6; If no, modify and make the transmission rate greater than the required rate.
6. Run the RNC MML command **LST IPLOGICPORT** to check whether the logic port has been configured. If no, skip the step; if yes, the bandwidth of the logic port configured must meet the test requirements, otherwise the bandwidth of the logic port configured must be modified.

Step 4 Determine whether packet loss exists at the Iub interface.

1. Determine whether the path exists based on the transmission mode of the Iub interface.

IF	Then
ATM transmission is applied over the Iub interface.	Go to 2.
IP transmission is applied over the Iub interface.	Go to 3.

2. Run the RNC MML command **PING IP**. Determine whether packet loss exists.
If yes, go to the procedure for handling packet loss under IP transmission.
If no, go to Step 5.
3. Run the RNC MML command **DSP AALVCCPFM** to determine whether packet loss or cell loss exists.
If yes, go to the procedure for handling packet loss under ATM transmission.
If no, go to Step 5.

Step 5 Perform the HSUPA service separately with the uplink rate limited to 1 Mbit/s and determine whether the rate is steady.

If yes, eliminate impact from the quality of the uplink air interface. Contact Huawei.

If no, go to RTWP abnormality handling.

Step 6 Contact Huawei.

If the problem still cannot be located, collect the following data on the site and deliver the data to Huawei for analysis.

NodeB WMPT logs

RNC CDT

NodeB CDT

UE LOG

RNC, NodeB License

RNC configuration files

5.4.7 Typical Cases

Case 1:

Fault Description

The DC service rate is low at an office (22 Mbit/s - 25 Mbit/s only).

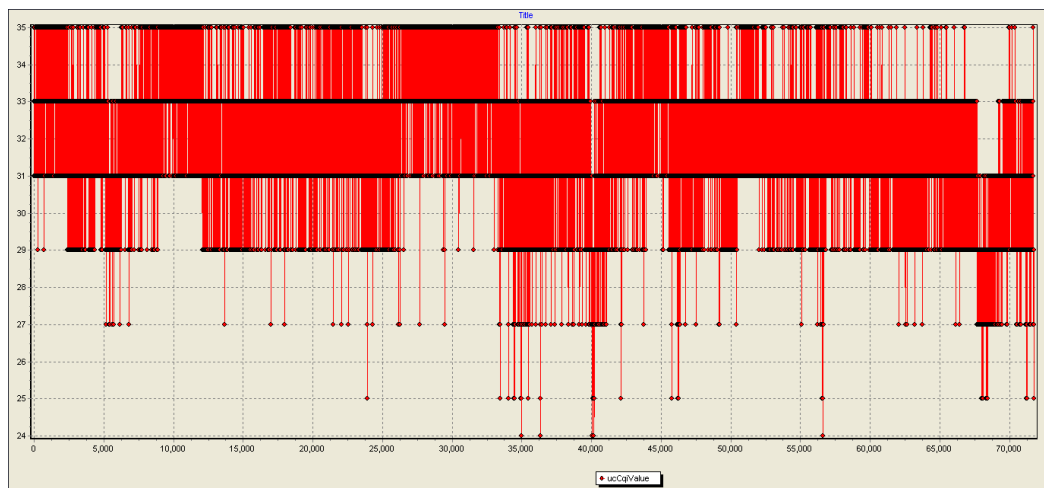
Possible Causes

Poor quality of the downlink air interface and insufficient data at the application layer result in a low DC rate. The DC rate is normal when the location is adjusted and a multithreading download tool is used.

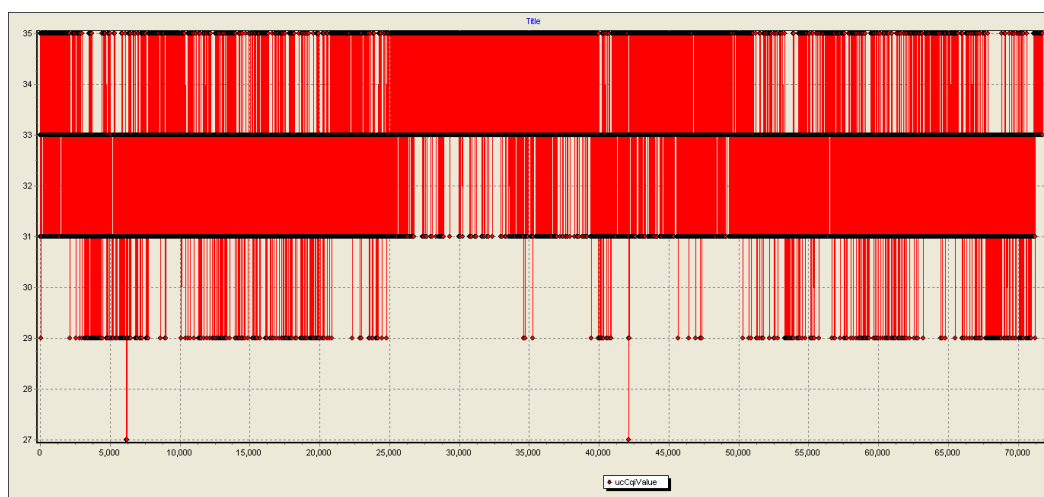
Fault Handling

1. Check the UE capability, CN assigned rate, RNC and NodeB license capabilities, and Iub transmission bandwidth, which are all normal.
2. Analyze the transmission at the Iub interface. Run the **Ping IP** (to NodeB) command on RNC to confirm no packet loss or abnormal delay exists.
3. Analyze the downlink signal quality at the air interface. Mainstream and sideline CQI values are both around 33 dB, which are low and fluctuate.

Mainstream CQI



Sideline CQI



4. Based on the analysis above, solve the poor quality of the downlink air interface. After position adjustment, the DC rate can steadily stay above 30 Mbit/s.
5. Run the **Auto Ping** command on RNC to make sure the target rate is reached. This suggests no problem exists below the RNC RLC layer.

Ensure sufficient data in the RNC buffer with multi-thread download. The DC rate steadily stays at 38 Mbit/s.

6 Troubleshooting SLC Faults

6.1 About This Chapter

This chapter describes the definition of a sleeping cell (SLC) and the troubleshooting procedure.

6.2 Definition of SLC Faults

No RRC connection setup request exist in the cell or certain subscribers cannot make calls if none of the following alarms are generated on the RNC.

Alarm ID	Alarm Name
22202	ALM-22202 UMTS Cell Unavailable
22214	ALM-22214 NodeB Unavailable
22206	ALM-22206 UMTS Cell Setup Failed

There are two types of SLC problems:

- No RRC connection setup requests are received.
- RRC connection setup fails.

6.3 SLC Problem Monitoring

SLC problems can be monitored through NodeB or M2000 alarms. For details, see Table 6-1.

Table 6-1 SLC problem monitoring

Monitoring Item/Network Element	NodeB Monitoring Condition	M2000 Monitoring Condition	Recommended Solution
The number of RRC connection setup requests is 0	The system reports ALM-28209 Cell No Traffic, and the system performs self-processing. Run the SET NOACCESSALMPARA command to set the alarm.	When ([VS.RRC.AttConnEstab.Cell]={0})&&([VS.Cell.UnavailTime.OM]={0})&&([VS.MeanRTWP]-[VS.MinRTWP])>={0.25}), an alarm is reported.	On the NodeB, select self-processing. The M2000 reports the alarm only without post-processing. Monitor the problem on the M2000.  NOTE This applies to the monitoring of cells with heavy traffic.
The RRC connection setup success rate is 0	/	When ([Number of RRC Connection Requests sent by the UE for cell]>{0})&&([Number of Successful RRC Connection Setups for Cell]/[Number of RRC Connection Requests sent by the UE for cell]<{0.1}), an alarm is reported.	On the M2000, monitor the problem that RRC requests are initiated while the service always fails to be set up.
The RB setup success rate is 0	/	When ([Number of RB Setup Attempts for Cell]>{0})&&([Number of Successful RB Setups for Cell]/[Number of RB Setup Attempts for Cell]<{0.1}), an alarm is reported.	On the M2000, monitor the problem that RAB requests are initiated while the service always fails to be set up.

6.4 Troubleshooting the Problem of No RRC Connection Request

6.4.1 Fault Description

The VS.RRC.AttConnEstab.Sum is 0. The IOS log contains no RRC CONNECT REQ signaling messages during the dialing test under the problematic site.

6.4.2 Possible Causes

- The data configuration is different from the physical configuration.
- No traffic exists (not a problem).
- The cell is barred.

6.4.3 Fault Handling Procedure

Step 1 (Optional: executed when cells under a new or relocated NodeB cannot be accessed).

1. Run the NodeB MML command **LST LOCELL** to check whether uplink and downlink frequencies are correct.
2. Run the RNC MML command **LST UCELL** on the and run the **LST LOCELL** command on the NodeB to check whether the frequencies of the RNC and NodeB are consistent.

If any abnormality exists, run the NodeB MML command **MOD LOCELL** or run the RNC MML command **MOD UCELL** to modify the configuration.

Check whether the problem is solved. If yes, the troubleshooting ends.

If no, go to Step 2.

If everything is normal, go to Step 2.

Step 2 (Optional: executed when cells under a relocated NodeB cannot be accessed).

Check for peripheral devices, such as Tower-Mounted Amplifiers (TMAs), which are exclusively used by another vendor. If any such devices exist, further check if they are incompatible with Huawei equipment. If yes, replace the TMA.

If no, go to Step 3.

Step 3 Check on the change in the number of successful RRC connection setups in the cell in the past month.

Check the RRC.SuccConnEstab.sum counter. If the value of the counter stays steady, go to Step 4; if the value of the counter fluctuates dramatically, check whether the service is available through the coverage of the cell, or check whether the cell is normal by initiating calls in the problematic cell. If yes, no problem occurs, and the troubleshooting ends. If no, go to Step 4.

Step 4 Check whether the cell is barred.

Run the RNC MML command **LST UCELLACCESSSTRICT** to check whether the operator reserved use indicator (**CellReservedForOperatorUse**) and the cell reservation extension indicator (**CellReservationExtension**) are RESERVED and whether access class 0 (**IsAccessClass0Barred**) through 15 (**IsAccessClass15Barred**) barring indicators and the idle cell access barring indicator (**IdleCellBarred**) are BARRED.

If no, go to Step 5.

If yes, run the RNC MML command **SET UCELLACCESSSTRICT** to modify the operator reserved use indicator (**CellReservedForOperatorUse**) and the cell reservation extension indicator (**CellReservationExtension**) into RESERVED and modify access class 0 (**IsAccessClass0Barred**) through 15 (**IsAccessClass15Barred**) barring indicators and the idle cell access barring indicator (**IdleCellBarred**) into NOT_BARRED. Check whether the problem is solved. If yes, the troubleshooting ends. If no, go to Step 5.

Step 5 Collect the following data and contact Huawei.

- Data to be collected before resetting the NodeB:

- Start pilot output power tracking on the RNC LMT which lasts 20 minutes during the problematic period.
- Start RRU output power monitoring on the NodeB LMT which lasts 20 minutes during the problematic period.
- Start cell RTWP and board RTWP real-time tracking on the NodeB LMT which lasts 20 minutes during the problematic period.
- Start cell tracking at the NodeB which lasts 20 minutes during the problematic period.

Note: The above tracking tasks can be launched and carried out simultaneously.

- Acquire RRU board logs.
- Acquire NodeB WMPT logs.
- Data to be collected after resetting the NodeB:
 - The original traffic statistics on the RNC side, which is the traffic statistics collected between the day immediately before the problem occurs and the time when the problem is solved.
 - Acquire RNC configuration files.
 - Acquire RNC CHR logs.

6.4.4 Typical Cases

Case 1:

Fault Description

An SLC problem occurred on a new site after swapping site, where the RRC-CONNECT-REQ message was not received, and the problem could not be solved by resetting the NodeB.

Fault Rectification

Before swapping, a competitor-customized TMA was used, which was incompatible with Huawei equipment. The problem was solved by replacing the TMA.

Fault Handling

- Step 1** Analyze the UE log from driving tests reported by the front line, finding that the RRC-CONNECT-REQ message was sent. However, according to the log on the NodeB, no uplink signal is detected.
- Step 2** Analyze other logs (output power, path delay, and path register), finding no abnormalities.
- Step 3** The front line and the customer found that the third-party device supplier had used a specially made TMA that was incompatible with Huawei equipment. Therefore, solve the problem by replacing the TMA.

Conclusion

Before the migration, the customer had used a specially made TMA that was incompatible with Huawei equipment. Finally the fault is rectified by replacing the TMA.

Case 2:

Fault Description

An office reported that an SLC problem had occurred on tens of sites in the live network. The symptom was that the RRC-CONNECT-REQ message was not received.

Fault Handling

1. These sites were new sites built during capacity expansion, without any neighboring cells specified.
2. No problems occurred during test calls on the site.
3. These were normal traffic-free sites, which were free of any SLC problem.

Conclusion

This was a normal traffic-free cell, which was free of any SLC problem.

6.5 Troubleshooting RRC Connection Setup Failures

6.5.1 Fault Description

While RRC CONNECT REQ signaling is present, the success rate of RRC-CONNECT-SETUP is 0, that is, all processes of setting up RRC connections fail. In this case, the RRC CONNECT REQ message is present in the IOS log, while the RRC-CONNECT-SETUP-COMPLETE message is absent.

6.5.2 Fault Handling Procedure

Follow the instructions below to collect data and contact Huawei.

- Data to be collected before resetting the NodeB:
 - Cell RTWP and board RTWP real-time tracking on the NodeB LMT
 - RNC IOS tracking. Run the RNC MML command **SET URRCTRLSWITCH** to enable complete tracing of CDT messages by selecting **CDT_MSG_FULL_TRACE** under **PROCESSSWITCH**.
 - User tracking on the NodeB side

**NOTE**

IOS tracking and NodeB CDT log tracking should proceed simultaneously when the problem appears.

- RRU board logs
- NodeB WMPT logs
- Data to be collected after resetting the NodeB:
 - Original traffic statistics on the RNC side, which is the traffic statistics between the day immediately before the problem occurs and the traffic statistics when the problem is solved.
 - RNC configuration files
 - CNC CHR log

7

Troubleshooting RRC Connection Setup Failures

7.1 Definition of RRC Access Failures

An RRC access failure refers to an RRC setup failure or the low RRC setup success rate.

7.2 Formula for the RRC Setup Success Rate

$$\text{VS.RRC.SuccConnEstab.Rate} = \text{RRC.SuccConnEstab.sum} / \text{VS.RRC.AttConnEstab.Cell}$$

The following causes are responsible for RRC access failures:

- No replies to the RRC setup requests. The RNC does not deliver the RRC CONNECTION SETUP or RRC CONNECTION REJ message. To address this problem, see section 7.6 "Troubleshooting Failures in Replying to RRC Connection Setup Requests."
- No replies to the RRC connection setup requests. The RNC cannot receive the message RRC CONNETION STEUP CMP from the UE after it sends an RRC CONNECTION SETUP message. To address this problem, see section 7.4 "Troubleshooting the Problem of No Replies to an RRC Connection Setup Request."
- Rejected RRC connection setup requests. The RNC sends an RRC CONNECTION REJ message after receiving an RRC CONNECTION SETUP REQUEST message. To address this problem, see section 7.5 "Troubleshooting Rejected RRC Connection Setup Requests."

7.3 Related Information

RRC access failure counters are as follows:

Causes	Counters	Description
RRC Connection Setup Rejected	VS.RRC.Rej.ULPower.Cong	Number of RRC Connection Rejects for Cell (UL Power Congestion)

Causes	Counters	Description
	VS.RRC.Rej.DLPower.Cong	Number of RRC Connection Rejects for Cell (DL Power Congestion)
	VS.RRC.Rej.ULIUBBand.Cong	Number of RRC Connection Rejects for Cell (UL Iub Bandwidth Congestion)
	VS.RRC.Rej.DLIUBBand.Cong	Number of RRC Connection Rejects for Cell (DL Iub Bandwidth Congestion)
	VS.RRC.Rej.ULCE.Cong	Number of RRC Connection Rejects for Cell (UL CE Resource Congestion)
	VS.RRC.Rej.DLCE.Cong	Number of RRC Connection Rejects for Cell (DL CE Resource Congestion)
	VS.RRC.Rej.Code.Cong	Number of RRC Connection Rejects for Cell (Code Resource Congestion)
	VS.RRC.Rej.RL.Fail	Number of RRC Connection Rejects for Cell (Radio Link Setup Failure)
	VS.RRC.Rej.TNL.Fail	Number of RRC Connection Rejects for Cell (Transmission Setup Failure on Iub Interface)
RRC Connection Setup No Reply	VS.RRC.FailConnEstab.NoReply	Number of RRC Connection Rejects Due to Timeout of RRC CONNECTION SETUP COMPLETE for Cell

The following causes are responsible for RRC access failures:

- No replies to the RRC connection setup requests. The RNC cannot receive the message RRC CONNECTION SETUP CMP from the UE after it sends an RRC CONNECTION SETUP message. To address this problem, see section 7.4 "Troubleshooting the Problem of No Replies to an RRC Connection Setup Request."
- Rejected RRC connection setup requests. The RNC sends an RRC CONNECTION REJ message after receiving an RRC CONNECTION SETUP REQUEST message. To address this problem, see section 7.5 "Troubleshooting Rejected RRC Connection Setup Requests."

7.4 Troubleshooting the Problem of No Replies to an RRC Connection Setup Request

7.4.1 Failure Description

When the RRC access success rate is high, the related signaling procedure shows that the UE does not respond to the RRC CONNECTION SETUP message sent by the RNC or the value of the VS.RRC.FailConnEstab.NoReply counter increases.

7.4.2 Fault Handling Procedure

Step 1 Locate the scope where the RRC access success rate decreases.

1. Check whether the RNC-level RRC access success rate decreases.
 - Check whether the values of the RNC-level counters listed in Table 7-1 decrease. If yes, go to Step 2.
 - If no, no more operations are required.

Table 7-1 Counters for analyzing the RNC-level RRC access success rate

KPIs	Counters
VS.RRC.AttConnEstab.RNC	VS.RRC.AttConnEstab.CellDCH.RNC
	VS.RRC.AttConnEstab.CellFACH.RNC
VS.RRC.SuccConnEstab.RNC	VS.RRC.SuccConnEstab.CellFACH.RNC
	VS.RRC.SuccConnEstab.CellDCH.RNC

2. Check the values of the counters listed in Table 7-2 to determine whether the problem mainly occurs on some CPUSs.
 - If yes, fix the exceptions in the problem CPUSs. If the exceptions are fixed, go to step 3. If the exceptions persist, contact Huawei Customer Service Center.
 - If no, go to Step 3.

Table 7-2 Counters for analyzing the RRC access success rate on the CPUS side

Counters	Description
VS.RRC.AttConnEstab.CPUS	Number of RRC Connection Requests for CPUS
VS.RRC.SuccConnEstab.CPUS	Number of Successful RRC Connection Establishments for CPUS

3. Check the values of the counters that are listed in Table 7-3 and related to cell-level RRC access success rate. Then, determine whether the problem mainly occurs in a single cell.
 - If yes, go to step 2.

- If no, the problem occurs in all the cells. Choose some typical cells to analyze and go to step 2.

Table 7-3 Counters for analyzing the RRC access success rate in the cell

Counters	Description
VS.RRC.AttConnEstab.Sum	Number of Processed RRC Connection Requests for Cell
RRC.SuccConnEstab.sum	Number of Successful RRC Connection Setups for Cell

Step 2 Analyze the trend of the counters one week before and one week after the failure based on the failure scope located in step 1. Check if the fluctuation of the counters is normal.

- If yes, no more operations are required.
- If no, locate the time when the RRC access success rate deteriorates and go to Step 3.

Step 3 Check whether any alarms are generated on the RNC or NodeB when the RRC access success rate decreases.

- If yes, clear the alarms according to the online help. If the alarms are cleared, no more operations are required. If the alarms persist, go to Step 4.
- If no, go to Step 4.

Step 4 Query RNC and NodeB operation logs to check whether any changes are falsely made to parameter settings after the problem occurs.

- If yes, check whether the changes are appropriate. If they are inappropriate, modify them and check whether the counters restore. If the counters restore, no more operations are required. If the counters do not restore, go to Step 5.
- If no, go to Step 5.

Step 5 Analyze the counters listed in Table 7-4 to check if the value of the VS MinRTWP is –106 dBm when no services are going on in the problem cell. (optional)

- If yes, there is no interference, go to step 5.
- If no, interference exists. Check whether the value of the counter is caused by external interferences.

Table 7-4 Counters for checking the value of VS MinRTWP

Counters	Description
VS.MeanRTWP	Average RTWP for Cell
VS.MaxRTWP	Maximum RTWP for Cell
VS.MinRTWP	Minimum RTWP for Cell

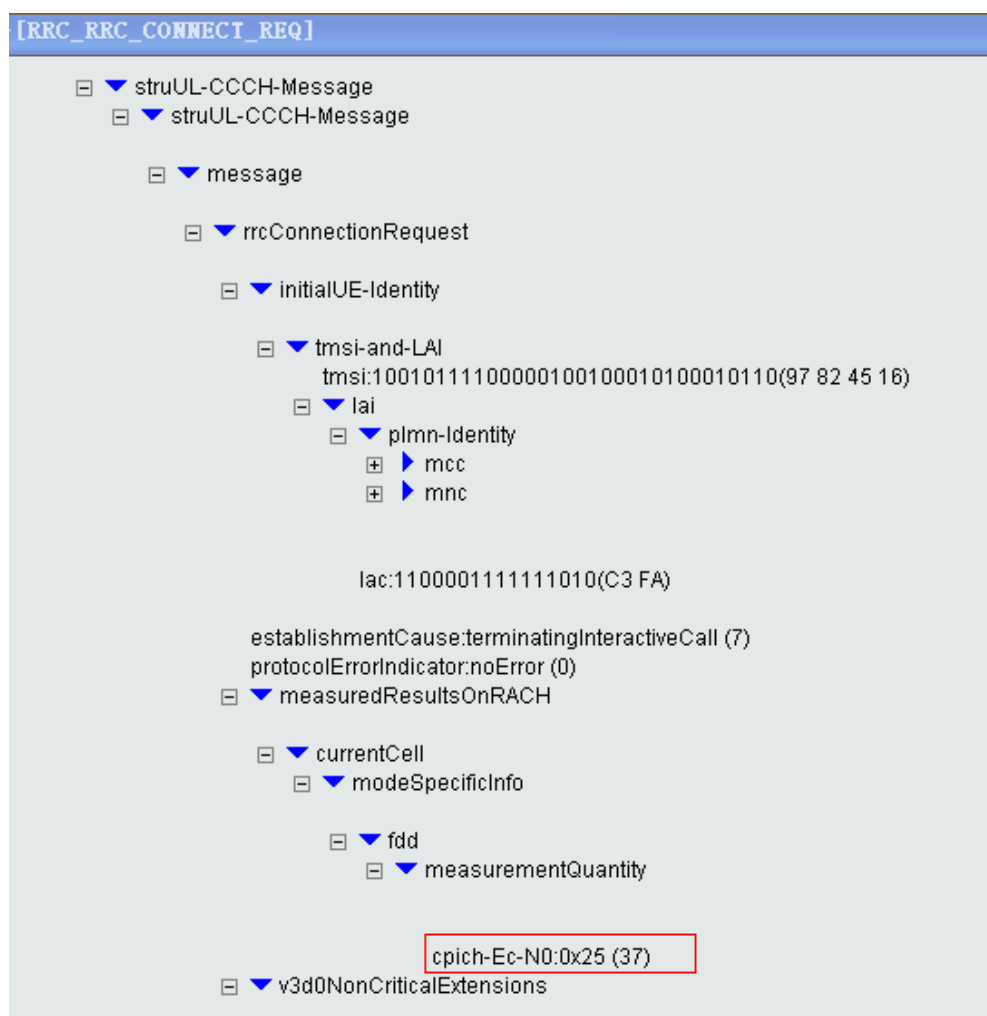
Step 6 Check whether the failure is caused by poor coverage. (optional)

Check whether the value of E_c/N_0 in the RRC CONNECT REQUEST message is lower than -13 dB. (If the value is lower than -13 dB, the downlink signal quality is poor.)

- If yes, the downlink coverage is poor. Upgrade the network to improve cell coverage. If the upgrade succeeds, no more operations are required. If the upgrade fails, go to Step 7.
- If no, the downlink coverage is sound. If the value of the counter is normal, go to Step 7.

The value of E_c/N_0 is shown in Figure 7-1.

Figure 7-1 Value of E_c/N_0



Step 7 If the access failure persists after the preceding steps are taken, contact Huawei Customer Service Center.

7.4.3 Typical Case 1

Failure Description

The RRC ASR decreases after an RNC is upgraded.

Possible Causes

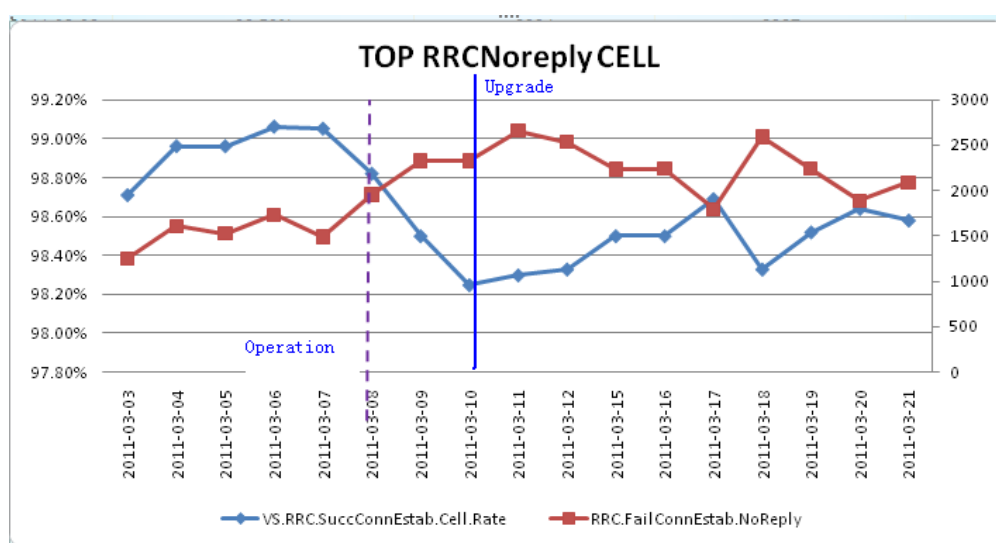
The problem may be caused by inappropriate changes in parameter settings.

Fault Handling Procedure

Statistics show the increase of the VS.RRC.FainConnEstab.NoReply counter is the main cause for the decrease of the RRC access success rate.

- Step 1** Check whether the RRC access success rate shown in Figure 7-2 decreases before the upgrade. The results show that the RRC access success rate decreased before the upgrade.
- Step 2** Analyze RNC and NodeB operation logs when the access failure rate is high. The results show that the **SET UCONNMODETIMER** command has been run and the N381 value is changed from D3 ms to D1 ms.

Figure 7-2 Results of operation logs

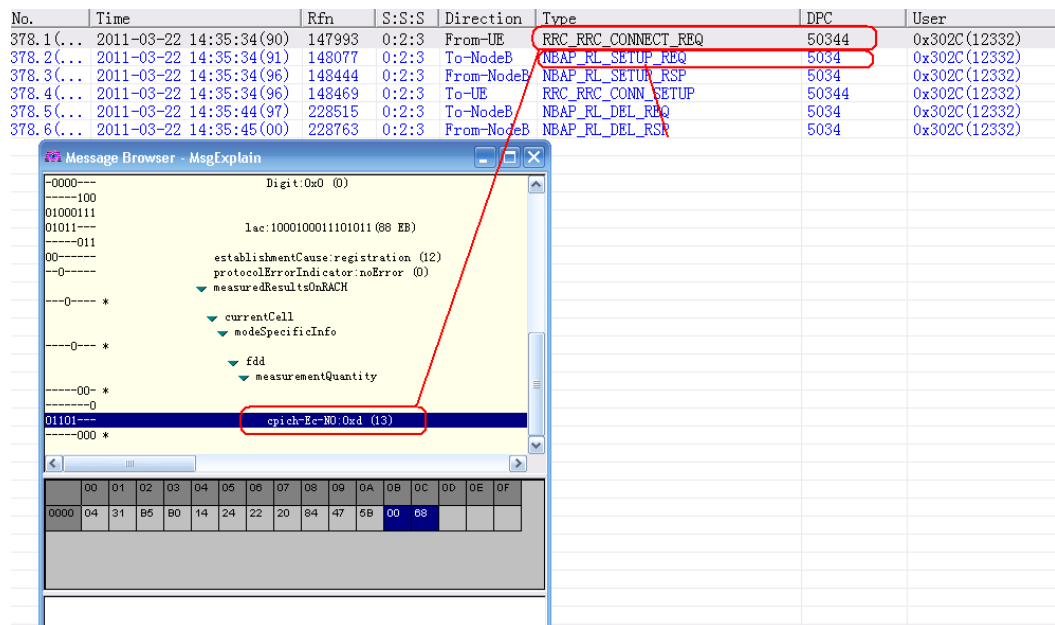


- Step 3** Change the N381 value to D0 ms and then check whether the RRC access success rate decreases. Related results show the RRC sends the CONNECTION SETUP message only once after the change. UEs on the cell edge experience RRC access failures, which cause the RRC access success rate to decrease, as shown in Figure 7-3.



NOTE

T381 is started after the RNC send the RRC CONNECTION SETUP message. If T381 expires and RNC does not receive an RRC CONNECTION SETUP COMPLETE message and the V381 value is smaller than the N381 value, RNC resends the RRC CONNECTION SETUP message and restarts the timer T381 and increases the V381 value. Currently N381 is set to D1 ms.

Figure 7-3 RRC access failure rate due to bad signal quality

7.4.4 Typical Case 2

Failure Description

The RRC access success rate fluctuates in a cell.

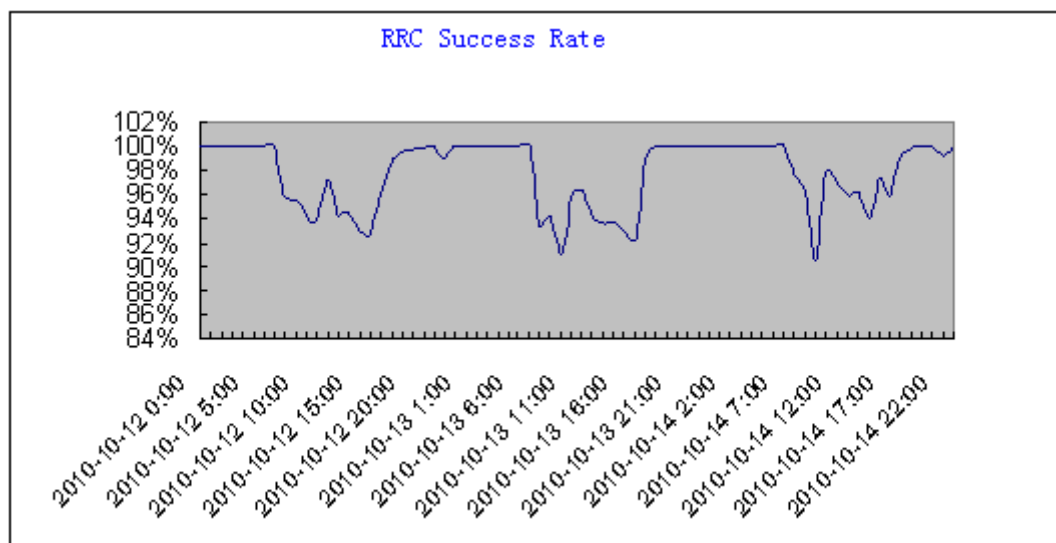
Possible Causes

Interference causes the sudden rise of the RTWP, leading to the increase of the VS.RRC.FailConnEstab.NoReply counter.

Fault Handling Procedure

Step 1 Analyze the values of cell-level counters.

The results show the RRC success rate fluctuates sharply, as shown in Figure 7-4.

Figure 7-4 Sharp fluctuation of RRC success rate

Step 2 Determine when the value of the VS.MaxRTWP counter fluctuates.

The results show the counter fluctuates sharply when the RTWP abnormally increases, and the counter is stable when the RTWP remains unchanged.

Then, analyze the relationship between the RTWP and the number of UEs camping on the problem cell. The results show the RTWP fluctuates sharply when there is a small number of UEs. It can be inferred that the rise of the RTWP is caused by external interference. Then check whether any external interference exists.

Step 3 Conduct an interference test.

The test results show external interference exists when the RTWP abnormally increases, which leads to the problem of no replies to an RRC connection setup request. After the interference is cleared the RTWP and the preceding counter restore.

7.5 Troubleshooting Rejected RRC Connection Setup Requests

7.5.1 Failure Description

The signaling procedure shows the RNC sends the RRC CONNECTION SETUP REJ message or statistics show the VS.RRC.FailConnEstab.Cong counter is increasing.

7.5.2 Handling Procedure

Step 1 Analyze the value of the counters listed in Table 7-5 to check what types of resources are congested.

Table 7-5 Counters for deciding what resources are congested

Counters	Description
VS.RRC.Rej.ULPower.Cong	Number of RRC Connection Rejects for Cell (UL Power Congestion)
VS.RRC.Rej.DLPower.Cong	Number of RRC Connection Rejects for Cell (DL Power Congestion)
VS.RRC.Rej.ULIUBand.Cong	Number of RRC Connection Rejects for Cell (UL Iub Bandwidth Congestion)
VS.RRC.Rej.DLIUBand.Cong	Number of RRC Connection Rejects for Cell (DL Iub Bandwidth Congestion)
VS.RRC.Rej.ULCE.Cong	Number of RRC Connection Rejects for Cell (UL CE Resource Congestion)
VS.RRC.Rej.DLCE.Cong	Number of RRC Connection Rejects for Cell (DL CE Resource Congestion)
VS.RRC.Rej.Code.Cong	Number of RRC Connection Rejects for Cell (Code Resource Congestion)

Step 2 To address the RRC.Rej.RL.Fail and VS.RRC.Rej.TNL.Fail counters, check if any transmission alarms are generated when the resources are congested.

1. If yes, clear the alarms by troubleshooting transmission problems. If the alarms are cleared, no more operations are required. If the alarms persist, go to Step 3.
2. If no, go to Step 3.

Step 3 Query RNC and NodeB operation logs to check whether any changes are falsely made to parameter settings when the congestion occurs.

1. If yes, check whether the changes are appropriate. If they are inappropriate, modify them and check whether the counters restore. If the counters restore, no more operations are required. If the counters do not restore, go to Step 4.
2. If no, go to Step 4.

Step 4 Analyze the value of the counters one week before and one week after the congestion. Check whether the resource congestion is caused by traffic bursts.

1. If yes, check whether the resources are sufficient. If the resources are insufficient, expand the network capacity. If the resources are sufficient, contact Huawei Customer Service Center.
2. If no, go to Step 5.

Step 5 If the problem persists after the preceding steps are taken, contact Huawei Customer Service Center.

7.6 Troubleshooting Failures in Replying to RRC Connection Setup Requests

7.6.1 Fault Description

The signaling process shows that the RNC does not return the RRC CONNECTION SETUP or RRC CONNECTION REJ message after receiving the RRC CONNECTIONREQ message.

7.6.2 Handling Procedure

Step 1 Determine whether the RNC discards the RRC connection setup requests due to flow control by doing as follows:

Check whether the VS.RRC.FC.Disc.Num counter increases.

- If yes, go to step 2.
- If no, go to step 3.

Step 2 Check whether a service burst occurs.

- If yes, change the parameters to reduce the probability of flow control.
- If no, go to step 3.

Step 3 Contact Huawei technical support.

8 Troubleshooting RAB Setup Faults

8.1 About This Chapter

This chapter describes how to locate and troubleshoot access faults.

8.2 Definition of RAB Setup Faults

An RAB setup fault can refer to a single RAB setup failure or a low RAB setup success rate as indicated by the related KPI.

8.2.1 RAB Setup Success Rate

The RAB setup success rate indicates the probability of successful CS/PS services setups after a UE finishes RRC signaling access. A low RAB setup success rate affects user experience. $\text{RAB setup success rate} = \text{Number of successful RAB setups} / \text{Number of RAB setup attempts}$

CS RAB and PS RAB setup success rates are as follows independently.

$$\begin{aligned} \text{VS.RAB.SuccEstCS.RNC.Rate} &= (\text{VS.RAB.SuccEstabCS.Conv.RNC} \\ &+ \text{VS.RAB.SuccEstabCS.Str.RNC}) \\ &/ (\text{VS.RAB.AttEstabCS.Conv.RNC} \\ &+ \text{VS.RAB.AttEstabCS.Str.RNC}) \end{aligned}$$

$$\begin{aligned} \text{VS.RAB.SuccEstPS.RNC.Rate} &= (\text{VS.RAB.SuccEstabPS.Bkg.RNC} + \\ &\text{VS.RAB.SuccEstabPS.Str.RNC} + \text{VS.RAB.SuccEstabPS.Conv.RNC} + \\ &\text{VS.RAB.SuccEstabPS.Int.RNC}) \\ &/ (\text{VS.RAB.AttEstabPS.Conv.RNC} + \text{VS.RAB.AttEstabPS.Bkg.RNC} + \\ &\text{VS.RAB.AttEstabPS.Int.RNC} + \text{VS.RAB.AttEstabPS.Str.RNC}) \end{aligned}$$

8.2.2 RAB Setup Procedure

1) The CN sends a RAB ASSIGNMENT REQUEST message to the RNC over the Iu-CS or Iu-PS interface.

2) After the RNC receives the RAB ASSIGNMENT REQUEST message, the RNC performs resource admission.

- If the resource admission fails, the RNC sends a RAB ASSIGNMENT RESPONSE message with failure cause to the CN.
- If the admission is successful, the RNC sends a RADIO BEARER SETUP message to the UE.

3) The UE launches the setup procedure of RADIO BEARER SETUP

- If the RB setup fails, which can be the RNC receives the RADIO BEARER SETUP FAILURE message from the UE or does not receive the respond message in time, the RNC writes the failure cause and then sends an RAB ASSIGNMENT RESPONSE message to the CN.
- If the RB setup is successful, the UE sends a RADIO BEARER SETUP COMPLETE message to the RNC. The RNC then return the RAB ASSIGNMENT RESPONSE message to the CN.

8.2.3 RAB Setup Failure Scenarios

1. The RNC fails in cell resources admission including code, CE, transmission or power resources.
2. The RNC sends a RADIO BEARER SETUP message to the UE but does not receive a RADIO BEARER SETUP COMPLETE or a RADIO BEARER SETUP FAILURE message from the UE.
3. The RNC sends a RADIO BEARER SETUP message to the UE but receives a RADIO BEARER SETUP FAILURE message.

8.3 Possible Causes

- The Uu does not respond.
Packet loss occurs on the SCTP link over the Iub interface.
- The traffic volume increases abnormally.
A certain type of UE is abnormal.
- Resources are congested.
 - The number of AAL2 path links configured on the Iub interface is insufficient.
 - The number of AAL2 path links configured on the Iu interface is insufficient.
 - The number of DSP resources on the DPU board is insufficient.
- The RNC configuration does not support RAB setup.
The service rate setting is incorrect.
- The network transmission is faulty.
The bandwidth of the IP PATH over the Iu-PS interface is insufficient.
- The physical channel is faulty.
- Two cells with different coverage areas are incorrectly set to be the neighboring cells for blind handovers.
- Others
 - The priority of services in the cell is configured incorrectly.
 - The RNC does not support multi-RAB.

8.4 Troubleshooting RAB Setup Failure

Step 1 Check whether the setup success rate drastically decreases from a certain time.

If yes, go to Step 2.

If no, record the period of time including the course of the decrease comparing to the working days and hours, and go to Step 4.

Step 2 View the operation logs and check whether related operations have been executed within 24 hours during this period.

If yes, go to Step 5 to contact Huawei to confirm the effects of the operations.

If no, go to Step 3.

Step 3 Check whether any alarms have been reported within 24 hours during this period.

If yes, troubleshooting the alarms faults.

If no, go to Step 4.

Step 4 Analyze the causes of setup failures.

As for the cell KPIs mentioned in the following sub-steps, the values of these KPIs must be accumulated before analysis.

1. Check whether the values of VS.RAB.FailEstabCS.UuNoReply or VS.RAB.FailEstabPS.UuNoReply increase noticeably.
 - If yes, see section 8.5 "Troubleshooting the Problem of Uu No Response."
 - If no, go to the next step.
2. Check whether the value of VS.RAB.FailEstabPS.Cong or VS.RAB.FailEstabCS.Cong increases noticeably.
 - If yes, go to the next step.
 - If no, go to the sixth step.
3. Check whether the numbers of CS RAB setup attempts and PS RAB setup attempts increase noticeably.
Number of CS RAB setup attempts = VS.RAB.AttEstabCS.Conv.RNC + VS.RAB.AttEstabCS.Str.RNC
Number of PS RAB setup attempts = VS.RAB.AttEstabPS.Bkg.RNC + VS.RAB.AttEstabPS.Conv.RNC + VS.RAB.AttEstabPS.Int.RNC + VS.RAB.AttEstabPS.Str.RNC
 - If yes, see section 8.6 "Troubleshooting Increased Traffic Volume."
 - If no, go to the next step.
4. Check whether the values of VS.RAB.FailEstabCS.ULIUBBand.Cong and VS.RAB.FailEstabPS.ULIUBBand.Cong increase noticeably.
 - If yes, see section 8.7 "Troubleshooting Iub Congestion."
 - If no, go to the next step.
5. Check whether the following counters increase noticeably.
 - If yes, go to step 5.
 - If no, see section 8.8 "Troubleshooting Other Congestions."

KPI	Counter
VS.RAB.FailEstabCS.Cong	VS.RAB.FailEstabCS.DLIUBBand.Cong VS.RAB.FailEstabCS.ULIUBBand.Cong VS.RAB.FailEstabCS.ULCE.Cong VS.RAB.FailEstabCS.DLCE.Cong VS.RAB.FailEstabCS.Code.Cong VS.RAB.FailEstabCS.ULPower.Cong VS.RAB.FailEstabCS.DLPower.Cong
VS.RAB.FailEstabPS.Cong	VS.RAB.FailEstabPS.DLIUBBand.Cong VS.RAB.FailEstabPS.ULIUBBand.Cong VS.RAB.FailEstabPS.ULCE.Cong VS.RAB.FailEstabPS.DLCE.Cong VS.RAB.FailEstabPS.Code.Cong VS.RAB.FailEstabPS.ULPower.Cong VS.RAB.FailEstabPS.DLPower.Cong

6. Check whether the values of VS.RAB.FailEstabCS.Unsp or VS.RAB.FailEstabPS.Unsp increases noticeably.
 - If yes, see section 8.9 "Troubleshooting the Problem of RAB Setup Not Allowed by the RNC Configuration."
 - If no, go to the next step.
7. Check whether the values of VS.RAB.FailEstabCS.TNL or VS.RAB.FailEstabPS.TNL increase noticeably.
 - If yes, see section 8.10 "Troubleshooting Transmission Network Faults."
 - If no, go to the next step.
8. Check whether the values of VS.RAB.FailEstabCS.PhyChFail or VS.RAB.FailEstabPS.PhyChFail increase noticeably.
 - If yes, see section 8.11 "Troubleshooting Physical Channel Faults."
 - If no, go to the next step.
9. Check whether the following KPIs increase noticeably.

CS KPI	PS KPI
VS.RAB.FailEstabCS.IubFail VS.RAB.FailEstabCS.RBIncCfg VS.RAB.FailEstabCS.RBCfgUnsup	VS.RAB.FailEstabPS.IubFail VS.RAB.FailEstabPS.RBIncCfg VS.RAB.FailEstabPS.RBCfgUnsup

- If yes, go to Step 5.
- If no, see section 8.12 "Miscellaneous."

Step 5 Contact Huawei technical support.

8.5 Troubleshooting the Problem of Uu No Response

8.5.1 Fault Description

The RNC RAB setup success rate decreases and the value of VS.RAB.FailEstabCS.UuNoReply or VS.RAB.FailEstabPS.UuNoReply increases noticeably.

8.5.2 Fault Handling Procedure

The following analysis is based on the period when the fault occurs.

Step 1 Analyze the values of VS.RAB.FailEstabCS.UuNoReply and VS.RAB.FailEstabPS.UuNoReply of each cell, and check whether they increase noticeably in some cells.

If yes, record the cell ID and go to Step 2.

If no, go to Step 6.

Step 2 Run the RNC MML command **LST UCELL** to query the NodeB name corresponding to the cell ID.

Step 3 Run the RNC MML command **LST UIUBCP** to locate the link number based on the NodeB name.

If...	Then...
The Iub interface adopts ATM transmission	Locate the SAAL link number
The Iub interface adopts IP transmission	Locate the SCTP link number.

Step 4 Check whether the following alarms are reported.

ALM-21541 SCTP Link Fault

ALM-21542 LAPD Link Fault

If yes, see section 14.3 "Troubleshooting SCTP Faults."

If no, go to Step 6.

Step 5 Check whether the value of VS.SCTP.RETX.PKGNUM changes noticeably.

If yes, see section 14.3 "Troubleshooting SCTP Faults."

If no, go to Step 6.

Step 6 Contact Huawei technical support.

8.5.3 Typical Cases

Fault Description

RNC CS and PS RAB setup success rates are both very low. The values of VS.RAB.FailEstabCS.UuNoReply and VS.RAB.FailEstabPS.UuNoReply increase noticeably.

Cause Analysis

Packet loss occurs on the Iub interface due to Iub transmission device faults. As a result, the RAB setup fails due to Uu no response. The problem is solved after troubleshooting transmission faults.

Fault Handling Procedure

- Step 1** Locate the cells where the values of VS.RAB.FailEstabCS.UuNoReply and VS.RAB.FailEstabPS.UuNoReply increase noticeably.
- Step 2** Identify the transmission mode of the problem cells. The problem cells use IP transmission. Locate the SCTP link number for the cell on the control plane.
- Step 3** View the counters for the SCTP link. The value of S.SCTP.RETX.RKGNUM increases noticeably.

RNCID	SCTPLNKId	SCTPLNKName	Time(As day)	VS.SCTP.TX.MAXBYTES	VS.SCTP.RX.MAXPKGNUM	VS.SCTP.TX.MAXPKGNUM	VS.SCTP.RETX.PKGNUM
130	0:2:152	SCTPLNK0:2:152	2009-10-21	3132	148	112	44
130	0:2:152	SCTPLNK0:2:152	2009-10-22	3279	172	131	536
130	0:2:152	SCTPLNK0:2:152	2009-10-23	3617	138	108	920
130	0:2:152	SCTPLNK0:2:152	2009-10-24	3791	155	121	1250
130	0:2:152	SCTPLNK0:2:152	2009-10-25	3423	157	125	1258
130	0:2:152	SCTPLNK0:2:152	2009-10-26	3753	163	131	1336
130	0:2:152	SCTPLNK0:2:152	2009-10-27	3052	141	111	827
130	0:2:152	SCTPLNK0:2:152	2009-10-28	3342	149	116	734
130	0:2:152	SCTPLNK0:2:152	2009-10-29	3200	133	109	558
130	0:2:152	SCTPLNK0:2:152	2009-10-30	4264	153	128	4132
130	0:2:152	SCTPLNK0:2:152	2009-10-31	3648	175	140	4846
130	0:2:152	SCTPLNK0:2:152	2009-11-1	3578	154	115	3397
130	0:2:155	SCTPLNK0:2:155	2009-10-20	2703	63	41	10
130	0:2:155	SCTPLNK0:2:155	2009-10-21	3132	65	45	12
130	0:2:155	SCTPLNK0:2:155	2009-10-22	2407	63	41	87
130	0:2:155	SCTPLNK0:2:155	2009-10-23	2781	63	40	101
130	0:2:155	SCTPLNK0:2:155	2009-10-24	2750	63	42	163
130	0:2:155	SCTPLNK0:2:155	2009-10-25	2772	65	44	156
130	0:2:155	SCTPLNK0:2:155	2009-10-26	3137	65	43	184
130	0:2:155	SCTPLNK0:2:155	2009-10-27	2720	68	45	180
130	0:2:155	SCTPLNK0:2:155	2009-10-28	2657	65	43	211
130	0:2:155	SCTPLNK0:2:155	2009-10-29	2930	65	40	83
130	0:2:155	SCTPLNK0:2:155	2009-10-30	2906	69	46	280
130	0:2:155	SCTPLNK0:2:155	2009-10-31	2817	71	46	627
130	0:2:155	SCTPLNK0:2:155	2009-11-1	3480	69	45	473
130	0:2:156	SCTPLNK0:2:156	2009-10-20	3110	145	108	25
130	0:2:156	SCTPLNK0:2:156	2009-10-21	3038	145	114	55
130	0:2:156	SCTPLNK0:2:156	2009-10-22	3406	136	106	722
130	0:2:156	SCTPLNK0:2:156	2009-10-23	3112	141	108	811
130	0:2:156	SCTPLNK0:2:156	2009-10-24	2887	152	107	1122
130	0:2:156	SCTPLNK0:2:156	2009-10-25	3106	143	117	830
130	0:2:156	SCTPLNK0:2:156	2009-10-26	3379	165	127	1083
130	0:2:156	SCTPLNK0:2:156	2009-10-27	3304	153	123	1094
130	0:2:156	SCTPLNK0:2:156	2009-10-28	3049	158	116	1091
130	0:2:156	SCTPLNK0:2:156	2009-10-29	3507	154	124	658
130	0:2:156	SCTPLNK0:2:156	2009-10-30	3759	171	135	2196
130	0:2:156	SCTPLNK0:2:156	2009-10-31	2907	168	127	3767
130	0:2:156	SCTPLNK0:2:156	2009-11-1	3487	150	111	2943

- Step 4** Troubleshoot the corresponding transmission link. Packet loss occurs over the Iub interface due to Iub transmission device faults. The RAB setup success rate increase after the problem is solved.

8.6 Troubleshooting Increased Traffic Volume

8.6.1 Fault Description

The RAB setup success rate decreases and the values of VS.RAB.FailEstabPS.Cong or VS.RAB.FailEstabCS.Cong increase noticeably. The number of CS RAB setup attempts or PS RAB setup attempts increases noticeably.

8.6.2 Fault Handling Procedure

The following analysis is based on the period when the fault occurs.

- Step 1** Analyze the number of online UEs. Check whether the values VS.CellDCHUEs.RNC and VS.CellFACHUEs.RNC increase noticeably.
- If yes, go to Step 2.
- If no, go to Step 5
- Step 2** Analyze the number of CS RAB setup attempts or PS RAB setup attempts in each cell. Check whether the numbers increase drastically in some cells.
- Number of CS RAB setup attempts = VS.RAB.AttEstabCS.Conv + VS.RAB.AttEstabCS.Str
- Number of PS RAB setup attempts = VS.RAB.AttEstabPS.Bkg + VS.RAB.AttEstabPS.Conv + VS.RAB.AttEstabPS.Int + VS.RAB.AttEstabPS.Str
- If yes, check whether mass gathering occurs in the site coverage area.
- If no, go to Step 3.
- Step 3** Check whether there are any network behaviors influencing the current traffic model.
- If yes, adjust the network according to the current traffic model.
- If no, go to Step 4.
- Step 4** Check whether the number of service requests initiated by a certain type of UE increases drastically on the CN side.
- If yes, troubleshoot the UE-related fault.
- If no, go to Step 5.
- Step 5** Contact Huawei technical support.

8.6.3 Typical Cases

Fault Description

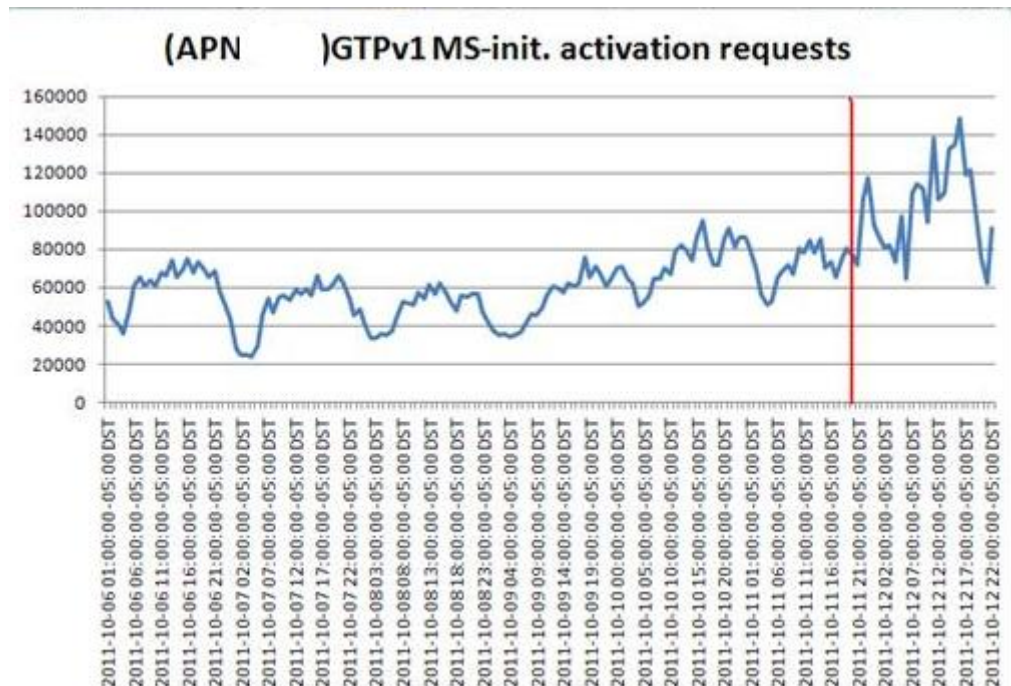
The RAB setup success rate decreases and the number of VS.RAB.FailEstabPS.Cong increases noticeably.

Cause Analysis

A large number of BlackBerry users initiate PS services at the same time, leading to resource congestion. As a result, the PS RAB setup fails.

Fault Handling Procedure

- Step 1** The number of PS RAB requests increase drastically.
- Step 2** Perform analysis on the GGSN side. Results show that the number of APN accesses initiated by BlackBerry users increases drastically. This is because the server of the RIM application layer is abnormal and rejects all the repeated PS service requests initiated by BlackBerry users.



8.7 Troubleshooting Iub Congestion

8.7.1 Fault Description

The RAB setup success rate decreases. The number of CS or PS RAB setup attempts remains unchanged, but the value of VS.RAB.FailEstabCS.ULIUBBand.Cong or VS.RAB.FailEstabPS.ULIUBBand.Cong increases noticeably.

8.7.2 Fault Handling Procedure

Step 1 Analyze the values of VS.RAB.FailEstabCS.ULIUBBand.Cong and VS.RAB.FailEstabPS.ULIUBBand.Cong for each cell. Check whether they increase noticeably in some cells.

If yes, record the cell ID and go to Step 2.

If no, go to Step 8.

Step 2 Check the transmission mode applied on the Iub interface in the cell.

If...	Then...
The Iub interface uses ATM transmission	Go to step 3.
The Iub interface uses IP transmission	Go to step 5.

Step 3 Check whether the CID resource for an AAL2 path is insufficient.

- Run the **RNC MML command LST UCELL** to query the NodeB name corresponding to the cell ID.
- Run the **RNC MML command LST ADJNODE** to query the ANI corresponding to the name of the adjacent node
- Analyze the value of VS.QAAL2.Act.Con with the measurement object ANI.
- Run the **RNC MML command LST AAL2PATH** to query the AAL2 path corresponding to the ANI, and record the number of links configured on the AAL2 path.
- Check whether the actual value exceeds the configured value.

Actual Value	Configured Value
VS.QAAL2.Act.Con	Number of paths x 248

If yes, the Iub bandwidth is insufficient. Add new AAL2 paths.

If no, go to Step 5.

Step 4 Check whether the total actual traffic of all AAL2 paths is far less than the allocated traffic.

If yes, that is the actual traffic of (AAL2PATH ID1+ AAL2PATH ID2+...AAL2PATH IDn) < the allocated traffic, execute the following steps to decrease the value of the activity factor.

1. Run the **RNC MML command LST ADJMAP** to query the FTI corresponding to the ANI.
2. Run the **RNC MML command MOD TRMFACTOR** to modify activity factor or **ADD TRMFACTOR** to add new activity factor list.

If no, go to Step 8.

	Path ID	Actual Traffic	Allocated Traffic
TX	AAL2PATH ID1	VS.AAL2PATH.PVCLAYER.TXBYTES*8	VS.QAAL2.AllocatedFwd.AAL2BitRate VS.QAAL2.AllocatedMaxFwd.AAL2BitRate. Value
	AAL2PATH ID2	VS.AAL2PATH.PVCLAYER.RXBYTES*8	
	...	...	
RX	AAL2PATH ID1	VS.AAL2PATH.PVCLAYER.RXBYTES*8	VS.QAAL2.AllocatedBwd.AAL2BitRate VS.QAAL2.AllocatedMaxBwd.AAL2BitRate. Value
	AAL2PATH ID2	VS.AAL2PATH.PVCLAYER.RXBYTES*8	
	...	...	

Step 5 Check whether the IP paths corresponding to the service exist.

- If yes, see section 3.5.1 "Troubleshooting Abnormal AAL2PATH or IPPATH."

Check whether the problem is solved. If yes, no further action is required. If no, go to Step 86.

- If no, go to Step 86.

Step 6 Check whether the bandwidth configured for the IP paths over the Iub interface is insufficient.

1. Run the **RNC MML command LST IPPATH** with the interface type specified to query the links configured for the Iub interface. Record the link numbers.
2. Analyze the following KPIs and record the transmit rate and receive rate of each link:
VS.IPPATH.IPLAYER.PEAK.TXRATE
VS.IPPATH.IPLAYER.MEAN.TX
VS.IPPATH.IPLAYER.PEAK.RXRATE
VS.IPPATH.IPLAYER.MEAN.RX
3. Run the **RNC MML command LST IPPATH** with **PATHID** specified to check the bandwidth configured for each path. Record the transmit bandwidth and receive bandwidth.
4. Check whether the actual rate of a path exceeds the configured one noticeably.

Path ID	Actual Rate	Configured Bandwidth
PATHID	VS.IPPATH.IPLAYER.PEAK.TXRATE VS.IPPATH.IPLAYER.MEAN.TX	Transmit bandwidth
PATHID	VS.IPPATH.IPLAYER.PEAK.RXRATE VS.IPPATH.IPLAYER.MEAN.RX	Receive bandwidth

If yes, adjust the bandwidth of the links or add new links. Check whether the problem is solved. If yes, no further action is required. If no, go to Step 8.

If no, go to Step 7.

Step 7 Check whether the actual traffic flow on an IP path is much less than the allocated one.

If yes, that is the actual traffic of (IPPATH ID1+ IPPATH ID2+...IPPATH IDn) < the allocated traffic, execute the following steps to adjust the value of activity factor.

1. Run the **RNC MML command LST ADJMAP** to find the FTI corresponding to the ANI.
2. Run the **RNC MML command MOD TRMFACTOR** to modify the value of activity factor or **ADD TRMFACTOR** to add a new activity factor.

	Path ID	Actual Traffic Flow	Allocated Traffic Flow
TX	IPPATH ID1	VS.IPPATH.IPLAYER.TXB YTES *8	OS.ANI.IP.AllocatedFwd
	IPPATH ID 2	VS.IPPATH.IPLAYER.TXB YTES *8	OS.ANI.IP.AllocatedFwd
	...	...	

TX	IPPATH ID 1	VS.IPPATH.IPLAYER.RXB YTES *8	OS.ANI.IP.AllocatedBwd
	IPPATH ID 2	VS.IPPATH.IPLAYER.RXB YTES *8	OS.ANI.IP.AllocatedBwd
	...	...	

If no, go to Step 8.

Step 8 Contact Huawei technical support.

8.7.3 Typical Cases

Fault Description

The RAB setup success rate decreases. The values of VS.RAB.FailEstabCS.ULIUBBand.Cong and VS.RAB.FailEstabPS.ULIUBBand.Cong increase noticeably.

Cause Analysis

The Iub congestion occurrences increase noticeably only in certain cells. With the increase in the number of HSPA users, the number of AAL2 paths becomes insufficient. Therefore, the Iub bandwidth admissions for CS and PS fail, leading to assignment failures.

Fault Handling Procedure

- Step 1** The Iub congestion increases noticeably only in certain cells.
- Step 2** The problem sites adopt ATM transmission, and check the number of AAL2 path links on the user plane.
- Step 3** Analyze the value of VS.QAAL2.Act.Con for the problem sites.
- Step 4** Check whether the value of VS.QAAL2.Act.Con exceeds the number of AAL2 path links multiplied by 248.
- Step 5** If the value of VS.QAAL2.Act.Con exceeds the number of AAL2 path links multiplied by 248, add new AAL2 path links.

8.8 Troubleshooting Other Congestions

8.8.1 Fault Description

The RAB setup success rate decreases. The value of VS.RAB.FailEstabPS.Cong or VS.RAB.FailEstabCS.Cong increases noticeably, but resource congestion occurrences do not increase noticeably.

8.8.2 Fault Handling Procedure

- Step 1** Check the transmission mode applied on the Iu-CS interface.
 1. Run the RNC MML command **LST ADJNODE** to query the ANI corresponding to the Iu-CS interface.

2. Analyze the value of VS.QAAL2.Act.Con with the measurement object being the ANI.
3. Run the RNC MML command **LST AAL2PATH** to query the AAL2 path corresponding to the ANI. Record the number of links configured on the AAL2 path.
4. Check whether the actual value of VS.QAAL2.Act.Con exceeds the number of links multiplied by 248.

Actual Value	Configured Value
VS.QAAL2.Act.Con	The number of links*248

If yes, the bandwidth of Iu-CS is insufficient, and therefore add new AAL2 path links.

If no, go to Step 2.

- Step 2** Analyze the value of VS.DSP.UsagePeak. Check whether the load of a DSP subsystem exceeds 90%.

If yes, it indicates that insufficient DSP capacity leads to the access failure. Check whether the load between DSP subsystems is balanced. If no, adjust the load sharing threshold on the user plane. If yes, expand the DPU capacity. For details about capacity expansion, go to Step 3.

Generally, if the value of VS.DSP.UsageAvg exceeds 60%, expand the DPU capacity.

If no, go to Step 33.

- Step 3** Contact Huawei technical support.

8.8.3 Typical Case 1

Fault Description

The CS RAB setup success rate decreases. The values of VS.RAB.FailEstabCS.Cong in most cells increase noticeably. The values of the following counters remain unchanged:

RAB.FailEstabCS.Cong.other = VS.RAB.FailEstabCS.Cong –

VS.RAB.FailEstabCS.DLIUBBand.Cong –

VS.RAB.FailEstabCS.ULIUBBand.Cong –

VS.RAB.FailEstabCS.ULCE.Cong –

VS.RAB.FailEstabCS.DLCE.Cong –

VS.RAB.FailEstabCS.Code.Cong –

VS.RAB.FailEstabCS.ULPower.Cong –

VS.RAB.FailEstabCS.DLPower.Cong

Cause Analysis

The number of AAL2 path links over the Iu-CS interface is insufficient. Applying for CID resource in busy hours fails, leading to the CS RAB setup failures.

Fault Handling Procedure

- Step 1** Analyze the KPIs. Only the CS KPIs are abnormal, whereas the PS KPIs are normal.

- Step 2** ATM transmission is applied on the Iu-CS interface, and check the number of configured AAL2 paths..

Step 3 Check whether the value of VS.QAAL2.Act.Con on the Iu-CS interface increases noticeably.

QAAL2Id	Time	VS.QAAL2.Act.Con
1995	2009-10-6 16:00	310.0056
1995	2009-10-6 16:30	275.4445
1995	2009-10-6 17:00	453.9528
1995	2009-10-6 17:30	454.4833
1995	2009-10-6 18:00	467.775
1995	2009-10-6 18:30	475.0695
1995	2009-10-6 19:00	438.1805

Step 4 Check the value of VS.QAAL2.Act.Con exceeds the number of AAL2 path links multiplied by 248.

Step 5 Add two AAL2 paths on the Iu-CS interface to solve the problem.

8.8.4 Typical Case 2

Fault Description

The CS and PS RAB setup success rates decreases. The values of VS.RAB.FailEstabPS.Cong increase noticeably and the value of VS.RAB.FailEstabCS.Cong increase slightly in certain cells. The resource congestion occurrences generally remain unchanged.

Cause Analysis

The traffic exceeds the configured DSP capacity for the DPU board, leading to the RAB setup failures.

Fault Handling Procedure

Step 1 Analyze the KPIs to check whether the problem cells are carried in one subrack.

Step 2 Analyze the value of VS.DSP.UsagePeak to check whether the DSP usages of some DPU boards in the subrack exceed 90%.

Step 3 Run the RNC MML command **SET UUSERPLNSHAREPARA** with **UserPlnSharingOutThd** set to **70** to decrease the inter-subrack load sharing threshold on the user plane to avoid the problem. Add new DPU boards to solve the problem.

8.9 Troubleshooting the Problem of RAB Setup Not Allowed by the RNC Configuration

8.9.1 Fault Description

The RAB setup success rate is very low. The value of VS.RAB.FailEstabCS.Unsp or VS.RAB.FailEstabPS.Unsp increases noticeably.

8.9.2 Fault Handling Procedure

- Step 1** Check whether the values of VS.RAB.FailEstabCS.Unsp and VS.RAB.FailEstabPS.Unsp increase drastically in some cells.
- If yes, go to Step 2.
- If no, go to Step 3.
- Step 2** Check whether the maximum rate assigned by the CN falls into the range of the RNC configuration.
1. Check the value of trafficClass and MaxBitrate information elements (IE) in the RANAP_RAB_ASSIGNMENT_REQUSET message.
 2. Run the RNC MML command **LST UTYPRAB** to check whether the maximum rates of the RNC and the CN are consistent according to the TrafficClass.
- If yes, go to Step 3.
- If no, adjust the maximum rate of the CN or of the RNC. Check whether the problem is solved. If the problem is solved, no further action is required. If the problem is not solved, go to Step 3.
- Step 3** Contact Huawei technical support.

8.9.3 Typical Cases

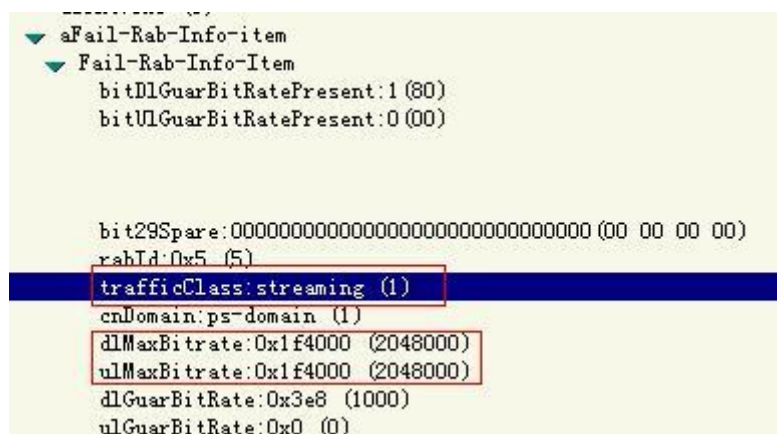
The PS RAB setup success rate decreases. The value of VS.RAB.FailEstabPS.Unsp increases noticeably. The value of VS.RAB.FailEstabPS.Cong generally remains unchanged.

Possible Causes

The Streaming services are registered at a rate larger than the maximum rate allowed by the RNC, leading to the RAB setup failures.

Fault Handling

- Step 1** Analyze the KPIs. Only the PS Streaming services fail to be set up.
- Step 2** Analyze the signaling to check the rate assigned by the CN for PS Streaming services. It is 2048 kbit/s.



- Step 3** Check the maximum rate for PS Streaming services configured on the RNC side. The maximum rate is 384 kbit/s, smaller than the rate assigned by the CN, which leads to the RAB setup failure.

ADD UTYPRABASIC:RABINDEX=21, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=8000, SSI
ADD UTYPRABASIC:RABINDEX=22, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=16000, S:
ADD UTYPRABASIC:RABINDEX=23, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=32000, S:
ADD UTYPRABASIC:RABINDEX=24, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=64000, S:
ADD UTYPRABASIC:RABINDEX=25, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=128000, S:
ADD UTYPRABASIC:RABINDEX=26, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=144000, S:
ADD UTYPRABASIC:RABINDEX=27, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=256000, S:
ADD UTYPRABASIC:RABINDEX=28, APPLIEDDIRECT=APPLIED_ON_BOTH, CNDOMAINID=PS_DOMAIN, TRAFFICCLASS=STREAMING, MAXBITRATE=384000, S:

- Step 4** Modify the registration rate on the CN side to solve the problem.

8.10 Troubleshooting Transmission Network Faults

8.10.1 Fault Description

The RAB setup success rate decreases. The value of VS.RAB.FailEstabCS.TNL or VS.RAB.FailEstabPS.TNL increases noticeably.

8.10.2 Fault Handling Procedure

The following analysis is based on the period when the fault occurs.

- Step 1** Run the RNC MML command **LST IPPATH** with the interface type set to Iu-PS to check the links configured for the Iu-PS interface. Record the link numbers.
- Step 2** Analyze the KPIs. Record the transmit rate and receive rate of each link.
- VS.IPPATH.IPLAYER.PEAK.TXRATE
VS.IPPATH.IPLAYER.MEAN.TX
VS.IPPATH.IPLAYER.PEAK.RXRATE
VS.IPPATH.IPLAYER.MEAN.RX
- Step 3** Run the RNC MML command **LST IPPATH** with the **PATHID** specified to check the configured bandwidth for each link. Record the transmit bandwidth and receive bandwidth.
- Step 4** Check whether the actual rate of a link exceeds the configured bandwidth noticeably.

Path ID	Actual Rate	Configured Bandwidth
PATHID	VS.IPPATH.IPLAYER.PEAK.TXRATE VS.IPPATH.IPLAYER.MEAN.TX	Transmit bandwidth
PATHID	VS.IPPATH.IPLAYER.PEAK.RXRATE VS.IPPATH.IPLAYER.MEAN.RX	Receive bandwidth

If yes, adjust the bandwidth of the links or add new links. Check whether the problem is solved. If the problem is solved, no further action is required. If the problem is not solved, go to Step 6.

If no, go to Step 5.

- Step 5** Check whether the user plane IP address carried in the RAB assignment request is consistent with that in the RNC configuration scripts by performing the following operation.

If...	Then...
The Iu-CS interface adopts ATM transmission	Check whether the transportLayerAddress field in the RAB ASSIGNMENTREQUEST message is consistent with the setting of the NSAP parameter for the corresponding ANI on the RNC side in the ADD AAL2RT command.
The Iu-CS/Iu-PS interface adopts IP transmission	Check whether the transportLayerAddress field in the RAB ASSIGNMENTREQUEST message is consistent with the setting of the PEERIPADDR parameter for the ANI on the RNC side in the ADD IPPATH command.

If not consistent, modify the parameters on the RNC side to keep them consistent with those of the CN.

If consistent, go to Step 6.

- Step 6** Contact Huawei technical support.

Typical Cases

Fault Description

The PS RAB setup success rate is very low. The value of VS.RAB.FailEstabPS.TNL increases noticeably.

Cause Analysis

The forward bandwidth and backward bandwidth configured for each IP path for the SGSN are small. The total bandwidth is less than PS traffic flow in busy hours, leading to the PS RAB setup failures.

Fault Handling Procedure

- Step 1** Check the number of IP paths configured on the Iu-PS interface and the forward bandwidth and backward bandwidth.

- Step 2** Analyze the transmit rate and receive rate by viewing IPPATH.IPLAYER.
- Step 3** Check whether the KPIs exceed the bandwidth configured for the path.
- Step 4** Increase the forward bandwidth and backward bandwidth of the IP paths on the Iu-PS interface to solve the problem.

8.11 Troubleshooting Physical Channel Faults

8.11.1 Fault Description

The RAB setup success rate decreases. The value of VS.RAB.FailEstabCS.PhyChFail or VS.RAB.FailEstabPS.PhyChFail increases noticeably.

8.11.2 Fault Handling Procedure

- Step 1** Check whether the values of VS.RAB.FailEstabCS.PhyChFail and VS.RAB.FailEstabPS.PhyChFail increase drastically in some cells.
- If yes, go to Step 2. If no, go to Step 5.
- Step 2** Check whether the DRD success rate decreases noticeably. $DRD.RBSetup.succRate = VS.DRD.RBSetup.SuccOut / VS.DRD.RBSetup.AttOut$
- Step 3** Check whether the problem cell is configured with multiple neighboring cells for blind handovers. Run the **LST UINTERFREQNCCELL** command to locate the record meeting the following requirements:
- The blind handover flag is "Yes."
 - The RNC ID is the same as the RNC ID of neighboring cells.
 - The Cell ID and neighboring cell ID show that the two cells belong to one site.
- If yes, identify which is the same-coverage cell and modify the blind handover flags of other cells to "No."
- If no, record the neighboring cell ID and go to Step 4.
- Step 4** Check the cell ID and neighboring cell ID and analyze whether they are same-coverage cells.
1. Run the **LST UCELLSETUP** command to locate the LOCELL corresponding to the cell ID.
 2. Locate the corresponding NodeB. Run the NodeB MML command **LST LOCELL** to check whether the two cells have the same SECNO.
- If no, the two cells are not the same-coverage cells, reconfigure blind handover neighboring cells.
- If yes, go to Step 5.
- Step 5** Contact Huawei technical support.

8.11.3 Typical Cases

Fault Description

The PS RAB setup success rate decreases. The value of VS.RAB.FailEstabPS.PhyChFail increases noticeably in some cells, and the DRD success rate is low.

Possible Causes

On the dual-carrier network, cells with different coverage areas are mistakenly set as the inter-frequency neighboring cells for blind handovers. The DRD to inter-frequency cells fails during PS service setup due to poor signal quality.

Fault Handling

- Step 1** Check whether the problem cell and multiple inter-frequency cells are configured as neighboring cells for blind handovers.
- Step 2** Set only the same-coverage cells as the neighboring cells of the problem cell for blind handovers.

8.12 Miscellaneous

8.12.1 Fault Description

The RAB setup success rate decreases, but the RAB setup failures due to a specific cause do not increase noticeably.

8.12.2 Fault Handling Procedure

- Step 1** Check whether the numbers of failed CS RAB setups and failed PS RAB setups increase noticeably in some cells.

	Number of Failed RAB Setups (All)	Number of Failed RAB Setups (Causes known)	Number of Failed RAB Setups (Others)
CS	$(VS.RAB.AttEstabCS.Conv + VS.RAB.AttEstabCS.Str)$ – $(VS.RAB.SuccEstabCS.Conv + VS.RAB.SuccEstabCS.Str)$	$VS.RAB.FailEstabCS.Unsp + VS.RAB.FailEstabCS.TNL + VS.RAB.FailEstabCS.IubFail + VS.RAB.FailEstabCS.UuFail$	$(VS.RAB.AttEstabCS.Conv + VS.RAB.AttEstabCS.Str)$ – $(VS.RAB.SuccEstabCS.Conv + VS.RAB.SuccEstabCS.Str)$ – $(VS.RAB.FailEstabCS.Unsp + VS.RAB.FailEstabCS.TNL + VS.RAB.FailEstabCS.IubFail + VS.RAB.FailEstabCS.UuFail)$

	Number of Failed RAB Setups (All)	Number of Failed RAB Setups (Causes known)	Number of Failed RAB Setups (Others)
PS	(VS.RAB.AttEstabPS.Bkg + VS.RAB.AttEstabPS.Conv + VS.RAB.AttEstabPS.Int + VS.RAB.AttEstabPS.Str) – (VS.RAB.SuccEstabPS.Bkg + VS.RAB.SuccEstabPS.Conv + VS.RAB.SuccEstabPS.Int + VS.RAB.SuccEstabPS.Str)	VS.RAB.FailEstabPS.Unsp + VS.RAB.FailEstabPS.TNL + VS.RAB.FailEstabPS.IubFail + VS.RAB.FailEstabPS.UuFail	(VS.RAB.AttEstabPS.Bkg + VS.RAB.AttEstabPS.Conv + VS.RAB.AttEstabPS.Int + VS.RAB.AttEstabPS.Str) – (VS.RAB.SuccEstabPS.Bkg + VS.RAB.SuccEstabPS.Conv + VS.RAB.SuccEstabPS.Int + VS.RAB.SuccEstabPS.Str) – (VS.RAB.FailEstabPS.Unsp + VS.RAB.FailEstabPS.TNL + VS.RAB.FailEstabPS.IubFail + VS.RAB.FailEstabPS.UuFail)

Step 2 Check whether the cells support the corresponding services.

1. Run the RNC MML command **LST UCELL** to locate the service priority group identity (SPG ID) corresponding to the cell ID.
2. Run the RNC MML command **LST USPG** to find the service priority list according to the SPG ID.

If the priority of the current service is 0, the cell does not support this service. Run the RNC MML command **MOD USPG** to modify the service priority first and check whether the problem is solved. If yes, no further action is required. If no, go to Step 3.

Step 3 Check whether the RNC supports multiple RAB services.

Check whether the value of VS.MultRAB.0CS.2PS.RNC or VS.MultRAB.0CS.3PS.RNC is 0.

If yes, run the RNC MML command **SET UCORRMALGOSWITCH** to enable **CFG_MULTI_RAB_SWITCH** in the **CfgSwitch** parameter. Check whether the problem is solved. If solved, no further action is required. If the problem is not solved, go to step 4.

If no, go to Step 4.

Step 4 Contact Huawei technical support.

8.12.3 Typical Case 1

Fault Description

The CS RAB setup success rate decreases, especially for some cells. The values of VS.RAB.FailEstabCS.RNL and VS.RAB.FailEstabCS.TNL do not increase noticeably.

Possible Causes

In the multi-carrier service layered network, the cell using frequency F1 is preferentially selected for camping, but the SPGs of cells using frequencies F2 and F3 do not support R99 real-time services. Therefore, the RAB assignment for CS services fails.

Fault Handling

- Step 1** Check the frequencies of the cells with a low CS assignment success rate. The cells use frequencies F2 and F3.
- Step 2** Check the configuration of the cells. The R99 real-time service priority of these cells is 0, indicating that these cells do not support R99 real-time services. Therefore, the CS services redirected from the cell using F1 to cells using F2 and F3 fail.
- Step 3** Modify the R99 real-time service priority in the SPG of cells using frequencies F2 and F3 to a value other than 0 to solve the problem.

8.12.4 Typical Case 2

Fault Description

The PS RAB setup success rate decreases, but the values of VS.RAB.FailEstabPS.TNL and VS.RAB.FailEstabPS.RNL remain unchanged.

Possible Causes

The multi-RAB switch is disabled and the PS domain does not support multi-RAB setup, leading to PS RAB assignment failures.

Fault Handling

- Step 1** Analyze the value of VS.MultRAB.0CS.2PS.RNC. It is 0.
- Step 2** Check the configuration to see whether the multi-RAB switch is disabled.

Run the **RNC MML command LST UCORRMALGOSWITCH** to check the setting of **CFG_MULTI_RAB_SWITCH**.

Enable the multi-RAB function to solve the problem.

9 Troubleshooting Call Drops

9.1 Definition of CDR

Call drop rate (CDR) refers to the proportion of abnormally dropped calls to the total calls initiated by the MS. The CDR indicates the retainability of CS services and it is one of the important KPIs that customers consider.

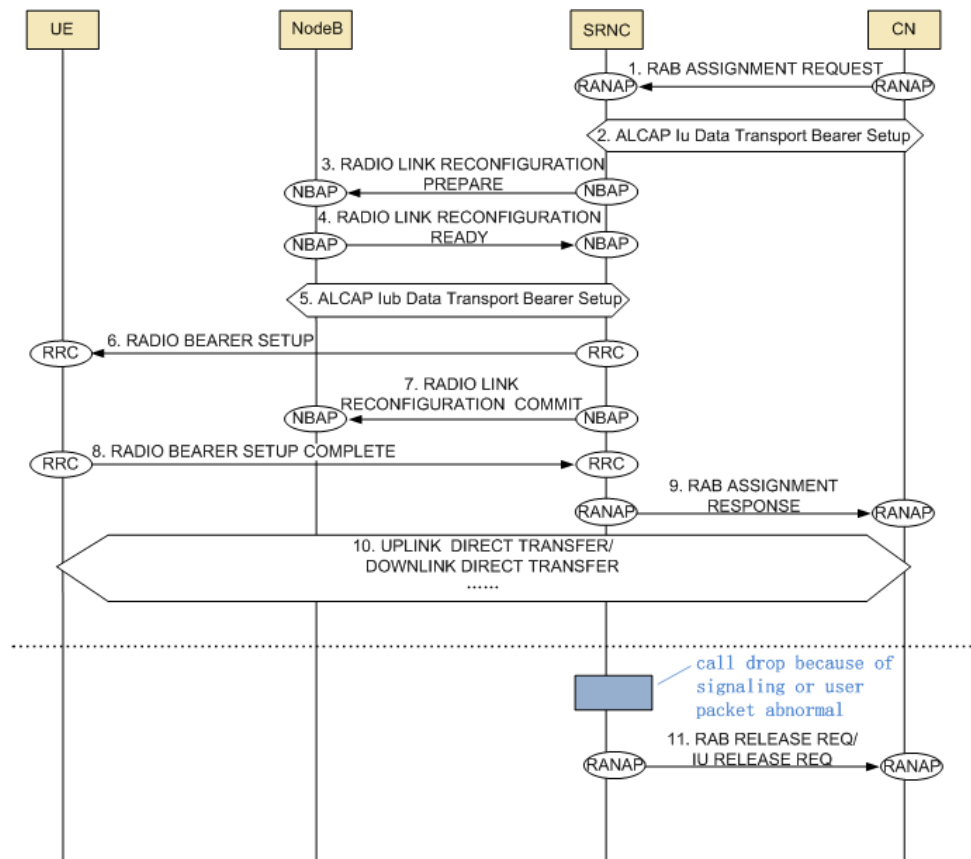
The higher the CDR is, the more it upsets the customers. CDR can be classified into CS CDR and PS CDR according to different service types in Core Network (CN) domain.

9.1.1 CDR Formulas

- The following formula is for CS CDR:
$$\text{VS.CS.Call.Drop.Cell.Rate} = \text{VS.RAB.AbnormRel.CS} / (\text{VS.RAB.AbnormRel.CS} + \text{VS.RAB.NormRel.CS})$$
- The following formula is for PS CDR:
$$\text{VS.PS.Call.Drop.Cell.Rate} = \text{VS.RAB.AbnormRel.PS} / (\text{VS.RAB.AbnormRel.PS} + \text{VS.RAB.NormRel.PS})$$

9.1.2 Signaling Procedure for a Call Drop

Figure 9-1 shows the signaling procedure for a call drop.

Figure 9-1 Signaling procedure for a call drop

9.2 Related KPIs for Call Drops

Table 9-1 lists cell-level KPIs for CS call drops.

Table 9-1 Cell-level KPIs for CS call drops

KPI	Counters	Remarks
VS.RAB.AbnormRel.CS	Number of RF call drops: VS.RAB.AbnormRel.CS.RF	All the sub-counters but the following: - VS.RAB.AbnormRel.CS.RF.ULSync - VS.RAB.AbnormRel.CS.RF.UuNoReply - VS.RAB.AbnormRel.CS.RF.SRBReset - Others
	Number of non-RF call drops: VS.RAB.AbnormRel.CS-VS	All the sub-counters but the following: VS.RAB.AbnormRel.CS.IuA

KPI	Counters	Remarks
	.RAB.AbnormRel.CS.RF	AL2 - VS.RAB.AbnormRel.CS.OM - VS.RAB.AbnormRel.CS.Prempt - VS.RAB.AbnormRel.CS.OLC - Others

Table 9-2 lists cell-level KPIs for PS call drops.

Table 9-2 Cell-level KPIs for PS call drops

KPI	Counters	Remarks
VS.RAB.AbnormRel.PS	Number of RF call drops: VS.RAB.AbnormRel.PS.RF	All the sub-counters but the following: - VS.RAB.AbnormRel.PS.RF.SRBReset - VS.RAB.AbnormRel.PS.RF.ULSync - VS.RAB.AbnormRel.PS.RF.UuNoReply - VS.RAB.AbnormRel.PS.RF.TRBReset - Others
	Number of non-RF call drops: VS.RAB.AbnormRel.PS- VS.RAB.AbnormRel.PS.RF	All the sub-counters but the following: - VS.RAB.AbnormRel.PS.GTPULoss - VS.RAB.AbnormRel.PS.OM - VS.RAB.AbnormRel.PS.Prempt - VS.RAB.AbnormRel.PS.OLC - Others

Table 9-3 lists Iur-interface-level sub-counters for the call drops at Iur-interface.

Table 9-3 Iur-interface-level sub-counters for call drops at Iur-interface

Description	Item
Number of abnormally released CS RABs according to different types of services on the SRNC IUR interface	- VS.RAB.AbnormRel.AMR.Iur - VS.RAB.AbnormRel.CS64.Iur - VS.RAB.AbnormRel.CS.Str.Iur - VS.RAB.AbnormRel.AMRWB.Iur
Number of RABs abnormally released on the Iur interface according to service types in PS domain	- VS.RAB.AbnormRel.PS.Conv.Iur - VS.RAB.AbnormRel.PS.Str.Iur - VS.RAB.AbnormRel.PS.BE.Iur

9.3 Troubleshooting Procedure

- Analyze the proportion of section 9.2 "Related KPIs for Call Drops" to the adding call drops. Decide the impact scopes. Generally, the faulty scope can be classified as the whole RNC cell, a set of cells containing Iur neighboring relationship, individual cell or site, a cell to which a subrack belongs, a cell to which an interface board belongs and a cell to which the CPUS corresponds. Then analyze and troubleshoot the problem according to different scopes.
 -If they occur in a single cell or site, see section 9.4 "Troubleshooting Call Drops in a Single Cell or Site".
 -If they occur in other areas, see section 9.5 "Troubleshooting Call Drops in the Entire Network".
- Please collect common fault information and the following information before you contact Huawei Customer Service Center.

Table 9-4 provides the information to be collected for fault locating before you contact Huawei Customer Service Center.

Table 9-4 Information to be collected for fault locating

NO.	Item	Description	Remarks
1	Detailed fault description	- Start and end time of the fault - Detailed fault description - Impact scopes (a cell, a NodeB, the whole RNC or other RNCs under the same MSC).	None
2	Operations taken before and after the fault occurs	Operations taken before and after the fault occurs, such as: - Board switchover - Software upgrade - Change of the clock source - Dynamic data configuration	None

NO.	Item	Description	Remarks
		- NodeB reset - RNC reset - MSC cutover - MSC data modification	
3	Version information of faulty NEs	Software versions of the related RNCs and NodeBs	For the method of collecting software versions, see Appendix "Methods to Collect Fault Information".
4	Data configuration script	Data configuration script used when the fault occurs	For the method of collecting a data configuration script, see Appendix "Methods to Collect Fault Information".
5	Historical alarms	Historical alarms generated seven days before and after the fault occurs	For the method of collecting historical alarms, see Appendix "Methods to Collect Fault Information".
6	Counter values	Values of the related counters obtained seven days before and after the fault occurs	For the method of collecting counter values, see Appendix "Methods to Collect Fault Information".
7	CALLFAULT, CHR and PCHR logs	CALLFAULT, CHR and PCHR logs (including all subrack logs) generated two hours before and after the fault occurs	For the method of collecting CALLFAULT, CHR and PCHR logs, see Appendix "Methods to Collect Fault Information".
8	Common debug logs	Common debug logs generated two days before and after the fault occurs	For the method of collecting common debug logs, see Appendix "Methods to Collect Fault Information".
9	Operation logs	Operation logs generated 10 days before and after the fault occurs	For the method of collecting operation logs, see Appendix "Methods to Collect Fault Information".
10	Results of IOS tracing	Results of IOS tracing in one or two faulty cells when the fault occurs	For the method of collecting IOS tracing results, see Appendix "Methods to Collect Fault Information".
11	NodeB logs	Logs of one or two faulty NodeBs	For the method of collecting NodeB logs, see Appendix "Methods to Collect Fault Information".

9.4 Troubleshooting Call Drops in a Single Cell or Site

9.4.1 Fault Description

The CS or PS call drop rate increases and the statistics show that call drops occur in a single cell or site.

9.4.2 Fault Handling Procedure

Step 1 Check the site to see whether any of the transmission alarms listed in Table 9-5 and Table 9-6 are generated.

1. If yes, clear the alarms according to the online help. Then, check whether the related KPIs restore. If the KPIs do not restore, go to Step 2. If the KPIs restore, no more operations are required.
2. If no, go to Step 2.

Table 9-5 RNC transmission alarms

Alarm ID	Alarm Name/Class
21541, 21542	SCTP link faults alarms
21531, 21232	SAAL link faults alarms
21345–21349, 21371, 21374, 21375, 21350, 21387	FEGE ports alarms
21251–21275, 21276–21291	Optical ports alarms
21201–21209	E1 transmission alarms

Table 9-6 NodeB transmission alarms

Alarm ID	Alarm Name/Class
21541, 21542	SCTP link faults alarms
21531, 21232	SAAL link faults alarms
25880–25900	FEGE ports alarms
25820–25834	ATM transmission alarms
25800–25807	E1 transmission alarms

Step 2 Check the site to see whether any of the device and clock alarms listed in Table 9-7 are generated.

1. If yes, clear the alarms according to the online help. Then, check whether the related KPIs restore. If the KPIs do not restore, go to Step 3. If the KPIs restore, no more operations are required.

2. If no, go to Step 3.

Table 9-7 NodeB device and clock alarms

NodeB Alarm ID	Alarm Name
25920–25938	Optical ports alarms
26200–26216	Board alarms
26501–26546	RF faults alarms
26751–26760	Antenna/TMA faults alarms
26260–26266	Clock alarms

Step 3 Collect the value of VS.MeanRTWP in the cells under the same site. If the value is larger than –95 dB, call drops may occur.

1. If yes, check if any interference exists. If the problem is solved, no more operations are required. If the counter remains large after the interference has been reduced, go to Step 4.
2. If no, go to Step 4.

Step 4 Collect and analyze the signaling messages traced by the IOS before call drops occur.

Check whether there are neighboring cells which are missed. It's RNC that cannot add cells with good signal quality to an active set after events 1A, 1C or 1D are reported.

1. If yes, add these cells to the active set. Then check whether call drops are cleared. If call drops are cleared, no more operations are required. If call drops persist, go to Step 5.
2. If no, go to Step 5.

Step 5 Collect the information for fault locating provided in Table 9-4. Then, contact Huawei Customer Service Center.

9.4.3 Typical Cases

Fault Description

After a NodeB is reparented from RNC1 to RNC2, the CS and PS call drop rates increase.

Possible Causes

Cells with good signal quality are not configured as neighboring cells for the problem cell.

When the NodeB is being reparented, the original bidirectional relationship between the problem cell and its neighboring cells becomes unidirectional. This leads to coverage holes and causes signal quality to deteriorate, leading to call drops.

Fault Handling

Note that cell 31509 is a problem cell in the following handling procedure.

Step 1 Analyze how a call drop occurs in cell 31509 by referring to the IOS tracing results.

The results show the RNC fails to initiate a soft handover to add related cells to the active set after events 1A and 1D are reported. As a result, the signal quality deteriorates, leading to call drops.

Step 2 Compare the RNC configuration files before and after the NodeB is reparented.

The results show the problem cell, cell 15429, and cell 35429 is configured as neighboring cells before the NodeB is reparented. However, the neighboring cell relationship is not configured after the NodeB is reparented, as shown in Figure 9-2.

Figure 9-2 Configuration files before and after NodeB is reparented

Step 3 Configure the three cells as neighboring cells to each other again.

9.5 Troubleshooting Call Drops in the Entire Network

9.5.1 Fault Description

The VS.RAB.AbnormRel.CS and VS.RAB.AbnormRel.PS KPIs provide the number of CS call drops and PS call drops, respectively. Statistics show that call drops occur in the entire network.

9.5.2 Fault Handling Procedure

Step 1 Query the operation logs to check whether parameter settings are changed when call drops occur.

1. If yes, check whether the parameter settings are appropriate. If some parameter settings are inappropriate, modify them and check whether the related KPIs restore. If the KPIs restore, no more operations are required. If the KPIs do not restore, go to Step 2.
2. If no, go to Step 2.

Step 2 Check whether any of the alarms listed in Table 9-8 and Table 9-9 are generated.

1. If yes, clear the alarms according to the online help. Then, check whether the related KPIs restore. If the KPIs do not restore, go to Step 3. If the KPIs restore, no more operations are required.
2. If no, go to Step 3.

Table 9-8 List of device alarms

Alarm ID	Alarm Name
20211	ALM-20211 DSP Time Synchronization Information Abnormal

Alarm ID	Alarm Name
20221	ALM-20221 Link Between GE Switching Boards Faulty
20222	ALM-20222 Communication Between GE Switching Boards Faulty
20224	ALM-20224 Broadcast Packet Overflow
20225	ALM-20225 GE Link on GE Switching Board Panel Faulty
20227	ALM-20227 Communication Between Subrack Faulty
20228	ALM-20228 GE Link Between GE Switching Board and Service Board Faulty
20232	ALM-20232 GE Interface Unit Fault
20233	ALM-20233 Board Voltage Abnormity Alarm
20234	ALM-20234 Board BIOS CRC Fault Alarm
20241	ALM-20241 Board Unavailable
20242	ALM-20242 Board Subsystem Unavailable
20243	ALM-20243 Board Hardware Fault
20244	ALM-20244 Subrack Reset
20248	ALM-20248 Incorrect Board Slot Information
20249	ALM-20249 Abnormal Information About DIP Switch of Subrack
20250	ALM-20250 Sub-board Status Abnormal
20251	ALM-20251 Board Temperature too High
20254	ALM-20254 DSP Unavailable
20256	ALM-20256 CPU Overload
20257	ALM-20257 Board Startup and Running Failure
20260	ALM-20260 Internal Communication Fault
20272	ALM-20272 Board Function Unavailable
20750	ALM-20750 CRC Value Inconsistency in Board Startup
22501	ALM-22501 DSP CPU Overload

Table 9-9 List of clock alarms

Alarm ID	Alarm Name
20204	ALM-20204 Clock Signal Inputs Faulty
20206	ALM-20206 Current System Clock Reference Source Status Abnormal

20209	ALM-20209 Faulty Phase-Locked Loop of the Board Clock
20210	ALM-20210 Current Clock Reference Source of Main Control Board Abnormal
20201	ALM-20201 1PPS State Abnormal
20202	ALM-20202 Time Information Reception Abnormal

Step 3 Check whether any of the transmission alarms listed in Table 9-10 are generated, especially transmission over the Iu and Iur interface. For Iub interface, check whether a large amount of new alarms is generated.

1. If yes, clear the alarms according to the online help. Then, check whether the related KPIs restore. If the KPIs do not restore, go to Step 4. If the KPIs restore, no more operations are required.
2. If no, go to Step 4.

Table 9-10 List of transmission alarms

Alarm ID	Alarm Name/Class
21541, 21542	SCTP link
21531, 21232	SAAL link
21551–21553	M3UA link set
21501–21506	MTP3B link set
21345–21349, 21371, 21374, 21375, 21350, 21387	FEGE ports
21251–21275, 21276–21291	Optical port transmission
21201–21209	E1 transmission

Step 4 If call drops persist after the preceding steps are taken, collect the information for fault locating before contact Huawei Customer Service Center.

Typical Case 1

Fault Description

The CS CDR rises suddenly in a site while the PS CDR remains unchanged.

Possible Causes

Changes in parameter settings cause the CS CDR to rise.

Fault Handling

Step 1 Analyze counter values.

The results show call drops do not occur in a single cell. In this case, it can be inferred that call drops occur in the entire network.

Step 2 Query operation logs.

The results show when call drops deteriorate, the MOD UCELLINTERFREQHOCOV reduces the CS 2D/2F threshold from -14/-12 dBm to -10/-8 dBm in cells with carrier frequency F2. That causes the CS to enter the compressed mode.

Step 3 Analyze power consumption.

More power is consumed when UEs operate in compressed mode. The Ec/N0 value is lower than that of the normal mode in same radio environment. As a result, call drops are more likely to occur.

Step 4 Restore the threshold for event 2D or 2F.

Typical Case 2

Fault Description

The CS CDR rises by 20% in a site. Statistics show that call drops are caused by none-RF reasons.

Possible Causes

Faults in the CN cause three paths over the Iu-CS interface to fail to work properly.

Fault Handling

Step 1 Check whether any alarm is generated.

It is found that no abnormal alarms are generated.

Step 2 Analyze the traffic volumes for the three paths.

The results show the three paths only transmit data.

Step 3 Perform an F5 CC loopback test by running the **ACT VCLCC** command.

The execution results indicate that the RNC is operating properly. The following is an example for the command:

```
ACT VCLCC: LNKT = AAL2PATH, ANI = xx, PATHID = xx, VCLTYPE = LOOPBACK;
```

Step 4 Check whether any exception occurs on the board on the CN side.

The result shows the board is faulty. Switch over the board and the data traffic on the path is steady. Call drops are cleared.

Typical Case 3

Fault Description

Both the CS and PS CDRs rise after the RNC is swapped.

Possible Causes

Transmission faults on the Iur interface cause congestion on the Iur links.

Fault Handling

The CS and PS CDR rise is shown in Figure 9-3.

Figure 9-3 Rise of CS and PS call drops

UCELL Group	Time(As day)	VS.PS.Call.Dro p.Cell.Rate	VS.CS.Call.D rop.Cell.Rate	VS.RAB.Abno rmRel.CS.RF	VS.RAB.Abno rmRel.CS	VS.RAB.Nor mRel.CS	VS.RAB.Abno rmRel.PS.RF	VS.RAB.Abno rmRel.PS	VS.RAB.Nor mRel.PS
ALLCELLV9R12	2011-04-19	5.16%	0.62%	358	360	58020	13375	13777	253142
ALL CELLV1R12	2011-04-12	3.47%	0.39%	242	2	62762	9033	678	270239

Step 1 Check the values of the related counters.

The results show call drops mainly occur in cells whose neighboring cells are controlled by a different RNC, as shown in Figure 9-4.

Figure 9-4 Rise of call drops on the Iur Interface

UCELL Group	Time(As day)	V.S.P.S.Call.D rop.Cell.Rate	V.S.CS.Call.D rop.Cell.Rate	VS.RAB.Abno rmRel.CS.RF	VS.RAB.Abno rmRel.CS	VS.RAB.Nor mRel.CS	VS.RAB.Abno rmRel.PS.RF	VS.RAB.Abno rmRel.PS	VS.RAB.Nor mRel.PS
ALLCELLV9R12	2011-04-19	5.16%	0.62%	358	360	58020	13375	13777	253142
IurCellM9R12	2011-04-19	11.77%	1.54%	185	185	11822	6069	6190	46385
ALL CELLV1R12	2011-04-12	3.47%	0.39%	242	2	62762	9033	678	270239
IurCell V1R12	2011-04-12	5.09%	0.31%	38	0	12268	2310	170	46198

Step 2 Analyze generated alarms and operation logs.

The results show no abnormal transmission alarms are generated or exceptions occur on devices. In addition, no changes are made to parameter settings.

Step 3 Analyze IOS tracing results specific to the problem cells.

The results show call drops occur when the signal is getting stronger in the DRNC.

Analyze the user-plane data.

The results show no frames are received from the DRNC.

Step 4 Check Iur-interface configurations.

The results show there are four paths between the SRNC and DRNC, but the configurations on the two RNCs are different. The differences are as follows:

- On the SRNC, links 1 and 2 are carried over a physical port; links 3 and 4 are carried over another physical port.
- On the DRNC, links 1 and 3 are carried over a physical port; links 2 and 4 are carried over another physical port.

Restore the links and call drops are cleared.

10 Troubleshooting Handover Faults

10.1 About This Chapter

This chapter describes the procedure for troubleshooting handover faults.

10.2 Definitions of Handover Faults

10.2.1 Handover Success Ratio Formula

Table 10-1 lists the handover success ratio formulas.

Table 10-1 Handover success ratio formulas

Soft Handover Success Ratio	Soft Handover Success Ratio (RNC) = $(VS.SHO.Succ.RNC / VS.SHO.Att.RNC) * 100\%$; Soft Handover Success Ratio (Cell) = $[(VS.SHO.SuccRLAdd + VS.SHO.SuccRLDel) / (VS.SHO.AttRLAdd + VS.SHO.AttRLDel)] * 100\%$
Inter-frequency Hard Handover Success Ratio	Inter-frequency Hard Handover Success Ratio (RNC) = $(VS.HHO.SuccInterFreq.RNC / VS.HHO.AttInterFreq.RNC) * 100\%$; Inter-frequency Hard Handover Success Ratio (Cell) = $(VS.HHO.SuccInterFreqOut / VS.HHO.AttInterFreqOut) * 100\%$
CS WCDMA-to-GSM Inter-RAT Handover Out Success Ratio	CS W2G Inter-RAT Handover Out Success Ratio (RNC) = $(VS.IRATHO.SuccOutCS.RNC / VS.IRATHO.AttOutCS.RNC) * 100\%$; CS W2G Inter-RAT Handover Out Success Ratio (Cell) = $(IRATHO.SuccOutCS / IRATHO.AttOutCS) * 100\%$
PS WCDMA-to-GSM Inter-RAT Handover Out Success Ratio	PS W2G Inter-RAT Handover Out Success Ratio (RNC) = $(VS.IRATHO.SuccOutPSUTRAN.RNC / VS.IRATHO.AttOutPSUTRAN.RNC) * 100\%$; PS W2G Inter-RAT Handover Out Success Ratio (Cell) = $(IRATHO.SuccOutPSUTRAN / IRATHO.AttOutPSUTRAN) * 100\%$
SRNC Relocation Success Ratio	SRNC Relocation Success Ratio = $[(VS.SRELOC.SuccExecUEInvolCS + VS.SRELOC.SuccExecUEInvolPS + VS.SRELOC.SuccExecUENonInvolCS + VS.SRELOC.SuccExecUENonInvolPS) / (RELOC.SuccPrepUEInvolCS + RELOC.SuccPrepUENotInvolCS + RELOC.SuccPrepUEInvolPS + RELOC.SuccPrepUENotInvolPS)] * 100\%$

10.2.2 Handover Signaling Procedure

For the signaling procedure for each type of handover, see the following description in the RAN feature Documentation.

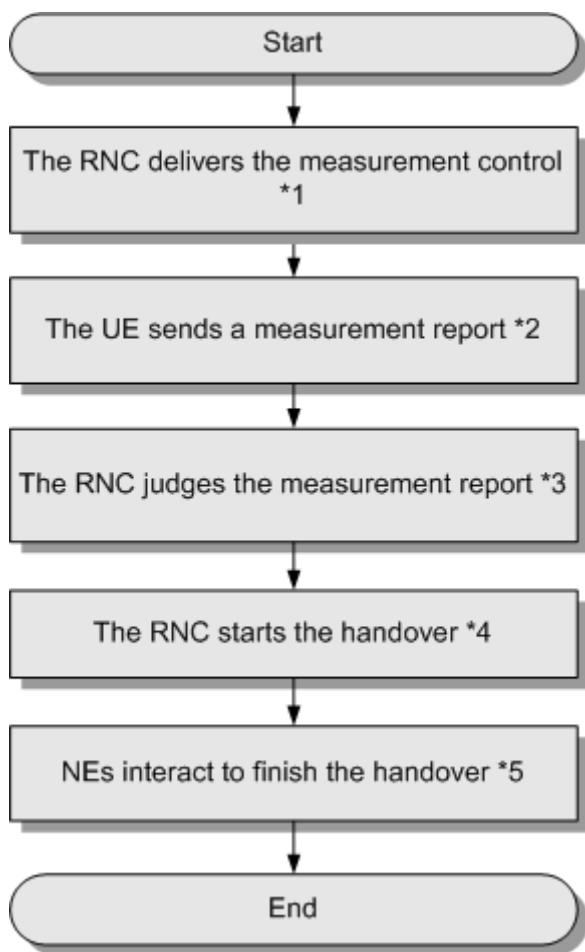
Table 10-2 lists the signaling procedure for each type of handover in the RAN feature Documentation.

Table 10-2 Signaling procedure for each type of handover

Signaling Procedures for Intra-Frequency Handover	Intra-NodeB Intra-Frequency Soft Handover Signaling Procedure
	Intra-RNC Inter-NodeB Intra-Frequency Soft Handover Signaling Procedure
	Inter-RNC Intra-Frequency Soft Handover Signaling Procedure
	Intra-RNC Inter-NodeB Intra-Frequency Hard Handover Signaling Procedure
	Inter-RNC Intra-Frequency Hard Handover Signaling Procedure
Signaling Procedures for Inter-Frequency Handover	Inter-Frequency Handover Within One RNC
	Inter-Frequency Handover Between RNCs
Signaling Procedures for Inter-RAT Handover	UMTS-to-GSM Handover in the CS Domain
	UMTS-to-GSM Handover in the PS Domain
	UMTS-to-GSM Handover in Both CS Domain and PS Domain
	GSM-to-UMTS Handover in the CS Domain
	GSM-to-UMTS Handover in the PS Domain

10.3 Handover Procedures

Figure 10-1 shows the handover procedure. When troubleshooting a fault according to the signaling procedure, first find the step where there is a fault, and then analyze the fault cause.

Figure 10-1 Handover procedure**NOTE**

The abnormal measurement control message is caused by the following reasons:

- The cell has no neighboring relationship with other cells.
- The neighboring cell parameter settings for the cell are incorrect.
- The corresponding handover switch is not turned on in the cell.

The measurement report may not be sent due to incorrect settings of the cell handover triggering conditions.

Check whether the cell supports the corresponding services.

The handover failure is caused by the following reasons:

- The radio link is not configured during the cross-Iur handover.
- The inter-RAT handover configuration is incorrect on the GSM side.

The handover failure is caused by the following reasons:

- The network side does not receive the handover completion message because of poor quality of the air-interface signal.
- The user equipment (UE) reports the handover failure message because the configuration does not support the handover or new cells cannot be synchronized.

10.4 Troubleshooting Handover Faults

10.4.1 Fault Description

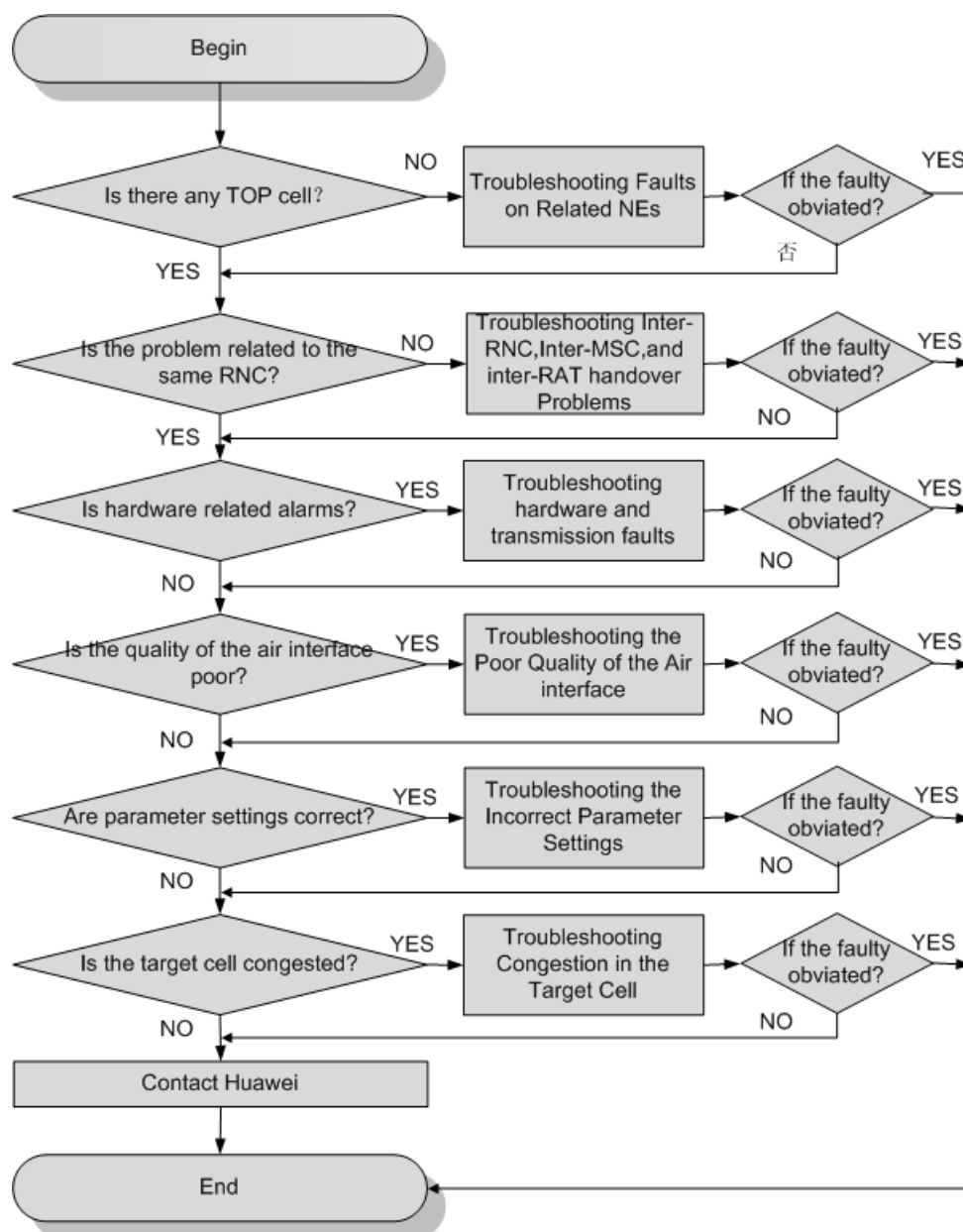
The handover success ratio is low.

10.4.2 Possible Causes

- First check whether the handover problem is caused by an RNC fault or a NodeB fault according to the range and background of handover failures.
 - If the handover problem is caused by an RNC fault, check the network level issues including the parameter settings on the mobile switching center (MSC) and radio network controller (RNC) and signaling interaction between the MSC and RNC.
 - If the handover problem is caused by a NodeB fault, check the parameter settings, air-interface signal quality, and TOP UE in the cell where the handover problem occurs. A TOP UE is a UE whose handover success ratio is much lower than others.
- The methods of locating handover faults are as follows:
 - Analyze the traffic measurement counters for the handover.
 - Locate the faults in the TOP cell..
 - Check the parameter settings.
 - Check for device and transmission alarms.
 - Check for problems related to the interference and coverage.
 - Check the neighbor relationship plan.

Figure 10-2 shows the common procedure for troubleshooting handover faults.

Figure 10-2 Procedure for handling handover faults



10.4.3 Fault Handling Procedure

Step 1 Analyze the handover success ratio and check whether there are TOP faulty cells.

- If yes, go to Step 2.
- If no, handle faults according to section 10.5 "Troubleshooting Faults on Related NEs."

Step 2 Check whether the source and target cells where the handover fails belong to the same RNC.

- If yes, go to Step 3.
- If no, handle faults according to section 10.6 "Troubleshooting Inter-RNC, Inter-MSC, and Inter-RAT Handover Problems."

Step 3 Check whether there is a hardware alarm in the cells where the handover fails.

- If no, go to Step 4.
- If yes, handle faults according to section 10.7 "Troubleshooting the Abnormal Handover Caused by Hardware and Transmission Faults."

Step 4 Check whether the air-interface quality is poor (low Ec/No or high RTWP).

- If yes, handle faults according to section 10.8 "Troubleshooting the Abnormal Handover Caused by Poor Quality."
- If no, go to Step 5.

Step 5 Check whether the handover parameter settings (including the neighboring cell capability, the handover threshold, and so on) is normal.

- If yes, go to Step 6.
- If no, handle faults according to section 10.9 "Troubleshooting the Abnormal Handover Caused by Incorrect Parameter Settings."

Step 6 Check whether there is a heavy congestion in the target cell.



NOTE

If the success ratio in the WCDMA-to-GSM inter-RAT handover is low, check the congestion ratio on the traffic channel (TCH) in the target neighboring GSM cells.

- If there is a heavy congestion in the target cell, handle faults according to section 10.10 "Troubleshooting Congestion in the Target Cell."
- If there is no heavy congestion in the target cell, go to Step 7.

Step 7 Contact Huawei Customer Service Center.

10.5 Troubleshooting Faults on Related NEs

10.5.1 Fault Description

10.5.2 The handover success ratio is low in most of cells, but there is no TOP cell which is quite low. Related Information

The clock exceptions cause the following problems:

- The UE cannot measure inter-frequency neighboring cells.
- The UE cannot send the measurement report.
- It is difficult to trigger handovers.

10.5.3 Fault Handling Procedure

Step 1 Run the RNC MML command **DSP CLK** to check whether the clock status is normal on each board. Select the clock board and check whether the clock reference source is normal on the RNC.

- If the phase-locked loop status of the current clock source on the clock board is tracing, and the radio frame number (RFN) state is normal on the SPU, DPU, and SCU boards, go to Step 2.

- If no, check for the alarms in Table 10-3. If the following alarms occur, handle the fault according to the alarm help.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 2.

Table 1-3 lists the clock alarms on each board.

Table 10-3 Clock alarms on each board

20201	ALM-20201 1PPS State Abnormal
20202	ALM-20202 Time Information Reception Abnormal
20203	ALM-20203 Clock Signal Outputs Faulty
20204	ALM-20204 Clock Signal Inputs Faulty
20205	ALM-20205 System Clock Reference Source Unavailable
20206	ALM-20206 Current System Clock Reference Source Status Abnormal
20207	ALM-20207 Failure in Locking System Clock Source
20208	ALM-20208 Clock Reference Source of Main Control Board Unavailable
20209	ALM-20209 Faulty Phase-Locked Loop of the Board Clock
20210	ALM-20210 Current Clock Reference Source of Main Control Board Abnormal
20211	ALM-20211 DSP Time Synchronization Information Abnormal

Step 2 Contact Huawei Customer Service Center.

10.6 Troubleshooting Inter-RNC, Inter-MS, and Inter-RAT Handover Problems

10.6.1 Fault Description

- The inter-RNC handover failure ratio is high in some cells.
- The inter-RAT handover failure ratio is high in some cells.

10.6.2 Possible Causes

- The parameter settings (CELL ID, RNC ID, and LAC) are incorrect in the cells related with the inter-MS handover.

- The parameter settings are incorrect in the cells related with the handover between target RNCs.
- The neighboring cell configuration is incorrect between systems in the network.
- The encryption process is faulty.
- The GSM clock is abnormal.
- The handover process is abnormal.

10.6.3 Fault Handling Procedure

Step 1 Run the RNC MML command **DSP CLK** to check whether the clocks on the source RNC, target RNC, source base station controller (BSC), and target BSC are normally synchronized with the clock on the MSC.

- If the phase-locked loop status of the current clock source on the clock board is tracing, go to Step 2.
- If no, perform troubleshooting to ensure the synchronization and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 2.

Step 2 Check whether neighboring cells are configured correctly on the source RNC, target RNC, source BSC, and target BSC.

According to the network plan and engineering parameters of the live network, compare the cell and neighboring cell configuration between the source and target cells to see whether all neighboring cells are configured or the cell ID and scrambling code is correct.

- If neighboring cells are configured correctly, go to Step 3.
- If neighboring cells are not configured correctly, reconfigure the neighboring cells and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 3.

Step 3 On the MSC, check whether the parameter settings related to the cells where the handover fails are correct. The parameters to be checked include CELL ID, RNC ID, and LAC.

- If the parameter settings are correct, go to Step 4.
- If the parameter settings are incorrect, reconfigure the parameters and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 4.

Step 4 Check whether the handover fails in the encryption process.

In the signaling handover process, the UE fails in accessing the cell controlled by the target RNC or BSC, and the RNC or BSC returns a physical channel failure, or the values of counters indicating physical channel failures, as listed in Table 10-4, increase.

Table 10-4 Counters for physical channel failures

Number	Counters for Physical Channel Failures
1	VS.HHO.FailOutInterRNCIur.PhyChFail.CS.NCell

Number	Counters for Physical Channel Failures
2	VS.HHO.FailOutInterRNCIur.PhyChFail.PS.NCell
3	IRATHO.FailOutCS.PhyChFail
4	IRATHO.FailOutPSUTRAN.PhyChFail
5	VS.IRATHO.FailRelocOutPS.PhyChFail
6	VS.U2LTEHO.FailOutPS.PhyChFail
7	VS.HHO.FailInterFreqOut.InterRNC.PhyChFail
8	VS.U2LTEHO.FailOutPS.PhyChFail
9	VS.SRELOC.FailExec.PhyChFail

- If yes, check whether the encryption algorithms are consistent on the MSC, RNC, and BSC.
 - If the encryption algorithms are consistent, go to Step 5.
 - If the encryption algorithms are inconsistent, modify the encryption process on the MSC or the encryption parameters or process on the RNC and BSC and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 5.

Step 5 Check whether the UMTS-to-GSM handover failure is caused by the abnormal clock on GSM base transceiver station (BTS).

- If yes, handle the fault and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 6.
- If no, go to Step 6.

Step 6 Trace the signaling of a user on the serving radio network controller (SRNC), drift radio network controller (DRNC), and BSC to check whether the signaling interaction is normal between the source RNC and the source MSC, the source MSC and the target MSC, the source RNC and the target BSC.

- If all the signaling processes are correct, go to Step 7.
- If any signaling process is incorrect, first analyze the NE that returns a failure message. For example, if an RNC returns a failure message, the personnel in charge of the RNC need to analyze the problem and then perform troubleshooting.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 7.

Step 7 Contact Huawei Customer Service Center.

10.6.4 Typical Cases

Typical Case 1

Fault Description

During the inter-RNC handover, after the SRNC sends a Relocation Required message to the CN, the CN responds with a Relocation Failure message. The cause value is "un-know RNC ID."

Possible Causes

Due to the incorrect DRNC configuration on the CN, the CN fails to find the correct DRNC after receiving a relocation request from the SRNC.

Fault Handling

1. The CN reports the failure, so the CN is checked first.
2. According to the message from the SRNC, the CN configuration is checked.
3. The DRNC configuration on the CN is incorrect. After the configuration is modified, the fault is rectified.

Typical Case 2

Fault Description

After a UMTS-to-GSM handover is triggered, the RNC on the UMTS side delivers the physical channel reconfiguration to a UE, but the UE reports a reconfiguration failure which is caused by a physical channel failure. Therefore, the handover fails.

After a GSM-to-UMTS handover is triggered, the UE sends the first message (HANDOVER TO UTRAN COMPLETE message) to the RNC on the UMTS side. The encryption algorithm used by the RNC on the UMTS side is not consistent with that on the GSM side. Therefore, the decryption fails, and the RNC does not receive the HANDOVER TO UTRAN COMPLETE message. As a result, the handover fails.

Possible Causes

The encryption algorithms used on the GSM and UMTS side are inconsistent. The message is encrypted by using the encryption algorithm (UEA1) on the UMTS side but it is not encrypted on the GSM side.

Fault Handling

1. The failure is analyzed as follows:
 - After a UMTS-to-GSM handover is triggered, the RNC on the UMTS side delivers the physical channel reconfiguration to a UE, but the UE reports a reconfiguration failure which is caused by a physical channel failure.
 - After a GSM-to-UMTS handover is triggered, the UE sends the first message (HANDOVER TO UTRAN COMPLETE message) to the RNC on the UMTS side. However, the RNC does not receive the HANDOVER TO UTRAN COMPLETE message.
2. The encryption policy is compared between the RNC and BSC to check whether the message is encrypted on the UMTS side but not on the GSM side. If yes, enable the encryption mode on the BSC.
3. After the encryption mode is enabled on the BSC, the troubleshooting ends.

Typical Case 3

Fault Description

During the GSM-to-UMTS handover, the RNC delivers the security mode after receiving an RRC_HO_UTRAN_CMP message from the UE, but the UE does not respond.

Possible Causes

The GSM clock fails to be synchronized with the MSC clock. Therefore, the UE cannot exchange information with the network after being handed over to the UMTS cell. As a result, the UE cannot respond to the Security Mode Cmd message delivered by the RNC.

Handling Process

1. The failure is analyzed as follows:
 - The GSM-to-UMTS handover process is completed.
 - The capability exchange is completed between the CN and the UE.
 - After the RNC delivers the security mode, the UE does not respond, and the RNC is released abnormally because of the timer expiration.
2. The GSM side is checked to see whether there is a clock alarm.
3. After the clock alarm on the GSM side is cleared, the troubleshooting ends.

10.7 Troubleshooting the Abnormal Handover Caused by Hardware and Transmission Faults

10.7.1 Fault Description

There are transmission alarms and the clock alarms.

Note:

If the parameter settings of the faulty cells and its neighboring cells are not modified recently, check whether the abnormal handover is caused by hardware and transmission faults first.

10.7.2 Related Information

The hardware fault alarms and IDs are as follows:

- ALM-21321 VCL CC Detection Failure
- ALM-21346 IP Connectivity Check Failure
- ALM-21581 Path Fault
- ALM-21252 SDH/SONET Loss of Signal
- ALM-21345 Ethernet Link Fault
- Alarms related to the clock source (ALM-20201 to ALM-20210).

10.7.3 Fault Handling Procedure

Step 1 Locate and clear the hardware fault alarm according to section 10.7.2 "Related Information."

- If the alarm is not cleared, go to Step 2.

- If the alarm is cleared, conduct the test again to check whether the handover counter is recovered.
 - If yes, the troubleshooting ends.
 - If no, go to Step 2.

Step 2 Contact Huawei Customer Service Center.

10.8 Troubleshooting the Abnormal Handover Caused by Poor Quality of the Air Interface

10.8.1 Fault Description

Table 10-5 shows that the handover failures increase obviously because the UE does not respond to the air interface message.

Table 10-5 Handover failure times

Times of the soft handover failure caused by poor quality of the air interface	VS.SoHO.FailRLAdd.NoReply VS.SHO.FailRLAdd.NoReply
Times of hard handover failure caused by poor quality of the air interface	VS.HHO.FailIntraFreqOut.NoReply VS.HHO.FailInterFreqOut.NoReply VS.HHO.FailIntraFreqOut.InterRNC.NoReply VS.HHO.FailInterFreqOut.InterRNC.NoReply

10.8.2 Related Information

- Common interferences include the uplink interference, downlink interference, antenna intermodulation interference, extranet inference, uplink intranet interference, and downlink intranet interference.
- If there is coverage difference between the uplink and downlink, the air interface will have a poor quality. As a result, signaling interaction will fail over the air interface.
- The MS reports the release abnormally because of the unspecified failure or timeout, protocol error, and others. They are usually caused by the poor quality of the air interface.

10.8.3 Fault Handling Procedure

- Step 1** Check whether there is interference in the cell by observing the counters such as the received total wideband power (RTWP), NodeB channel quality indication (CQI), and the Ec/No when users are accessing the cell. The Ec/No value is obtained from the RRC CONNECTION REQ message.
- If there is no interference in the cell go to Step 2.
 - If there is interference in the cell, clear the interference source or change the interfered frequency and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.

- If the fault persists, go to Step 2.

Step 2 Check the quality of the air interface by observing the counters such as the RTWP, NodeB CQI, and the Ec/No when users are accessing the cell. The Ec/No value is obtained from the RRC CONNECTION REQ message.

- If the quality of the air interface is good, go to Step 3.
- If the quality of the air interface is poor, perform network optimization to improve the quality of the air interface and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 3.

Step 3 Contact Huawei Customer Service Center.

10.8.4 Typical Cases

Fault Description

The radio coverage difference between the uplink and downlink causes a delay in the soft handover. As a result, handover fails, and therefore a call drop occurs. **The IOS tracing results shows that a soft handover fails.**

Possible Causes

When a soft handover starts, the radio quality in the serving cell and target cell is poor. The radio quality is worsening continuously. After delivering an **Active Set Update message**, the timer for the RNC waiting for the **Active Set Update Cmp** message from the UE expires. And then the handover fails, which causes a call drop.

Fault Handling

1. The UE reports event 1A. According to event 1A, the cell scrambling code to be added to the active set is 327.
2. The RNC delivers an **Active Set Update** message.
3. The measurement report from the UE shows that Ec/No reduces from -6.5 dB to -17.5 dB in 1s in the serving cell.
4. The RNC does not receive the **Active Set Update Cmp** message from the UE, so a CS call drop occurs.

10.9 Troubleshooting the Abnormal Handover Caused by Incorrect Parameter Settings

10.9.1 Fault Description

- The drive test and signaling monitoring results show that the signal strength or quality is poor in the serving cell of the UE, and the signal quality reaches the handover decision threshold in its neighboring cells, but the handover is difficult to trigger. Therefore, the call drop rate is high.
- The signal quality in the neighboring cells is almost the same as that in the serving cell, but handovers are frequently triggered. As a result, the conversation quality is poor, and call drops are easily caused.

- The PS WCDMA-to-GSM handover occurs frequently, but the handover success ratio is low.

10.9.2 Related Information

- Incorrect setting of the threshold for triggering the handover
If the handover time threshold and hysteresis for triggering events 1A, 2A, and 3A are set incorrectly, handovers are difficult to trigger or frequently triggered, and call drops are caused. For more details, see the description about parameters in the **SET UINTRAFFREQHO**, **SET UINTERFREQHOCOV**, and **SET UINTERRATHOCOV** commands.
- Incorrect cell parameter settings
If a cell and its neighboring cell have the same scrambling code, the RNC will start a handover to an incorrect cell after the UE sends the measurement report. Therefore, the UE cannot be synchronized with the target cell, which causes a handover failure and a call drop.
- Incorrect neighboring cell configuration
The signal quality is good in the neighboring cells. However, if the neighboring relationship is not configured or the neighboring cell configuration is incorrect, the UE will not report its neighboring cells or will report incorrect neighboring cells. As a result, the UE cannot start a handover or it is difficult to start a handover.

10.9.3 Fault Handling Procedure

- Step 1** Check the neighboring cell configuration to see whether all neighboring cells are configured, there is any redundant neighboring cell, the neighboring cell configuration are correct, and there is any cell whose frequency and scrambling code are same as its neighboring cells.



NOTE

Check the neighboring cells according to the network plan and engineering parameters of the live network.

- If the neighboring cell configuration is correct, go to Step 2.
- If the neighboring cell configuration is incorrect, reconfigure neighboring cells and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 2.

- Step 2** Optional: If the problem is related to inter-frequency or inter-RAT handovers, check whether parameter settings of the compressed mode are correct by running the **LST UHOCOMM**, **LST UCMCF**, **LST UCELLCMCF**, and **LST UCORRMALGOSWITCH** commands on the BSC.

- If parameter settings of the compressed mode are correct, go to Step 3.
- If parameter settings of the compressed mode are incorrect, run corresponding commands to reconfigure the parameters and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 3.

- Step 3** Check the handover parameter settings in the cell by running the **LST UCELLINTERFREQHOCOV**, **LST UCELLINTERFREQHONCOV**, **LST UCELLINTERRATHOCOV**, **LST UCELLINTERRATHONCOV**, and **LST**

UCCELLINTRAFREQHO commands on the BSC. Compare the parameter settings with those in the cells where the handover is normal to check for improper parameter settings.

- If parameter settings are proper, go to Step 4.
- If parameter settings are improper, run corresponding commands to reconfigure the parameters and conduct the test again.
 - If the fault is rectified, the troubleshooting ends.
 - If the fault persists, go to Step 4.

Step 4 Contact Huawei Customer Service Center.

Typical Cases

Fault Description

The PS relocation fails. As shown in the Iu interface trace result, after the RNC delivers a Relocation Required message and the core network (CN) delivers a Relocation Command message, the RNC delivers a Relocation Cancel message to terminate the relocation.

The cause value is "Iu transport connection failed to establish."

Possible Causes

According to the analysis of the fault symptom, the possible causes are as follows:

- The GTPU (Tunneling Protocol for the user plane) IP path for the DRNC is not configured or configured improperly. GTPU is short for the GPRS Tunneling Protocol for the user plane.
- The GTPU IP route (IPRT) to the DRNC is not configured or configured improperly.
- The target RNC does not accept the relocation.

Fault Handling

1. According to the Relocation_Command message delivered by the CN over the Iu interface, it is found that the GTPU address identified by the information element (transportLayerAddress-First) is 0C 11 0A 0D which becomes 12.17.10.13 after being changed into decimal, and then this address is confirmed to be the GTPU address of the DRNC.
2. The parameter settings of the RNC are checked. It is found that the SRNC cancels the relocation, because the IP path to the DRNC is not configured.
3. The IP path and IPRT are configured according to "Configuring a Path for Static SRNC Relocation" in the *BSC6900 UMTS initial configuration Guide*. Then the fault is rectified.

10.10 Troubleshooting Congestion in the Target Cell

10.10.1 Fault Description

The handover failures increase obviously in a cell because sources congestion in the target cell. Table 10-6 lists the number of failures in resource assignment during handovers in the cell.

Table 10-6 Number of failures in resource assignment during handovers in the cell

Number of Failures in Resource Assignment During Soft Handovers	Number of Failures in Resource Assignment During Hard Handovers
VS.RAC.SHO.Fail.ULCE.Cong	VS.RAC.HHO.Fail.ULCE.Cong
VS.RAC.SHO.Fail.ULPower.Cong	VS.RAC.HHO.Fail.ULPower.Cong
VS.RAC.SHO.Fail.DLPower.Cong	VS.RAC.HHO.Fail.DLPower.Cong
VS.RAC.SHO.Fail.Code.Cong	VS.RAC.HHO.Fail.ULIUBBand.Cong
VS.RAC.SHO.Fail.ULIUBBand.Cong	VS.RAC.HHO.Fail.DLIUBBand.Cong
VS.RAC.SHO.Fail.DLIUBBand.Cong	VS.RAC.HHO.Fail.HSDPANum.Cong
VS.RAC.SHO.Fail.HSUPANum.Cong	VS.RAC.HHO.Fail.HSUPANum.Cong
VS.RAC.SHO.Fail.DLCE.Cong	VS.RAC.HHO.Fail.Code.Cong
	VS.RAC.HHO.Fail.DLCE.Cong

10.10.2 Possible Causes

During some meetings or activities, subscribers increase sharply in a cell.

10.10.3 Fault Handling Procedure

Step 1 Check whether VS.RAB.FailEstabCS.Cong or VS.RAB.FailEstabPS.Cong in the UMTS target cell and the TCH congestion in the target GSM cell increase obviously.

- If yes, check whether the traffic increases.
 - If the traffic increases, modify the network to relieve the congestion.
 - If the traffic does not increase, go to Step 2.
- If no, go to Step 2.

Step 2 Contact Huawei Customer Service Center.

Typical Cases

Fault Description

The inter-RAT handover success ratio is quite low in a NodeB and much lower at busy hours.

Possible Causes

According to the analysis of the fault symptom, the possible causes are as follows:

- The target cell coverage becomes smaller and the coverage hole appears.
- The NodeB hardware is faulty.
- The TOP UE behavior causes the handover failure.
- UEs cannot access neighboring GSM cells because resources are unavailable in the target cell.

Fault Handling

The channel status of the target neighboring GSM cell is checked. It is found that all TCHs are occupied in the cell. When a TCH is available in the cell, the UE can be handed over.

11 Troubleshooting Paging Faults

11.1 About This Document

This document describes how to troubleshoot paging faults in terms of the definition and analysis of paging faults.

11.2 Definition of Paging Faults

The Iu paging success rate is low and the RRC setting success rate is normal. Answering calls is abnormal and making calls is normal.

11.3 Related Information

11.3.1 Paging Scenario

NOTE

This section describes how to troubleshoot paging faults of the PAGING TYPE1 in **IDLE** mode.

If the network needs to contact UEs in IDLE, CELL_PCH, URA_PCH, CELL_FACH, and CELL_DCH mode, paging needs to be initiated.

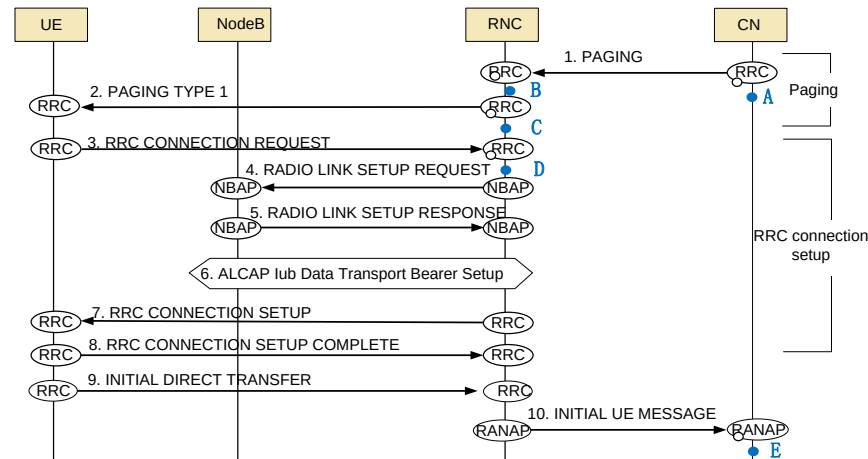
Paging messages are classified into two types: PAGING TYPE1 and PAGING TYPE2. The UTRAN determines the type of the paging message sent to the UE. The PAGING TYPE1 pages the UEs in IDLE, CELL_PCH, and URA_PCH mode through the PCCH logical channel. PAGING TYPE2 pages the UEs in CELL_FACH and CELL_DCH mode through the DCCH.

The network initiates paging in one of the following scenarios:

- The network receives UE paging requests.
- The UE needs to be notified of information updates in the cell system.
- The UE needs to be notified of PRC status changes.

11.3.2 Paging Procedure and Performance Counters

Figure 11-1 shows the called UE paging procedure in idle mode.

Figure 11-1 Called UE paging procedure in idle mode

NOTE

- In RRC CONNECTION REQUEST, establishmentCause is terminatingConversationalCall.
- In INITIAL UE MESSAGE, rr-msg-type is paging response.

Table 11-1 lists performance counters.

Table 11-1 Performance counters

Description of Measured Points	Performance Counters
Point A: The CN counts the number of times of sending PAGING.	See the number of paging attempts on the CN.
Point B: number of times of receiving PAGING on the RNC	VS.RANAP.Paging.Att.IdleUE
Point C: number of times of delivering PAGING on the RNC	none
Between point B and point C: number of times of RNC losing PAGING	Number of times of loss at the RNC level VS.RANAP.CsPaging.Loss + VS.RANAP.PsPaging.Loss
	Number of times of loss at the CPUS level VS.Paging.FC.Disc.Num.CPUS
	Number of times of loss at the cell level VS.RRC.Paging1.Loss.PCHCong.Cell
Point D: number of times of RNC receiving PAGING answered by the UE	VS.RANAP.Paging.Succ.IdleUE
Point E: number of times of CN receiving PAGING of callee setting success	For details, see number of success times of paging on the CN.

11.3.3 Difference Between Paging Success Rates on the RNC and on the CN

Idle paging success rate on the RNC in idle mode =

$$\text{VS.RANAP.Paging.Succ.IdleUE} / \text{VS.RANAP.Paging.Att.IdleUE}$$

The paging success rate on the CN is the paging success rates of the CS and PS domains.

Paging success rate of the CS domain = Number of paging attempts on the MSC / Number of paging success times on the MSC

Paging success rate of the PS domain = Number of paging attempts on the SGSN / Number of paging success times on the SGSN

The paging success rate on the CN stands for the rate of setting normal called-related services. The paging success rate on the RAN is just for reference. Table 11-2 describes comparison analysis on the paging success rates on the RNC and CN.

Table 11-2 Comparison analysis on the paging success rates on the RNC and CN

Performance Specifications	Statistics Method on the RNC	Statistics Method on the CN	Result
Number of paging requests	Including paging messages sent again	The same paging message can be regarded as one request in calculation.	$\text{RNC} \geq \text{CN}$
	Including the number of paging times of the CS domain and PS domain	The MSC and SGSN count the number of paging times of the CS and PS domains.	$\text{RNC} \geq \text{CN}$
	When the CN performs paging on the entire network, the RNC that the UE does not belong to counts the number of paging attempts.	When the CN performs paging on the entire network and the RNC is not the RNC that the UE belongs to, the CN does not count the number of paging attempts.	$\text{RNC} \geq \text{CN}$
Number of successful paging times	When the RRC CONNECTION REQUEST message is received, paging succeeds.	When the initial direct transmission message about the paging response type is received, paging succeeds.	$\text{RNC} \geq \text{CN}$
	When the CN performs paging on the entire network, the RNC that the UE does not belong to does not count the number of paging attempts.	When the CN performs paging on the entire network and the RNC is not the RNC that the UE belongs to, the CN does not count the number of paging attempts.	$\text{RNC} = \text{CN}$

11.3.4 Related Paging Handling

1. When the CPU usage of the RNC SPU subsystem exceeds the set paging flow control threshold, the RNC enables the flow control to paging services and protects system stability. The settings of the paging flow control threshold are as follows: SET FCCPUTHD: BRDCLASS = XPU, SMPAGECTHD = 70, SMPAGERTHD = 60, SLPAGECTHD = 80, SLPAGERTHD = 70, CPAGECTHD = 90, CPAGERTHD = 80.
2. The air interface PCH capacity is limited. Paging messages will be lost if the number of UEs being paged at the same time exceeds the system handling capacity. Currently, the PCH transmission block supported by the MACC is 240 bit and the coded paging message supported by each frame does not exceed 240 bit. Based on the information element structure of paging type1 and ASN.1 PER coding rules, if the UE labels of paging messages are IMSI, a maximum of three UEs are supported at each paging and if the UE labels are TMSI or PTMSI, a maximum of five UEs are supported.
3. The RTWP is too high. The UE may have received the paging message but the NodeB cannot parse the RRC CONNECTION REQ message. This results in paging failure.

11.4 Possible Causes

When troubleshooting paging faults, locate faults based on the Table 1-3 and analyze possible causes.

Table 11-3 describes possible causes of paging faults.

Table 11-3 Possible causes of paging faults

Possible Cause	Phase	Symptom
Group short messages are sent and the paging is performed on the entire network. These are caused by improper paging policies.	Exceptions occur when KPIs are monitored.	The paging success rate on the CN is normal but the paging success rate on the RNC is low.
The high RNC CPU usage performs flow control on paging messages.	Paging messages are not delivered at the air interface.	VS.Paging.FC.Disc.Num.CPUS increases abnormally.
Blocked paging channels cause a large number of paging messages to be lost.		VS.RRC.Paging1.Loss.PCHCong.Cell increases abnormally.
Other causes: High RTWP in cells results in failure to parse RRC CONNECTION REQ. The overflow paging channel ratio of cells causes paging messages not be received by the UE and results in blind coverage	After paging messages are delivered at the air interface.	The UE does not receive paging messages or receives wrong paging messages.

Possible Cause	Phase	Symptom
areas, mobile phone performance problems.		

11.5 Troubleshooting Paging Faults

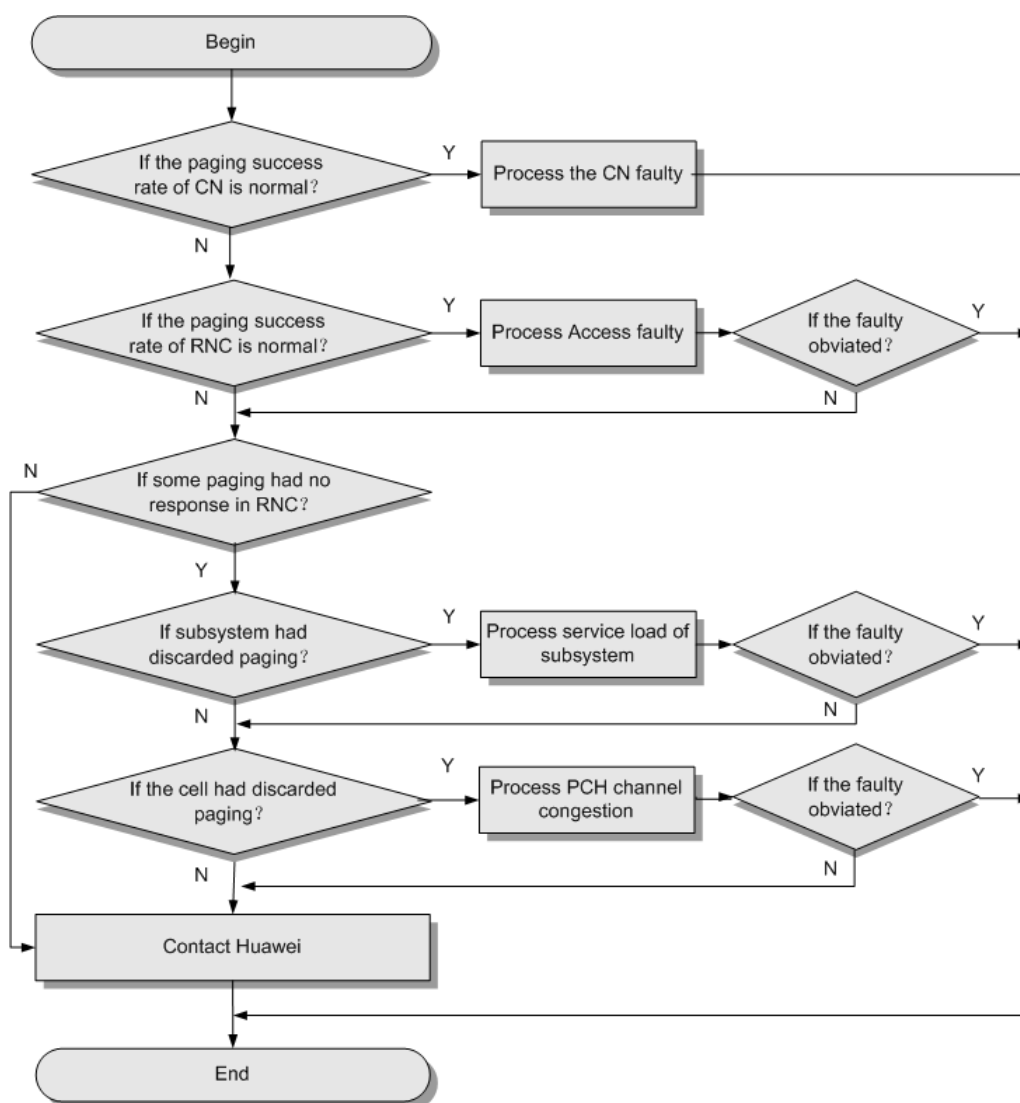
11.5.1 Fault Description

The paging success rate decreases.

11.5.2 Fault Handling Flowchart

Figure 11-2 shows the fault handling flowchart for paging faults.

Figure 11-2 Fault handling flowchart for paging faults



11.5.3 Fault Handling Procedure

Step 1 Determine whether the CN paging success rate is normal.

If yes, the paging success rate of end-to-end paging services is normal. The CN needs to analyze whether improper configurations exist.

If no, go to Step 2.

Step 2 Determine whether the RNC paging success rate is normal.

If yes, RRC setting failure results in terminal paging failure. For details, see Chapter 2 "Access Problems".

If no, the CN and RNC paging success rates are low. Go to Step 3.

Step 3 Determine whether paging messages without responses exist under the RNC.

Check whether the VS.RANAP.CsPaging.Loss /VS.RANAP.PsPaging.Loss increases sharply.

If yes, go to Step 4.

If no, go to Step 6.

Step 4 Determine whether the subsystem loses paging messages.

Check whether the VS.Paging.FC.Disc.Num.CPUS increases sharply.

If yes, the corresponding heavily-loaded CPUS subsystem results in paging loss. Determine whether the fault is rectified after performing the following operations in Table 1-4. If yes, no further action is required. If no, go to step 5. Table 1-4 Related operations

MML Command	Parameter	Operation
LST/SET URRCTRLSWITCH URRCTRLSWITCH	RNC_SHARE_SWITCH of PROCESSSWITCH	If the switch is turned off, turn on the switch.
LST/SET FCCPUTHD	CPU usage flow control threshold of the XPU board	Determine whether the threshold needs to be adjusted.

If no, go to Step 5.

Step 5 Determine whether the cell loses paging messages.

Check whether the VS.RRC.Paging1.Loss.PCHCong.Cell increases sharply.

If yes, the PCH channel is congested. Determine whether the fault is rectified after performing the following operations. If yes, no further action is required. If no, go to step 6.

- Number of times for resending the optimized CN paging message on the CN
- The LAC is split on the RNC and paging areas are reduced.
- Number of times for resending the optimized UTRAN paging message on the RNC

- The DRX paging period on the RNC is added. The side impact is that the paging cycle becomes long.

If no, go to Step 6.

Step 6 If the following information needs to be collected, contact Huawei technical support.

- Paging policy on the CN
- CN traffic
- RNC traffic statistic files
- RNC scripts

Typical Cases

Case 1:

Incorrect CN configurations result in normal paging success rates counted by the CN and abnormal paging success rates counted by the RNC.

Fault Description

The RNC paging success rate on site I is 50%.

Fault Handling

1. The CN paging success rate is about 9X% (within the normal range). This indicates that the terminal paging is normal and improper configurations exist.

2. Analyze further causes of exceptions.

Trace the standard signaling at the Iu interface and discover that the LAC/RAC in many paging messages received by the RNC belongs to other RNCs instead of the local RNC.

The CN checks configurations and discovers incorrect LAC configurations on the MSC. The number of LACs/RACs configured on the MSC/SGSN is greater than the number of LAC cells on the RNC. This causes the RNC to receive correct paging messages and the number of attempts of RNC receiving paging messages to be too large.

Fault Rectification

The CN modifies LCA and RAC configurations.

Case 2:

Paging messages are sent suddenly and the PCH is congested. This results in reduced paging success rates.

Fault Description

Paging success rates decrease in T project on site I.

Fault Handling

1. The paging success rates counted by the CN and RNC are reduced and tend to be the same.

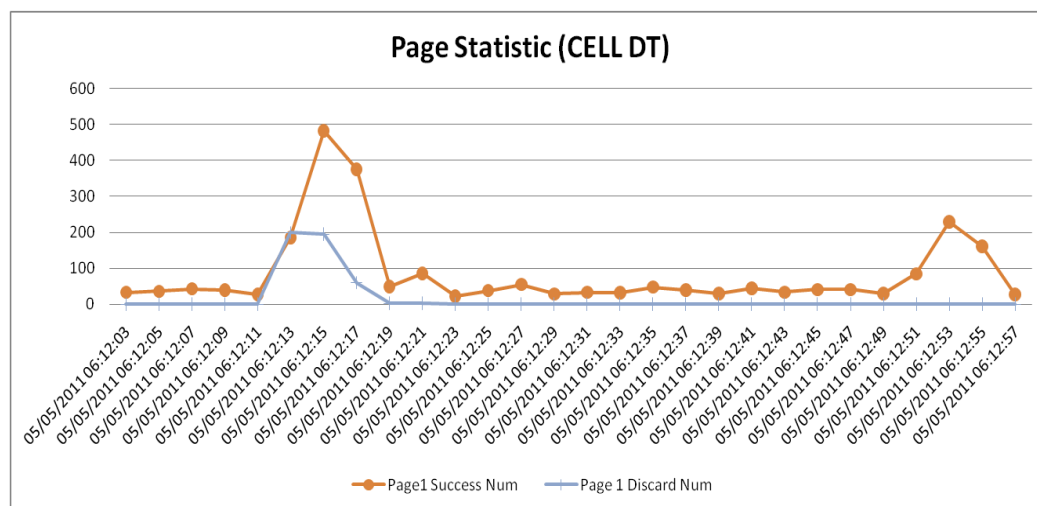
2. There is paging loss caused by CPU flow control.

3. PCHs are congested in some cells and the VS.RRC.Paging1.Loss.PCHCong.Cell is greater than 0.

4. Based on the result of checking the number of paging attempts of cells (for 60 or 15 minutes), when the number of paging attempts is small in the morning, paging congestion increases sharply, as shown in Figure 11-3.

5. The RNC CELLDT signaling tracing is collected in the morning and the number of pagings per second is greater than 500. This indicates paging bursts occur in the morning and exceeds air interface capacity of the PCH.

Figure 11-3 Page statistics



Fault Rectification

The LAC is split.

12 Troubleshooting OM Faults

12.1 OM Faults Definition

The data of the OM terminal such as the OMU and M2000 is not proper, the performance counters are abnormal, and alarms fail to be reported.

Note: This chapter only describes configuration data synchronization failure.

12.2 Context

After quick configuration is enabled, configuration objects fail to be synchronized on NEs and the M2000/CME in real time.

If no files are transmitted between the RNC and M2000 for a consecutive half minute, the M2000 may interrupt the connection forcibly.

12.3 Troubleshooting Configuration Data Synchronization Faults

12.3.1 Fault Symptom

After data is configured on the RNC or the NodeB LMT, data on the M2000/CME fails to be synchronized with that on NEs.

12.3.2 Possible Causes

Quick configuration is enabled on the RNC.

12.3.3 Troubleshooting Steps

Step 1 Check whether quick configuration is enabled.

Run **LST QUICKCFG** to check whether the **Configuration Mode** is **On**.

If yes, the fault is identified. Run **SET QUICKCFG** to set the **MODE** to **OFF** to disable quick configuration.

If no, go to step 2.

Step 2 Contact Huawei Customer Service Center.

12.3.4 Typical Cases

Fault Symptom

After cells are configured on the RNC LMT, no configured cells exist on the M2000/CME.

Fault Rectification

Disable quick configuration and synchronize configuration objects on NEs with that on the M2000/CME

Locating Faults

Step 1 Analyze the operation log and run **SET QUICKCFG: MODE=ON**.

Step 2 Run **SET QUICKCFG: MODE=OFF** to disable quick configuration.

12.4 Troubleshooting Counter Abnormalities

12.4.1 Fault Symptom

There are no performance statistics on the M2000 or the counter value is abnormal.

12.4.2 Possible Causes

1. The FTP transmission between the RNC and the M2000 is faulty.
2. The M2000 fails to deliver the measurement task file.

12.4.3 Troubleshooting Steps

Step 1 (Optional. This step is applicable to the scenarios where no performance statistics exist on the M2000) Check whether performance statistics file on the RNC exists when faults occur.

For example: check for the A20110815.1530+0200-1600+0200_EMS-NORMAL.mrf file in the **bam\common\MeasResult** directory on the RNC.

If yes, go to step 2.

If no, go to step 3.

Step 2 Analyze the ftp_server.log file in the RNC OMU log, and check whether RNC uploads files to the M2000.

For example: 2011-08-15 16:01:16[0xa08] Message MSG: {**data transfer failed**, error:The operation completed successfully.

in connect:711935.} File:ftp_session_worker.cpp,line:211

If yes, transmission from the RNC to the M2000 is abnormal, and therefore files are transmitted unstably. Troubleshoot transmission abnormality and check whether the fault is cleared. If the fault is cleared, no further action is required. If the fault persists, go to step 5.

If no, go to step 3.

Step 3 Check whether the M2000 fails to deliver a measurement task.

If yes, retransmit the measurement task and check whether the fault is cleared. If the fault is cleared, no further action is required. If the fault persists, go to step 5.

If no, go to step 4.

Step 4 (Optional. This step is applicable to the scenarios where the counter is 0) Check whether the actual counter value 0 is normal based on the counter meaning.

If yes, no further action is required. If the fault persists, go to step 3.

For example: the performance counter is not 0 only when inter-RAT neighboring cells handover under UCELL_GCELL.

Step 5 Contact Huawei Customer Service Center.

12.4.4 Typical Cases

Fault Symptom

No performance statistics from 15:30 to 24:00 on Aug. 15 on a RNC exists on the M2000.

Fault Rectification

Manually copy the performance counter statistics on the OMU to the corresponding directory on the M2000.

Locating Faults

Step 1 Check for the A20110815.1530+0200-1600+0200_EMS-NORMAL.mrf file in the **bam\common\MeasResult** directory on the RNC.

Step 2 Analyze the ftp_server.log file in the RNC OMU log, and check whether RNC uploads files to the M2000. If yes, transmission from the RNC to the M2000 is abnormal, and therefore files are transmitted unstably. Troubleshoot transmission abnormality to clear the fault.

2011-08-15 16:01:16[0xa08] Message MSG:
{download:D:\mbisc\bam\common\ems\10.149.104.20\pm\ne.3221229568.3221278720.3221287242\A20110815.1530+0200-1600+0200_EMS-NORMAL.mrf.bz2 failed in

connect:711935 error:**An existing connection was forcibly closed by the remote host..}**
File:ftp_transfer.cpp,line:245

2011-08-15 16:01:16[0xa08] Message MSG: {**data transfer failed**, error:The operation completed successfully.

in connect:711935.} File:ftp_session_worker.cpp,line:211

13

Troubleshooting ATM Transmission Faults

13.1 Procedure for Troubleshooting ATM Transmission Faults

13.1.1 Determining ATM Transmission Fault Type

ATM transmission faults consist of application layer abnormalities, poor ATM transmission QoS and bottom-layer transmission abnormalities. It is recommended that troubleshoot faults after determining faults type.

ATM Transmission Fault Type	Troubleshooting
Application layer abnormalities	Troubleshooting SAAL faults Troubleshooting AA2 path faults
Poor ATM transmission QoS	Troubleshooting packet loss in ATM transmission Troubleshooting delay and jitter in ATM transmission Troubleshooting packet error in ATM transmission Troubleshooting transient interruption in ATM transmission
Bottom-layer transmission abnormalities	Troubleshooting E1T1 and IMA faults(physical layer) Troubleshooting PVC faults(ATM layer)

13.1.2 Measures to Troubleshoot ATM Transmission Faults

Common measures to troubleshoot ATM transmission faults include a layer-by-layer check and a segment-by-segment check. Usually, find out the faults by a segment-by-segment check, then determine the fault type by a layer-by-layer check, and finally locate the root cause.

Layer-by-Layer Check

Check whether the layer where faults occur is abnormal.

- If yes, rectify the fault and then check whether the fault is cleared.

If yes, the fault is cleared.

If no, check whether the next layer is abnormal.

- If no, check whether the next layer is abnormal.

If yes, check the fault layer by layer (from the present layer to bottom layer).

If no, the fault occurs at this layer.



NOTE

In actual scenarios, locate faults from the upper or bottom layer and then the middle layer. For example, if you check each node on the network for PVC faults at the ATM layer, locate faults from the bottom layer or the upper layer and then the PVC layer.

Segment-by-Segment Check

Divide an end-to-end network into segments, and check a fault segment by segment.

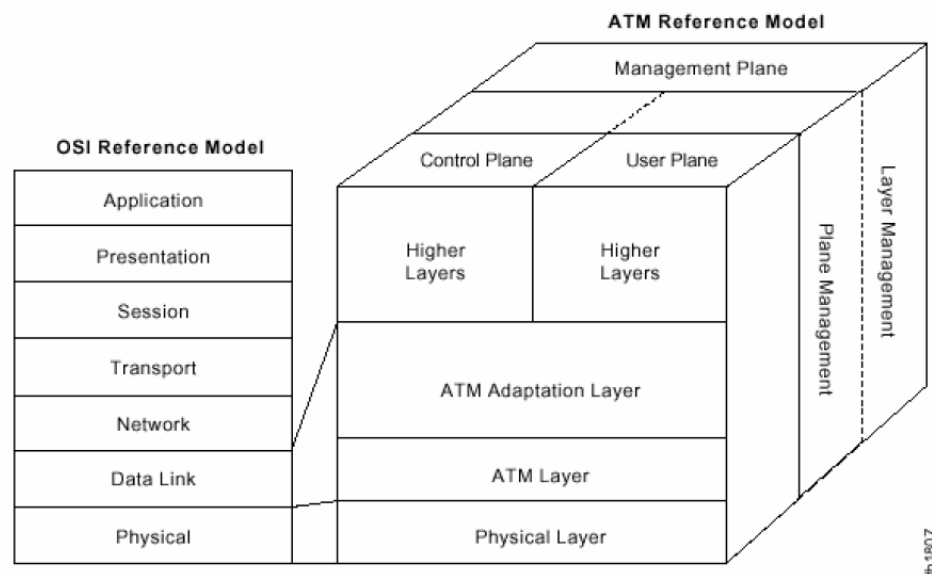
13.2 Basic knowledge of ATM Transmission

13.2.1 Characteristics of ATM Transmission Faults

An upper layer of the TCP/IP model works only when its lower layers are available.

Faults occurred on the ATM layer or the physical layer will result in the following problems: ATM transmission failure, poor ATM transmission QoS and the application layer abnormalities. Troubleshoot such faults layer by layer.

13.2.2 Introduction to the ATM Layer

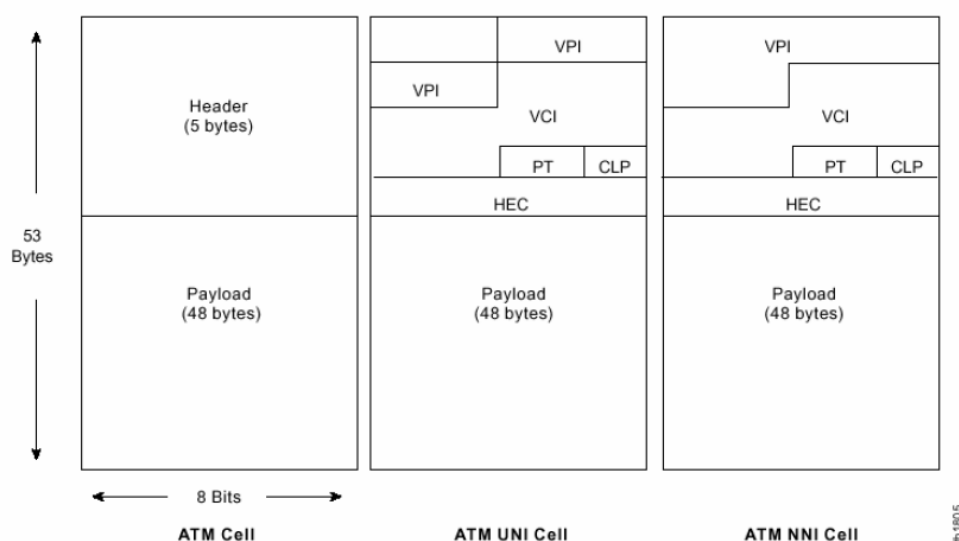


The ATM layer is above the physical layer and it is not related to the type of the physical layer media, the physical layer implementation, or the transmitted service type. Actually, the ATM layer communicates with the peer layer through IEs based on the services provided by the physical layer. The ATM layer implements multiplexing, demultiplexing, header operations, and flow control.

13.2.3 ATM Cell Architecture

Two ATM cells exist, as listed below:

- **UNI headers:** used on private networks for communication between ATM terminals and ATM switching nodes.
- **NNI headers:** used for communication between ATM switching nodes.



NOTE

An ATM cell consists of a 48-byte payload and a 5-byte header. The preceding figure shows that no GFC exists in the NNI cell for GFC is expanded to VPI.

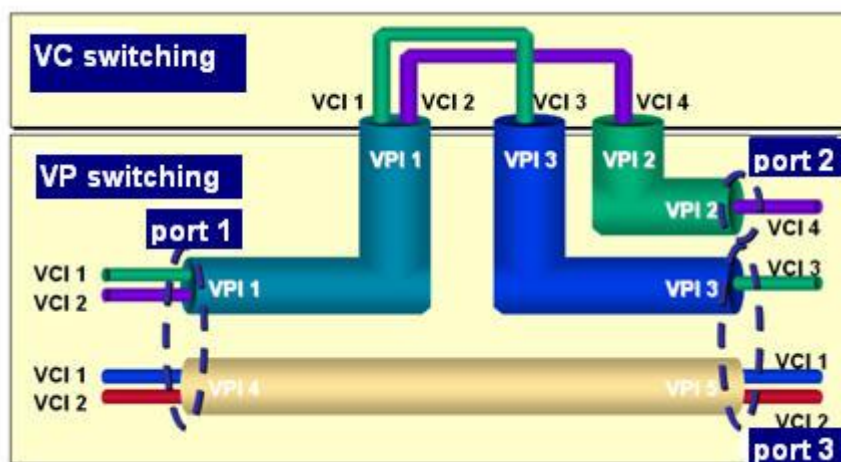
13.2.4 VP/VC Switching

In ATM communications, VP switching and VC switching is achieved as described below: According to the inputted cell VPI/VCI mark and routing table resulted from connection, ATM switch exchanges cells to the corresponding output port and changes the VPI/VCI values of these cells.



Common Cases:

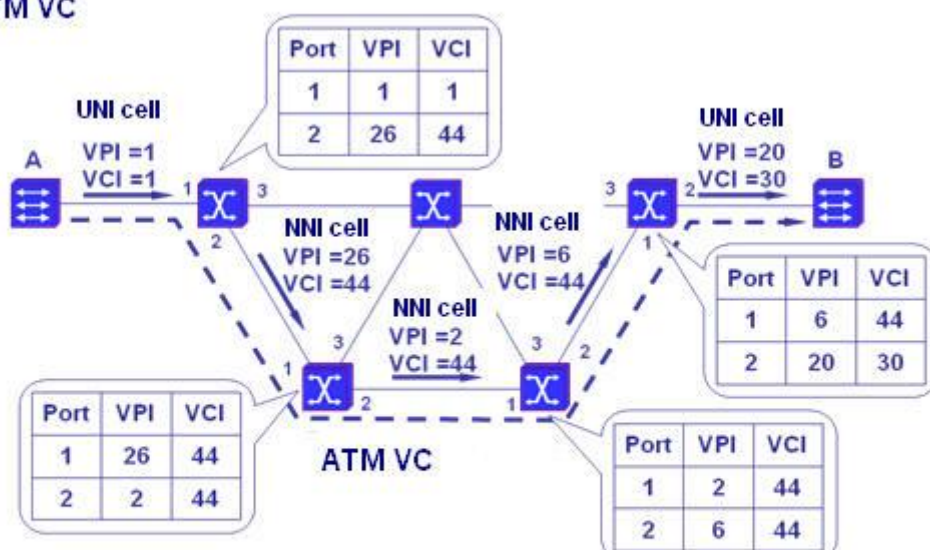
1. In VP switching, only VPI value is changed.
2. In VC switching, both VPI value and VCI value are changed.



13.2.5 ATM VCL

ATM virtual connection links (VCL) include SAAL LNK, AAL2 path and IPOA PVC.

ATM VC



If A needs to transmit data to B, series of switching tables are created on the ATM nodes which cells pass through to ensure that the cells arrive B after being forwarded. After the creation of the switching tables, the cell path from A to B remains unchanged, at least in one call, this path is called an ATM virtual connection.

13.3 Troubleshooting SAAL Faults

13.3.1 Fault Symptom

An SAAL fault occurs when any of the following appears:

The following alarms are displayed:

ALM-21531 SAAL Link Failure

ALM-21532 SAAL Link Congestion

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates; control plane transmission is abnormal.

13.3.2 Possible Causes

1. SAALNK parameters are incorrect.
2. The QoS of ATM transmission is poor.
3. The processing of the SAALNK module on the NE side is abnormal.

13.3.3 Troubleshooting Procedure

Check for SAALNK configuration faults.

Check for bottom-layer configuration faults or transmission faults.

13.3.4 Troubleshooting Steps

It is recommended that troubleshoot faults by fault type

If...	Then...
Packet loss occurs during using VCLCC to check for link faults Packet loss occurs during using VCLPM to check for abnormal links	Troubleshoot packet loss in ATM transmission
Large delay occurs during using VCLCC to check for link delays	Troubleshoot delay and jitter in ATM transmission
Error packets occur during performing VCL link performance query Error packets occur during using VCLPM to check for abnormal links	Troubleshoot packet error in ATM transmission
Transient transmission interruption occurs during performing VCL link performance query Transient transmission interruption occurs during using VCLCC to check for link faults Transient transmission interruption occurs during using LOP VCL to check for link faults or link delays	Troubleshoot transient interruption in ATM transmission
Other abnormalities	Go to step 2

Step 1 Check whether upper-layer application links are configured at both ends.

If...	Then...
Iub interface	Run LST UIUBCP on the RNC to check whether the SAAL link number is in use. Run LST IUBCP on the NodeB to check whether the SAAL link number is in use.
Iu-CS/Iu-PS interface	Run LST MTP3LNK on the RNC to check whether the SAAL link number is in use.

If yes, go to step 2.

If no, configure the upper-layer application links. If the fault is cleared, no further action is required. If no, go to step 2.

Step 2 Check whether the configurations at both ends are consistent.

Run **LST SAALLNK** on the RNC, and record the link transmission indexes (TXTRFX and RXTRFX).

Run **LST ATMTRF** on the RNC to check the values of **ST**, **PCR** and **CDVT** when transmission indexes are TXTRFX and RXTRFX.

Check the configurations.

ST: Is the ST consistent at both ends?

PCR: Is the PCR higher than the transmission network at both ends?

CDVT: Is the CDVT greater than the transmission network at both ends?

If yes, go to step 3.

If no, modify the parameter setting to meet the preceding conditions. If the fault is cleared, no further action is required. If the fault persists, go to step 4.

Step 3 Check whether faults occur on a bottom layer.

For details, see "Troubleshooting PVC Faults (ATM layer)."

If the fault is rectified, no further action is required.

If the fault persists, go to step 4.

Step 4 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.4 Troubleshooting AAL2 Path Faults

13.4.1 Fault Symptom

An AAL2 path fault occurs when any of the following appears:

The following alarms are displayed:

ALM-21581 Path Failure

ALM-21582 Path Congestion.

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates; control plane transmission is abnormal.

13.4.2 Possible Causes

1. Physical port fault
2. Incorrect configuration

13.4.3 Troubleshooting Procedure

Check the status of physical ports which bears the AAL2 path.

Check E1 link status

Check QoS of ATM transmission

13.4.4 Troubleshooting Steps

Step 1 It is recommended that troubleshoot faults by fault type.

If...	Then...
Packet loss occurs during using VCLCC to check for link faults Packet loss occurs during using VCLPM to check for abnormal links	Troubleshoot packet loss in ATM transmission
Large delay occurs during using VCLCC to check for link delays Large delay occurs during performing node synchronization detection to check for transmission delay and jitter on the user plane	Troubleshoot delay and jitter in ATM transmission
Error packets occur during performing VCL link performance query Error packets occur during using VCLPM to check for abnormal links	Troubleshoot packet error in ATM transmission
Transient transmission interruption occurs during performing VCL link performance query Transient transmission interruption occurs during using VCLCC to check for link faults Transient transmission interruption occurs during using LOP VCL to check for link faults or link delays	Troubleshoot transient interruption in ATM transmission
Transient transmission interruption occurs during performing VCL link performance query Link failure occurs during using VCLCC to check for link faults Link failure occurs during using LOP VCL to check for link faults and link delays	Troubleshoot PVC faults(ATM layer)
Other abnormalities	Go to step 2

If the fault is rectified, no further action is required.

If the fault persists, go to step 2.

Step 2 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.5 Troubleshooting Packet Loss in ATM Transmission

13.5.1 Fault Symptom

Packet loss in ATM transmission occurs when any of the following appears:

1. Packet loss occurs during using VCLCC to check for link faults.
2. Packet loss occurs during using VCLPM to check for abnormal links.



NOTE

Users feel that the voice quality is poor, and call drops even occur. The HSPA rate is affected. The OM channels transmit commands slowly and the results of the ping test conducted on OM channels show that some packets are lost.

13.5.2 Possible Causes

1. The transmission media on the physical layer are abnormal. For example, the E1/T1 cable or fiber is faulty or improperly connected; line interference occurs; link bit errors occur.
2. Interconnecting parameters are inconsistent, which are described as follows:
 - The service types or bandwidths configured on the PVC layer are inconsistent. The interconnecting parameters configured over IMA layer are inconsistent.
 - Configurations, such as the E1/T1 encoding mode, scrambling mode, frame format, impedance, and timeslot are incorrect.
 - The interconnecting parameters of optical interfaces are inconsistent.
3. The QoS policy configured on the transmission network is incorrect, or the transmission network is congested, or packet loss occurs.
4. A device is faulty

13.5.3 Troubleshooting Procedure

1. Identify the fault symptom.
2. Isolate abnormal NE devices.
3. Analyze a faulty NE based on the protocol stack.
4. Investigate the cause for packet loss.

13.5.4 Troubleshooting Steps

Step 1 Analyze abnormal sites distribution rules.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how abnormal sites are distributed according to configurations to collect data about whether faulty sites mainly occur on the specific ports, interface boards, and subsystems of the CPUS.

If yes, collect the data collected in the previous steps and contact Huawei for technical support.

If no, go to step 2.

Step 2 Optional. Take this step in the bit errors scenario. Perform a loopback segment by segment and conduct an E1/T1 port bit error test to check whether bit errors exist.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view the bit error test result. If no bit errors are detected, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment where bit errors occur.

If the faulty segment is detected, troubleshoot transmission bit errors.

If the faulty segment is not detected, go to step 3.



NOTE

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the E1/T1 configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 3 Check the parameter settings on the PVC layer at both ends.

ST: Is the service type consistent?

PCR: Is the PCR consistent?

SCR: Is the SCR consistent?

RCR: Is the RCR consistent?

MCR: Is the MCR consistent?

CDVT: Is the CDVT interconnected with NEs smaller than the transmission layer?

Note:

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the PVC configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 4 Analyze how faulty links are distributed on the transmission network.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how faulty links are distributed according to transmission network adjustment to collect data about whether faulty links mainly occur on the specific transmission nodes.

If yes, troubleshoot transmission network abnormality.

If no, go to step 5.

Step 5 Check whether the transmission network is abnormal.

Check whether traffic shaping is performed on the transmission network and whether the transmission network is congested. If a transmission device is configured with a QoS policy, check whether the QoS policy is proper.

If yes, troubleshoot transmission network abnormality.

If no, go to step 6.

Step 6 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.6 Troubleshooting Delay and Jitter in ATM Transmission

13.6.1 Fault Symptom

Delay and jitter in ATM transmission occurs if any of the following appears:

1. Large delay occurs during using VCLCC to check for link faults.
2. Large delay occurs during performing the IP over ATM OMCH continuity check.
3. Large delay occurs during performing node synchronization detection to check for transmission delay and jitter on the user plane.

13.6.2 Possible Causes

1. The transmission network is congested.
2. The QoS policy is improper.
3. A device is faulty.

13.6.3 Troubleshooting Procedure

1. Identify the fault symptom.
2. Isolate faulty NEs and the protocol layer.
 - Analyze NE distribution rules.
 - Locate the faulty layer based on the protocol stack.
3. Investigate the root cause.
4. Make analysis policies based on the actual situation.

13.6.4 Troubleshooting Steps

Step 1 Analyze abnormal sites distribution rules.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how abnormal sites are distributed according to configurations to collect data about whether faulty sites mainly occur on the specific ports, interface boards, and subsystems of the CPUS.

If yes, go to step 5.

If no, go to step 2.

Step 2 Check the parameter settings on the PVC layer at both ends.

ST: Is the service type consistent?

PCR: Is the PCR consistent?

SCR: Is the SCR consistent?

RCR: Is the RCR consistent?

MCR: Is the MCR consistent?

CDVT: Is the CDVT interconnected with NEs smaller than the transmission layer?

Note:

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the PVC configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 3 Analyze how faulty links are distributed on the transmission network.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how faulty links are distributed according to transmission network adjustment to collect data about whether faulty links mainly occur on the specific transmission nodes.

If yes, troubleshoot transmission network abnormality.

If no, go to step 5.

Step 4 Check whether the transmission network is abnormal,
and whether the transmission network is congested.

If yes, troubleshoot transmission network abnormality.

If no, go to step 5.

Step 5 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.7 Troubleshooting Packet Error in ATM Transmission

13.7.1 Fault Symptom

Error packets in ATM transmission occur when any of the following appears:

1. Error packets occur during performing VCL link performance query.
2. Error packets occur during using VCLPM to check for abnormal links.

13.7.2 Possible Causes

1. The transmission line quality is poor, and transmission is affected by interference.
2. If E1/T1 transmission is used, inconsistent configurations cause error bits.
3. A transmission device is faulty.

13.7.3 Troubleshooting Procedure

1. Identify the fault symptom.

2. Isolate faulty NEs and the protocol layer.
 - Analyze NE distribution rules.
 - Locate the faulty layer based on the protocol stack.
3. Investigate the cause.
4. Make analysis policies based on the actual situation.

13.7.4 Troubleshooting Steps

Step 1 Analyze abnormal sites distribution rules.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how abnormal sites are distributed according to configurations to collect data about whether faulty sites mainly occur on the specific ports, interface boards, and subsystems of the CPUS.

If yes, collect the preceding results and contact Huawei for technical support.

If no, go to step 2.

Step 2 Optional. Take this step in the bit errors scenario. Perform a loopback segment by segment and conduct an E1/T1 port bit error test to check whether bit errors exist.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view the bit error test result. If no bit errors are detected, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment where bit errors occur.

If the faulty segment is detected, troubleshoot transmission bit errors.

If the faulty segment is not detected, go to step 3.



NOTE

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the E1/T1 configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 3 Analyze how faulty links are distributed on the transmission network.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how faulty links are distributed according to transmission network adjustment to collect data about whether faulty links mainly occur on the specific transmission nodes.

If yes, troubleshoot transmission network abnormality.

If no, go to step 4.

Step 4 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.8 Troubleshooting Transient Interruption in ATM Transmission

13.8.1 Fault Symptom

Transient interruption in ATM transmission occurs if any of the following appears:

1. Transient transmission interruption occurs during performing VCL link performance query.
2. Transient transmission interruption occurs during using VCLCC to check for link faults.
3. Transient transmission interruption occurs during using LOP VCL to check for link faults or link delays

13.8.2 Possible Causes

1. The transmission media on the physical layer are abnormal. For example, the E1/T1 cable or fiber is faulty or improperly connected; line interference occurs; link bit errors occur.
2. Interconnecting parameters are inconsistent, which are described as follows:
 - The service types or bandwidths configured on the PVC layer are inconsistent.
 - The interconnecting parameters configured over IMA layer are inconsistent.
 - Configurations, such as the E1/T1 encoding mode, scrambling mode, frame format, impedance, and timeslot are incorrect.
 - The interconnecting parameters of optical interfaces are inconsistent.
3. The QoS policy configured on the transmission network is incorrect, or the transmission network is congested, or packet loss occurs.
4. A device is faulty.

13.8.3 Troubleshooting Procedure

1. Identify the fault symptom.
2. Isolate faulty NEs and the protocol layer.
 - Analyze NE distribution rules.
 - Locate the faulty layer based on the protocol stack.
3. Investigate the cause.
4. Make analysis policies based on the actual situation.

13.8.4 Troubleshooting Steps

Step 1 Further analyze one faulty NE or several faulty NEs if no obvious rules can be found after the preceding detection. You can analyze abnormal sites distribution rules layer by layer based on the protocol stack.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how abnormal sites are distributed according to configurations to collect data about whether faulty sites mainly occur on the specific ports, interface boards, and subsystems of the CPUS.

If yes, collect the preceding results and contact Huawei for technical support.

If no, go to step 2.

Step 2 Check whether E1/T1 configuration is consistent with the peer end configuration.

1. Run **DSP E1T1** on the RNC to check whether the parameter is set to the same value as that of the peer end. for example:
DIP balance mode
Scrambling mode attribute
Frame format (sending and expected receiving frame format)
Encoding (transmitting line code mode, receiving line code mode)
Impedance
2. Run **DSP E1T1** on the NodeB to check whether the parameter is set to the same value as that of the peer end. for example:
Work mode
Frame format
Line code

Step 3 Optional. Take this step in the bit errors scenario. Perform a loopback segment by segment and conduct an E1/T1 port bit error test to check whether bit errors exist.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view the bit error test result. If no bit errors are detected, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment where bit errors occur.

If the faulty segment is detected, troubleshoot transmission bit errors.

If the faulty segment is not detected, go to step 4.



NOTE

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the E1/T1 configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 4 Check the parameter settings on the PVC layer at both ends.

ST: Is the service type consistent?

PCR: Is the PCR consistent?

SCR: Is the SCR consistent?

RCR: Is the RCR consistent?

MCR: Is the MCR consistent?

CDVT: Is the CDVT interconnected with NEs smaller than the transmission layer?



NOTE

Identify the fault segment by segment transversely and locate the segment where faults occur.

Vertically compare the PVC configuration of normal sites and abnormal sites to check whether the configuration is incorrect.

Step 5 Analyze how faulty links are distributed on the transmission network.

Analyze the alarm objects or detected link No. to obtain the list of abnormal sites.

Analyze how faulty links are distributed according to transmission network adjustment to collect data about whether faulty links mainly occur on the specific transmission nodes.

If yes, troubleshoot transmission network abnormality.

If no, go to step 6.

Step 6 Check whether the transmission network is abnormal,

for example, check whether traffic shaping is performed on the transmission network and whether the transmission network is congested. If a transmission device is configured with a QoS policy, check whether the QoS policy is proper.

If yes, troubleshoot transmission network abnormality.

If no, go to step 7.

Step 7 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.9 Troubleshooting PVC Faults (ATM layer)

13.9.1 Fault Symptom

PVC disconnection occurs in ATM transmission if any of the following appears:

1. Transient transmission interruption occurs during performing VCL link performance query.
2. Link failure occurs during using VCLCC to check for link faults.
3. Link failure occurs during using LOP VCL to check for link faults and link delays.

13.9.2 Possible Causes

1. The E1/T1 cable or optical fiber is faulty.
2. The configurations on the PVC layer are incorrect

13.9.3 Troubleshooting Procedure

Check the configurations of each node on the PVC layer. Generally check whether faults occur on the physical layer and then check whether faults occur on the PVC layer.



NOTE

In actual scenarios, you can check whether PVC faults occur, and then check whether faults occur on the physical layer.

13.9.4 Troubleshooting Steps

Step 1 For details, see "Troubleshooting IMA Faults (physical layer)."

If the fault is rectified, no further action is required. If the fault persists, go to step 2.

Step 2 For details, see "Troubleshooting E1/T1 Faults (physical layer)."

If the fault is rectified, no further action is required.

If no, go to step 3.

Step 3 Check whether the VPI/VCI configurations of each node on the PVC layer at both ends are correctly set.

Check the value of each node and whether each node is correctly configured. The query methods vary with link types, which are described as follows:

1. Run **LST AAL2PATH** on the RNC or the NodeB to query the carried VPI and VCI.
2. Run **LST SAALLNK** on the RNC or the NodeB to query the carried VPI and VCI.
3. Run **LST IPOAPVC** on the RNC to query the carried VPI and VCI.

If yes, go to step 4.

If no, modify the information. After that, if the fault is rectified, no further action is required. If the fault still remains, go to step 4.

Step 4 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.10 Troubleshooting E1/T1 Faults (physical layer)

13.10.1 Fault Symptom

Alarms are generated on RNC/NodeB as follows:

1. E1/T1 Excessive Bit Error
2. E1 Excessive Bit Error
3. E1/T1 Signal Loss
4. E1/T1 Alarm Indication Signal

13.10.2 Possible Causes

1. The E1/T1 cable or fiber is faulty or improperly connected; line interference occurs.
2. Configurations such as the E1/T1 encoding mode, scrambling mode, frame format, impedance, and timeslot are incorrect.
3. A device is faulty.

13.10.3 Troubleshooting Procedure

1. Check whether E1/T1 parameters are configured properly.
2. Check the physical cable connection.
3. Perform a loopback detection.

13.10.4 Troubleshooting Steps

Step 1 Check whether E1/T1 status is normal.

Run **DSP E1T1** on the RNC to check whether the port status is **Normal**.

Run **DSP E1T1** on the RNC to check whether the link status is **Available**.

Step 2 Check whether E1/T1 configuration is consistent with the peer end configuration.

1. Run **DSP E1T1** on the RNC to check whether the parameter is set to the same value as that of the peer end, for example:
DIP balance mode
Scrambling mode attribute
Frame format (sending and expected receiving frame format)
Encoding (transmitting line code mode, receiving line code mode)
Impedance
2. Run **DSP E1T1** on the NodeB to check whether the parameter is set to the same value as that of the peer end. for example:
Work mode
Frame format
Line code

Step 3 Checking whether the connections between the RNC and the NodeB are correct.

If yes, go to step 5.

If no, go to step 4.

Step 4 Perform a loopback segment by segment to detect the segment where faults occur.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view whether ALM-25807 E1/T1 loopback alarm is generated on the NodeB (cause value: physical loopback). If no alarms, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment that causes the fault.

If the faulty segment is detected, troubleshoot transmission faults.

If the faulty segment is not detected, go to step 5.

Step 5 Optional. Take this step in the bit errors scenario. Perform a loopback segment by segment and conduct an E1/T1 port bit error test to check whether bit errors exist.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view the loopback result. If no bit errors are detected, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment where bit errors occur.

If the faulty segment is detected, troubleshoot transmission bit errors.

If the faulty segment is not detected, go to step 6.

Step 6 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

13.11 Troubleshooting IMA Faults (physical layer)

13.11.1 Fault Symptom

If no abnormality occurs on the E1 layer, the following alarms may be generated:

21221	ALM-21221 IMA Link Out of Frame
21222	ALM-21222 IMA Link Out of Delay
21227	ALM-21227 IMA/UNI link Loss of Cell Delimitation
21229	ALM-21229 IMA Group Configuration Failure

13.11.2 Possible Causes

The IMA interconnecting parameters are improper.

Transmission faults occur.

13.11.3 Troubleshooting Steps



NOTE

The two ends means ends where IMA protocol is interconnected. If both RNC and NodeB complies with the IMA protocol, the two ends are the RNC and NodeB. If the RNC does not comply with the IMA protocol while the NodeB complies with the IMA protocol, the two ends are the NodeB and transmission devices connected to the NodeB.

Step 1 Check whether timeslot 16 is used at both ends.

Run **LST IMAGRP** on the NodeB to check whether **Timeslot 16 Support** is **ENABLE** and **Scramble Mode** is **ENABLE**.

Run **LST E1T1** on the RNC to check whether **Timeslot 16 Support Switch** is **ON** and **Scramble Switch** is **ON**.

Step 2 Check whether IMA parameters at both ends are configured consistently and check for IMA group configuration failure.

Run **LST IMAGRP** on the NodeB or RNC to check whether the following parameter settings.

1. IMA protocol version: The IMA protocol versions configured at both ends must be the same.
2. IMA symmetric mode: The fixed configuration on the RNC and NodeB is symmetric mode.
3. The IMA TX frame length does not need to be configured to the same value at both ends. However, confirm that the peer device supports the frame length because the device of some version may not support the frame lengths other than 128.
4. The sending clock mode does not need to be configured to the same value at both ends. However, confirm whether the peer device supports the mode because many devices do not support the ITC mode. The default sending clock mode of Huawei RNC is CTC, and the default sending clock mode of Huawei NodeB is CTC or ITC.

- Step 3** Optional. Take this step in the bit errors scenario. Perform a loopback segment by segment and conduct an E1/T1 port bit error test to check whether bit errors exist.

Networking sample: RNC---A---B---C---D---NodeB

Perform a loopback from transmission device A to the NodeB and view the loopback result. If no bit errors are detected, terminate the loopback.

Continue to perform a loopback from transmission device B, C, D to the NodeB until you detect the segment where bit errors occur.

If the faulty segment is detected, troubleshoot transmission bit errors.

If the faulty segment is not detected, go to step 4.

- Step 4** Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14

Troubleshooting IP Transmission Faults

14.1 Procedure for Troubleshooting IP Transmission Faults

14.1.1 Determining IP Transmission Fault Type

IP transmission faults consist of the application layer abnormalities, poor IP transmission QoS and IP transmission failure. It is recommended that troubleshoot faults after determining faults type.

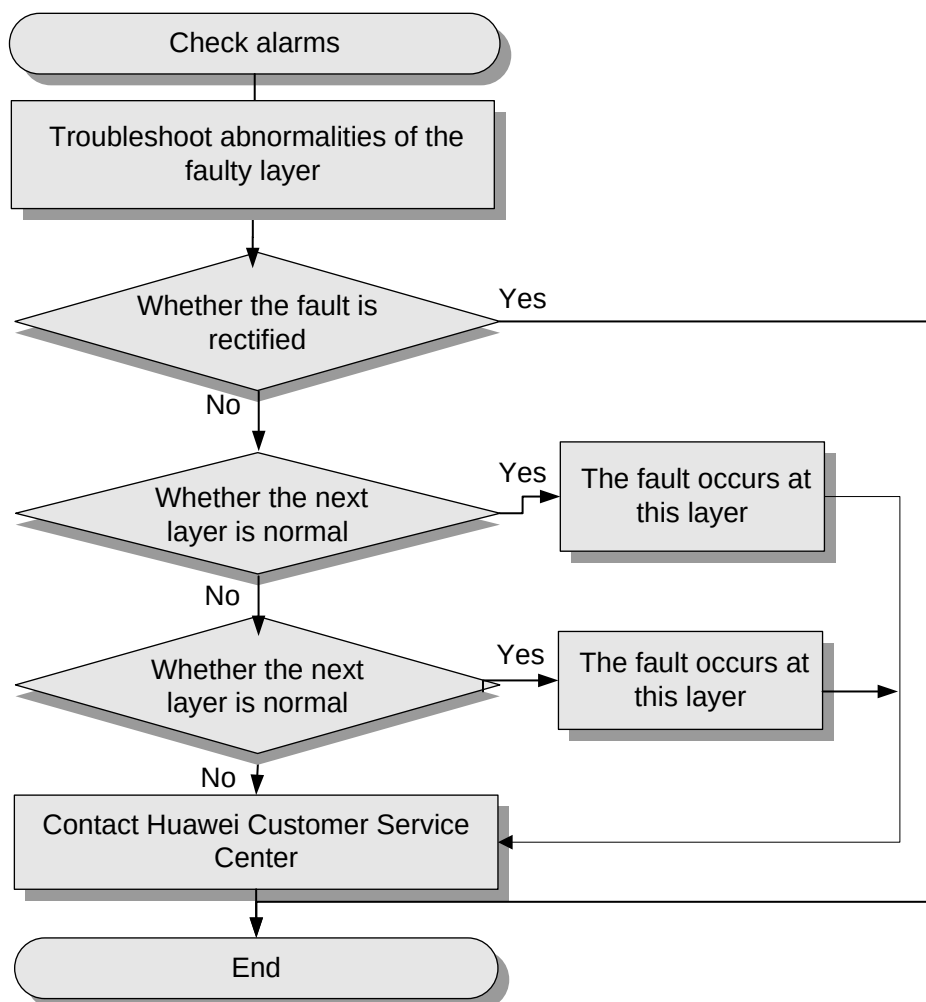
IP Transmission Fault Type	Troubleshooting
Application layer abnormalities	Troubleshooting SCTP faults Troubleshooting IP Path faults
Poor IP transmission QoS	Troubleshooting packet loss in IP transmission Troubleshooting delay and jitter in IP transmission Troubleshooting packet errors in IP transmission Troubleshooting transient interruption in IP transmission
IP transmission failure	Troubleshooting IP over FE/GE interface disconnection Troubleshooting MP/PPP link failure in IP over E1 mode

14.1.2 Measures to Troubleshoot IP Transmission Faults

Common measures to troubleshoot IP transmission faults include a layer-by-layer check and a segment-by-segment check. Usually, find out the faults by a segment-by-segment check, then determine the fault type by a layer-by-layer check, and finally locate the root cause.

Layer-by-Layer Check

As shown in the following figure, check a fault layer by layer (from the present layer where faults occur to the bottom layer), isolate the fault and finally locate the fault and the layer where the fault occurs.



Segment-by-Segment Check

Divide an end-to-end network into segments, and check a fault segment by segment.

14.2 Basic Knowledge of IP Transmission

Characteristics of IP Transmission Faults

An upper layer of the TCP/IP model works only when its lower layers are available.

Faults occurred on the IP layer or the link layer or the physical layer will result in the following problems: IP transmission failure, poor IP transmission QoS and the application layer abnormalities. Troubleshoot such faults layer by layer.

FE/GE Port Negotiation Parameters

Port negotiation parameters mainly include the port speed and duplex mode. The two ends must negotiate these parameters and keep them the same.

Take the speed as an example. If the rate at one end is 100 Mbit/s, the rate at the other end must also be 100 Mbit/s. If the rate at one end is set to AUTO, the speed at the other end must also be set to AUTO. The duplex mode at both ends must also be the same. Table 14-1 shows the recommended configurations.

Table 14-1 Recommended configurations for negotiation parameters

	Port Rate	Duplex Mode
Recommended configuration 1	100 M (1000M for GE)	FULL
Recommended configuration 2	100 M (1000M for GE)	AUTO
Recommended configuration 3	AUTO	AUTO

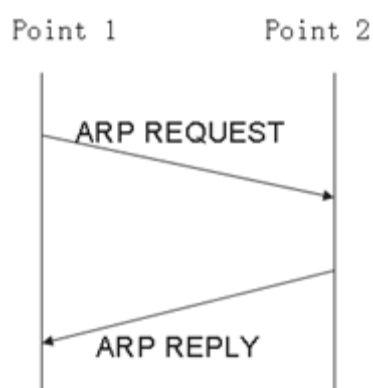
Overall Process of Sending ARP Request Packets

The BSC and NodeB process data packets in the same way. That is, they query the corresponding next-hop MAC address based on the IP route entries. Packets (ICMP packets, SCTP packets or UDP packets and so on) can be sent only when ARP entries exist.

The local end sends an ARP request broadcast packet to the peer end. If the transmission is available, the peer end sends an ARP reply back with the next-hop MAC address.

Figure 14-1 shows the process of sending an ARP request.

Figure 14-1 Flowchart for sending an ARP request



NOTE

ARP packets are broadcast packets transmitted between two layer 2 communication nodes.

If layer 2 networking is applied to the BSC and NodeB, the ARP request is sent to the NodeB or BSC. If layer 3 networking is applied, the ARP request is sent to its own gateway.

Introduction to the PPP/MP Technology

1. Introduction to PPP

The Point-To-Point Protocol (PPP) is applied on layer 2 (link layer) of the TCP/IP protocol stack. This protocol supports point-to-point data transmission over full-duplex synchronous and asynchronous links.

PPP is applied to the Iub interface in IP over E1 mode.

2. Introduction to ML-PPP

MultiLink PPP (ML-PPP) is also abbreviated as MP. It bundles multiple MP links as one logical path MPGRP, which is a link for the network layer to increase the bandwidth.

MP is applied to the Iub interface in IP over E1 mode.

Introduction to SCTP

The Stream Control Transmission Protocol (SCTP) is a reliable transmission protocol operating on top of a connectionless network (such as IP network). SCTP is applied to the IP-based Iub interface, Iu-CS interface and Iu-PS interface.

Process of Establishing an SCTP Link

Common types of SCTP messages are as follows:

Type of SCTP Messages	Purpose of Messages
INIT, INITACK, COOKIEECHO, COOKIEECHOACK	Four-way handshake link setup process (initiated by the client)
HEARTBEAT, HEARTBEATAck	Heartbeat messages (initiated by the client or the server)
DATA, SACK	Data interaction (initiated by the client or the server)
ABORT, SHUTDOWN, ERROR	Initiated when the client or server is abnormal

As shown in Figure 14-2, the first four messages are about a four-way handshake link setup process, the last four messages are heartbeat messages and the data interaction is in the middle.

Figure 14-2 Information interaction during the process of establishing an SCTP link

23/09/2010 22:30:41	61	27609	INIT	0:0:0:3	45 C0 00 48 67 87 00 00 FF 84 2F 29 0A AD 87 55 0...
23/09/2010 22:30:41	61	27665	INITACK	0:0:0:3	45 FF 02 0C 10 F3 00 00 FE 84 84 BA 0A AD 87 12 0...
23/09/2010 22:30:41	61	27667	COOKIEECHO	0:0:0:3	45 C0 01 E4 67 88 00 00 FF 84 2D 8C 0A AD 87 55 0...
23/09/2010 22:30:41	63	27777	COOKIEEACK	0:0:0:3	45 00 00 24 10 F4 00 00 FE 84 87 A0 0A AD 87 12 0...
23/09/2010 22:30:41	63	27781	DATA	0:0:0:3	45 C0 00 58 67 89 00 00 FF 84 2F 17 0A AD 87 55 0...
23/09/2010 22:30:41	71	28465	DATA	0:0:0:3	45 02 00 38 10 F5 00 00 FE 84 87 89 0A AD 87 12 0...
23/09/2010 22:30:41	71	28466	SACK	0:0:0:3	45 C0 00 30 67 8A 00 00 FF 84 2F 3E 0A AD 87 55 0...
23/09/2010 22:30:41	71	28469	DATA	0:0:0:3	45 C0 00 40 67 8B 00 00 FF 84 2F 2D 0A AD 87 55 0...
23/09/2010 22:30:41	72	28561	SACK	0:0:0:3	45 02 00 30 10 F6 00 00 FE 84 87 90 0A AD 87 12 0...
23/09/2010 22:30:41	80	29137	DATA	0:0:0:3	45 02 00 60 10 F7 00 00 FE 84 87 5F 0A AD 87 12 0...
23/09/2010 22:30:41	80	29138	SACK	0:0:0:3	45 C0 00 30 67 8C 00 00 FF 84 2F 3C 0A AD 87 55 0...
23/09/2010 22:30:43	64	43844	HEARTBEAT	0:0:0:3	45 C0 00 7C 67 8D 00 00 FF 84 2E 6B 0A AD 87 59 ...
23/09/2010 22:30:43	64	43874	HEARTBEATAK	0:0:0:3	45 02 00 7C 76 C6 00 00 FE 84 20 F0 0A AD 87 92 0...
23/09/2010 22:30:44	5	47137	HEARTBEAT	0:0:0:3	45 02 00 3C 76 C7 00 00 FE 84 21 2F 0A AD 87 92 0...
23/09/2010 22:30:44	5	47139	HEARTBEATAK	0:0:0:3	45 C0 00 3C 67 8E 00 00 FF 84 2E AA 0A AD 87 59 0...

Introduction to IP Path

An IP path is a logical link with virtual bandwidth and is carried by the physical links on an IP transmission network.

An IP path only carries the user plane data, not the signaling plane data or the OM plane data.

An IP path is defined by the source and destination IP addresses and the path type (corresponding to PHB type).

Admission control is performed during service establishment according to the service type and the bandwidth of the corresponding IP path.

14.3 Troubleshooting SCTP Faults

14.3.1 Fault Symptom

An SCTP fault occurs when any of the following appears:

An SCTP fault occurs when you run **DSP SCTPLNK** on the RNC, but in the command output, the **Operation Status** is **Unavailable** or **Congested**, or the following alarms are displayed.

Alarm ID	Alarm Name
21541	ALM-21541 SCTP Link Failure
21542	ALM-21542 SCTP Link Congestion
22915	EVT-22915 SCTP Link Path Switchover

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates; control plane transmission is abnormal.

14.3.2 Possible Causes

1. The transmission is faulty. The configurations at the two ends are improper.
2. The NE is faulty.

14.3.3 Troubleshooting Procedure

Check whether a transmission fault occurs under the IP layer. If yes, troubleshoot the fault, and then check whether the configurations at both ends are proper.

14.3.4 Troubleshooting Steps

Step 1 It is recommended that troubleshoot faults by fault type.

If...	Then...
Packet loss occurs in the control plane	Troubleshoot packet loss in IP transmission
Delay and jitter occur in the control plane	Troubleshoot delay and jitter in IP transmission
Error packets occur in the control plane	Troubleshoot error packets in IP transmission
Transient interruption occurs in the control plane	Troubleshoot transient interruption in IP transmission
Link failure or other abnormalities occur in the control plane	Go to step 2

Step 2 Perform the **Ping** operation to check the IP-layer connectivity and end-to end connectivity.

If the ping operation fails, troubleshoot link faults.

If...	Then...
IP over FE/GE	Troubleshoot IP over FE/GE interface disconnection
IP over E1	Troubleshoot MP/PPP link failure in IP over E1 mode

If the ping operation succeeds, go to step 3.

Step 3 Optional. If SCTP link abnormal disconnection occurs, re-establish the link and check whether the fault is rectified.

If...	Then...
Iub interface	Configure the Control Plane over the Iub Interface (over IP) by referring to the <i>UMTS Initial Configuration Guide</i> , and delete an SCTP link and re-configure an SCTP link.

Iu-CS/Iu-PS interface	Configure the Control Plane over the Iu-CS Interface (over IP) by referring to the <i>UMTS Initial Configuration Guide</i>, and delete an SCTP link and re-configure an SCTP link. Configure the Control Plane over the Iu-PS Interface (over IP) by referring to the <i>UMTS Initial Configuration Guide</i>, and delete an SCTP link and re-configure an SCTP link.
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If the fault is rectified, no further action is required.

If the fault persists, go to step 4.

Step 4 Check whether the VLAN configuration on the RNC is correct.

Run **LST VLANID** and **LST SCTPLNK** on the RNC to check whether the VLAN ID is configured as the transport network requires.

If yes, go to step 5.

If no, modify the VLAN configuration. After that, if the fault is rectified, no further action is required. If the fault persists, go to step 5.

Step 5 Check whether the MTU value at both ends is less than that of the transport network.

1. Run **LST SCTPLNK** on the RNC to check whether the MTU value is less than that of the transport network.
2. Run **LST ETHPORT** on the NodeB to check whether the maximum transfer unit is less than that of the transmission network.

If yes, go to step 6.

If no, modify MTU setting. If the fault is rectified, no further action is required. If the fault persists, go to step 6.

Step 6 Check whether upper-layer application links are configured at both ends.

If...	Then...
Iub interface	Run LST UIUBCP on the RNC to check whether the SCTP link number is in use. Run LST IUBCP on the NodeB to check whether the SCTP link number is in use.
Iu-CS/Iu-PS interface	Run LST M3LNK on the RNC to check whether the SCTP link number is in use.

If yes, go to step 7.

If no, configure the upper-layer application links. If the fault is rectified, no further action is required. If the fault persists, go to step 7.

Step 7 Use SCTP tracing to determine the faulty NEs.

Perform an Iub/Iu-CS/Iu-PS interface SCTP tracing on the RNC LMT.

According to the process of establishing an SCTP link, locate the faulty NEs. For example, the RNC sends **INIT ACK** and fails to receive the packets **COOKIE ECHO** returned by the MSC.

If yes, check the faulty NEs. If the fault is rectified, no further action is required. If the fault persists, go to step 8.

If no, go to step 8.

- Step 8** Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.3.5 Typical Cases

Fault Symptom

An operator performs an IP reconstruction for the Iu interface. After the data of a signaling point is configured, the link is disconnected and its status is abnormal.

Locating Faults

- Step 1** After confirmation, the RNC board is configured completely and no board hardware fault alarms are generated.
- Step 2** Contact maintenance personnel for the core network to query the interconnecting data, and it is found that the local port number of the SCTP link configured on the core network is incorrect.
- Step 3** The SCTP link is in normal status after the configuration of the core network is modified.

Fault Rectification

Data is configured incorrectly, and modify configurations of the core network.

14.4 Troubleshooting IP Path Faults

14.4.1 Fault Symptom

An IP path fault occurs if any of the following appears:

An IP path fault occurs when you run **DSP IPPATH** on the RNC, but in the command output, **Operation Status** is **Unavailable**, or the following alarms are displayed.

Alarm ID	Alarm Name
21581	ALM-21581 Path Failure
21582	ALM-21582 Path Congestion.
21352	ALM-21352 IPPATH Excessive Packet Loss Rate

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates; transmission between location and the user plane is abnormal.

14.4.2 Possible Causes

1. The transmission is faulty.
2. The configurations at the two ends are improper.

14.4.3 Troubleshooting Procedure

Check whether the IP path configuration is correct.

Then check whether any transmission faults under the IP layer occur. If yes, troubleshoot such faults.

If no, check whether the configurations at the two ends are proper.

14.4.4 Troubleshooting Steps

Step 1 It is recommended that troubleshoot faults by fault type.

If...	Then...
Packet loss occurs in the user plane	Troubleshoot packet loss in IP transmission
Delay and jitter occurs in the user plane	Troubleshoot delay and jitter in IP transmission
Packet loss occurs in the user plane	Troubleshoot packet error in IP transmission
Transient interruption occurs in the user plane	Troubleshoot transient interruption in IP transmission
Other abnormalities	Go to step 2

Step 2 Check whether the IP path configuration is proper.

Run **LST IPPATH** on the RNC to check whether the IP address of the local end and the IP address of the peer end are properly set.

If yes, go to step 2.

If no, correct the configuration.

Step 3 Optional. Check whether the IP route is correctly set in layer 3 networking.

Run **LST IPRT** on the NodeB or RNC to check whether the route is configured.

If yes, go to step 2.

If no, add IP routes. If the fault is rectified, no further action is required. If the fault persists, go to step 3.

Run **DSP IPRT** on the NodeB or RNC to check whether correct destination IP address, subnet mask and next hop IP address exist.

If yes, go to step 3.

If no, modify the route configuration. If the fault is rectified, no further action is required. If the fault persists, go to step 3.

- Step 4** Perform the **Ping** operation to check the IP-layer connectivity and end-to end connectivity. If the ping operation fails, troubleshoot link faults.

If...	Then...
IP over FE/GE	Troubleshoot IP over FE/GE interface disconnection
IP over E1	Troubleshoot MP/PPP link failure in IP over E1 mode

If the ping operation succeeds, go to step 4.

- Step 5** Optional. Run **LST IPPATH** on the RNC. If the VLAN ID is a valid value, check whether VLAN is configured properly on the RNC.

Run **LST VLANID** and **LST IPPATH** on the RNC to check whether the VLAN ID is configured as the transport network requires.

If yes, go to step 5.

If no, modify VLAN settings. If the fault is rectified, no further action is required. If the fault persists, go to step 5.

- Step 6** Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.4.5 Typical Cases

Fault Symptom

An operator performs an IP reconstruction for the Iu interface. After the data is configured, the signalings are correct but call connection fails, and the RNC returns assignment failures to the core network.

Locating Faults

- Step 1** After confirmation, the BSC boards are configured completely and no board hardware fault alarms are generated.
- Step 2** Query the status of the IP path and confirm that the IP path is unavailable.
- Step 3** Query the data configuration and find out configurations of routes to the peer core network are lost.
- Step 4** Add routes to clear the fault.

Fault Rectification

Data is configured incorrectly. Add routes to troubleshoot faults.

14.5 Troubleshooting IP over FE/GE Interface Disconnection

14.5.1 Fault Symptom

Run **DSP ETHPORT** on the RNC. In the command output, the **Link Availability Status** is **Unavailable** or the following alarms are generated.

ALM-21345 Ethernet Link Fault

ALM-21347 IP Address Conflict

ALM-21389 MAC over the FE Port Excessive Frame Error Rate

14.5.2 Possible Causes

1. IP-layer configurations (such as DSCP, MTU, and routing configurations) are incorrect.
2. Link-layer configurations such as virtual local area network (VLAN) configurations are incorrect.
3. Physical layer configurations (such as Ethernet port configurations) are incorrect..
4. The transport network is disconnected.
5. The network cables or optical fibers are faulty or connected improperly.
6. A port is faulty or a device is abnormal.

14.5.3 Troubleshooting Procedure

Locate the fault layer by layer. The sequence is IP layer > link layer > physical layer.

14.5.4 Troubleshooting IP Layer Faults

Step 1 Perform the **Ping** operation to check the end-to-end connectivity and gateway connectivity.

If the ping to ends fail, go to step 2.

If the ping to the gateway fails, see "Troubleshooting Data Link Layer Faults."

If the ping operation succeeds, troubleshoot application layer faults (upper-layer faults).

Step 2 Perform an trace operation to detect faulty transmission nodes, and record the IP address of the last hop. Then go to step 3.

Step 3 Check route configurations.

Run **DSP IPRT** on the NodeB or RNC to check whether correct destination IP address, subnet mask and next hop IP address exist.

If yes, troubleshoot abnormal transmission on the IP devices. If the fault is rectified, no further action is required. If the fault persists, troubleshoot data link faults.

If no, modify the route configuration. If the fault is rectified, no further action is required. If the fault persists, troubleshoot data link faults.

Note: Run **DSP IPRT** to query active routes and run **LST IPRT** to query configured routes.

Step 4 Collect the data collected in the previous steps and contact Huawei for technical support.

14.5.5 Troubleshooting Data Link Layer Faults

Step 1 Perform ARP query and check whether the link is bidirectionally connected.

Step 2 Run **DSP ARP** on the NodeB or RNC to check whether the gateway IP address of the next hop is gained.

Step 3 Perform ARP query on the router gateway to check whether the IP address of the NEs which are directly connected are gained in the reverse direction.

If both NEs and routers receive the IP address, the link is bidirectionally connected. If faults are generated, collect the data collected in the previous steps and contact Huawei for technical support.

If both NEs and routers fail to receive the IP address, go to step 2.

Step 4 Check whether the VLAN configurations on the RNC or NodeB are correct.

1. Run **LST VLANMAP** on the NodeB to check whether the configured VLAN ID and VLAN priority are consistent with those of transmission devices which are directly connected. (If the VLAN group ID is a valid value, run **VLANCLASS** on the LST.)
2. Run **LST VLANID** on the RNC to check whether the VLAN ID is configured as the transport network requires.

If yes, troubleshoot physical layer faults.

If no, modify VLAN settings. If the fault is rectified, no further action is required. If the fault persists, troubleshoot physical layer faults.

14.5.6 Troubleshooting Physical Layer Faults

Step 1 Check whether the board indicator is in normal state.

1. Optional. If FE/GE interface boards are used, check whether the LINK indicator is normal.

If yes, go to step 2.

If no, replace the network cables or boards.

2. Optional. If optical interface boards are used, check whether the LINK indicators are normal. (If the optical interface indicator is on, the link is normal.)

If yes, go to step 2.

If no, check whether the optical module and the fiber are plugged properly and replace the transmitter and receiver of the optical fiber and the board.

Step 2 Check whether parameter settings of the Ethernet port are consistent between the transmission devices that are directly connected.

Run **LST ETHPORT** on the RNC to check whether the port rate and the auto-negotiation parameter settings are consistent with those of the transmission devices that are directly connected to the RNC.

Run **LST ETHPORT** on the NodeB to check whether the port rate and the duplex mode settings are consistent with those of the transmission devices that are directly connected to the NodeB.

If yes, go to step 3.

If no, correct the configuration.

Step 3 Optional. If FE/GE interface boards are used, check whether the NEs are faulty or ports of transmission devices which are directly connected are abnormal.

1. Connect a PC to the network port of faulty NEs (RNC or NodeB) to check whether the alarm is cleared.

If yes, the port of directly connected transmission devices is faulty.

2. Connect a PC to transmission device ports of faulty NEs (RNC or NodeB) to check whether the indicator of the network interface card (NIC) is on.

If yes, RNC ports or NodeB ports are faulty. Run **RST ETHPORT** and **RST BRD** on the RNC or the NodeB, or replace interface boards.



CAUTION

You must run commands to reset interfaces or boards. Be cautious that running **RST BRD** to reset the interface board interrupts all services under the interface board.

If no, go to step 4.

Step 4 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.5.7 Typical Cases

Fault Symptom

An operator performs an IP reconstruction for interface A, but services are abnormal.

Locating Faults

Step 1 Check data configuration and no incorrect configuration is detected.

Step 2 Check alarms. The Ethernet link fault alarms are generated. Check the network cable and the cable is correctly connected.

Step 3 Run **DSP ETHPORT** on the RNC to query the status of the Ethernet port. In the command output, the **Working Mode** of the Ethernet port on the BSC is **Half Duplex**, and the **Automatic Negotiation Mode** is **Enabled**. This may indicates that the forced mode is configured at the peer end.

Step 4 Check configurations of the peer device. The port is the forced mode. The rate is 100 Mbit/s and the mode is full duplex. Modify the Ethernet port mode and then the fault is rectified.

Fault Rectification

1. If data is configured incorrectly, modify configurations.
2. FE/GE transmission is faulty.

14.6 Troubleshooting MP/PPP Link Failure in IP over E1 Mode

14.6.1 Fault Symptom

A fault occurs if an MP/PPP link is configured on the RNC or NodeB, run **DSP PPPLNK** and **DSP MPGRP** or **DSP MPLNK**, but in the command output, the link status is **Down** or **Inactive**, or run **LST MPGRP** and **LST PPPLNK** on the RNC, any of the following alarms are generated:

ALM-21344 MLPPP Group Failure

ALM-21343 PPP/MLPPP Link Failure

ALM-21201 E1T1 Loss of Signal

ALM-21203 E1T1 Alarm Indication Signal

ALM-21204 E1T1 Alarm Indication Signal

14.6.2 Possible Causes

1. IP-layer configurations (such as DSCP, MTU and routing configurations) are incorrect.
2. Link-layer configurations (such as PP/MPGRP configurations and VLAN configurations) are incorrect.
3. Physical-layer configurations such as E1T1 configurations are incorrect.
4. The transport network is disconnected.
5. The E1/T1 cables or optical fibers are faulty or connected improperly.
6. A port is faulty or a device is abnormal.

14.6.3 Troubleshooting Procedure

Locate the fault layer by layer. The sequence is IP layer > physical layer > link layer.

14.6.4 Troubleshooting IP Layer Faults

For details, see "Troubleshooting IP Layer Faults."

14.6.5 Troubleshooting E1T1 Faults (physical layer)

For details, see "Troubleshooting IP Layer Faults."

14.6.6 Troubleshooting Data Link Layer Faults

Step 1 Run **DSP MPGRP** to check the status. In the command output, if the status is **Down**, check whether the MP negotiation parameters are consistent with those of transmission devices which are directly connected. MPGRP negotiations parameters are as follows:

Maximum-Receive-Unit, Async-Control-Character-Map, Authentication-Protocol, Magic-Number, Protocol-Field-Compression, Address&Control-Field-Compression, Short Sequence, Endpoint Discriminator

If yes, go to step 2.

If no, correct the configuration.

- Step 2** Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.7 Troubleshooting Packet Loss in IP Transmission

14.7.1 Fault Symptom

Perform the **Ping** operation to check the IP-layer connectivity and packet loss is displayed.

Run **LST IPPATH** on the RNC, the IP path checkflag is displayed as a valid value (follow "Using the Ping Operation to Check the IP Path Status") and the VS.IPPATH.PING.MeanLOST counter is greater than 2%.

Run **DSP IPPM** on the RNC, the IPPM status is normal (follow "Performing IP PM Detection to Check IP Path Performance on the Iub Interface") and the forward average packet loss ratio of the VS.IPPM.Forward.DropMeans IPPM counter is high.



NOTE

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates.

14.7.2 Possible Causes

1. If Ethernet ports are faulty, the possible cause is that the port negotiation modes are inconsistent.
2. If the E1/T1 configurations are improper, the possible cause is that negotiation parameters such as the encoding mode and impedance are inconsistent.
3. The QoS policy is improper.
4. The bandwidth is limited or the speed limit function is used.
5. The cable quality is poor and signal interference occurs..
6. The network is faulty or a device is abnormal.

14.7.3 Troubleshooting Steps

- Step 1** Check whether parameter settings of the Ethernet port are consistent between the transmission devices that are directly connected.

Run **LST ETHPORT** on the RNC to check whether the port rate and the auto-negotiation parameter settings are consistent with those of the transmission devices that are directly connected to the RNC.

Run **LST ETHPORT** on the NodeB to check whether the port rate and the duplex mode settings are consistent with those of the transmission devices that are directly connected to the NodeB.

If yes, go to step 3.

If no, correct the configuration.

- Step 2** Perform gateway **Ping** operations to check the IP-layer connectivity and collect data about the packet loss ratio.

Perform ping operations from NEs at both ends to the gateway respectively.

1. If no packet loss occurs, it indicates that packet loss occurs in the intermediate transmission network. Contact transmission engineers to troubleshoot the fault.
2. If packet loss occurs only when some DSCP values are used or large packets are used, modify configurations to troubleshoot the fault.
3. If packet loss always occurs on a certain NE, contact NE and transmission engineers to troubleshoot the fault.

If the fault persists, collect the data collected in the previous steps and contact Huawei for technical support.

If the fault is rectified, no further action is required.

Step 3 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.8 Troubleshooting Delay and Jitter in IP Transmission

14.8.1 Fault Symptom

Large delay is displayed when you perform the **Ping** operation to check the IP-layer connectivity.

Large delay is displayed when you perform IP loopback to detect faulty network nodes.

Run **LST IPPATH** on the RNC, the IP PATH checkflag shows a valid value (follow "Using the Ping Operation to Check the IP Path Status") and the VS.IPPATH.PING.MeanDELAY counter shows large delay.

Run **DSP IPPM** on the RNC, the IPPM status is normal (follow "Performing IP PM Detection to Check IP Path Performance on the Iub Interface") and the average RTT delay of the VS.IPPM.Rtt.Means IPPM counter shows large delay.



NOTE

When delay and jitter exceed the thresholds during packet exchange between communication devices, users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates.

14.8.2 Possible Causes

1. The transmission network is congested.
2. The QoS policy is improper.
3. A device is abnormal.

14.8.3 Troubleshooting Procedure

Isolate the fault segment by segment.

14.8.4 Troubleshooting Steps

Step 1 Perform a trace operation to detect faulty transmission nodes, and gain all the IP addresses on the end-to-end path.

- Step 2** Perform the ping operation to check the IP-layer connectivity and analyze the point where delay and jitter occur.

Perform ping operations from NEs at both ends to the gateway respectively. Ping the nearest router from the RNC. If the result is successful, ping the next router. In this way, you can locate the delay and jitter.

1. If no delay and jitter occur, it indicates that the fault occurs in the intermediate transmission network. Contact transmission engineers to troubleshoot the fault.
2. If delay and jitter occur only when some DSCP values are used or large packets are used, modify configurations to troubleshoot the fault.
3. If delay and jitter always occurs on a certain NE, contact NE and transmission engineers to troubleshoot the fault.

If the fault persists, collect the data collected in the previous steps and contact Huawei for technical support.

If the fault is rectified, no further action is required.

- Step 3** Check whether the physical bandwidth is sufficient.

Compare the maximum allocated physical bandwidth on the transmission network (value A) and the maximum configured bandwidth (value B). Ensure that A is larger than B. Reserve bandwidth to prevent congestion and larger delay/jitter so that the service quality can be ensured.

In this case, value A needs to be provided by the datacom engineers.

- Step 4** Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.9 Troubleshooting Packet Errors in IP Transmission

14.9.1 Fault Symptom

Perform an Ethernet port query to detect the working status of the port, and packet errors are displayed or the following alarms are generated:

Unavailability alarms such as SCTP link congestion

IP path packet loss exceeds threshold



NOTE

Users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates.

14.9.2 Possible Causes

1. The transmission line quality is poor, and transmission is affected by interference.
2. If E1/T1 transmission is used, inconsistent configurations cause error bits.
3. If the fault occurs on the Ethernet, inconsistent port negotiation causes error packet.
4. A transmission device is faulty.

14.9.3 Troubleshooting Procedure

Locate the fault layer by layer (from bottom to top) based on the protocol stack.

Locate the fault on the transport network segment by segment.

14.9.4 Troubleshooting Steps

Step 1 Check the link status, clock status, Ethernet configuration and E1 configuration to rule out configuration faults.

Perform the following operations:

Run the **DSP ETHPORT** command. In the command output, check whether the **Link Availability Status** is **Available** and whether the link is activated.

Run the **DSP CLKSTAT** command. In the command output, check whether the clock is locked.

Run the **LST ETHPORT** and **DSP ETHPORT** commands. In the command output, check the duplex mode and negotiation parameters of the Ethernet ports. Ensure that the settings at both ends are consistent.

Run the **LST E1T1** and **DSP E1T1** commands. Check the E1 frame format, encoding mode and scrambling mode. Ensure the setting at both ends are consistent.

Step 2 Check the cables. For example, replace the network cable, E1 cable, and optical module.

Step 3 Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

14.10 Troubleshooting Transient Interruption in IP Transmission

14.10.1 Fault Symptom

SCTP unavailability alarms and path fault alarms (under the circumstance that IP path ping is in operation) are generated randomly or any of the following appears:

Transmission is interrupted transiently when you perform the **Ping** operation to check the IP-layer connectivity.

Packet loss ratio is high randomly when you perform IP loopback to detect faulty network nodes for multiple times.

Run **LST IPPATH** on the RNC, the IP PATH checkflag shows a valid value (follow "Using the Ping Operation to Check the IP Path Status") and the VS.IPPATH.PING.MeanDELAY counter shows large delay randomly.

Run **DSP IPPM** on the RNC, the IPPM status is normal (follow "Performing IP PM Detection to Check IP Path Performance on the Iub Interface") and the VS.IPPM.Forword.DropMeans IPPM counter shows high packet loss ratio randomly.

**NOTE**

When delay and jitter exceed the thresholds during packet exchange between communication devices, users feel that the voice quality becomes poorer and the call drop rate becomes higher. The HSPA rate is relatively low and fluctuates.

14.10.2 Possible Causes

1. If Ethernet ports are used, the possible cause is that the port negotiation modes are inconsistent.
2. If the E1/T1 configurations are used, the possible cause is that negotiation parameters such as the encoding mode and impedance are inconsistent.
3. The quality of the transport network is poor.

14.10.3 Troubleshooting Procedure

Isolate the fault segment by segment.

14.10.4 Troubleshooting Steps

If transient interruption in IP transmission occurs, perform the following operations:

Step 1 Perform the ping operation to check the transient interruption and obtain the transient interruption rules (Does transient interruption occur only when the transmission is busy? Does transient interruption occur in a fixed time every day?) Isolate the scope where the transient interruption occurs and gradually narrow the fault location scope. For details about manual ping operations and analysis, see II. "Ping" in 1.1.7 "Maintenance and Test Methods in IP Transmission."

Step 2 Perform the ping operation to check the IP-layer connectivity and analyze the point where the transient interruption occurs.

Perform ping operations from NEs at both ends to the gateway respectively. Ping the nearest router from the RNC. If the result is successful, ping the next router. In this way, you can locate the delay and jitter.

1. If no delay and jitter occur, it indicates that the fault occurs in the intermediate transmission network. Contact transmission engineers to troubleshoot the fault.
2. If delay and jitter occur only when some DSCP values are used or large packets are used, modify configurations to troubleshoot the fault.
3. If delay and jitter always occurs on a certain NE, contact NE and transmission engineers to troubleshoot the fault.

If the fault persists, collect the data collected in the previous steps and contact Huawei for technical support.

If the fault is rectified, no further action is required.

Collect common fault information and the data collected in the previous steps, and contact Huawei Customer Service Center.

15

Appendix: Common Methods of Collecting Fault Information

When a fault cannot be rectified using the methods described in this troubleshooting guide, ask Huawei technical support personnel to rectify the fault and provide them with associated information to locate the fault immediately. This section describes how to collect various information for locating faults.

Table 15-1 Common methods of collecting fault information on the RNC

Information to Be Collected	Collection Method
Version information of the faulty NE	Run the LST VER command on the RNC LMT to query the BSC software version.
Configuration script	Run the EXP CFGMML command to export the BSC configuration script. After the command is executed, obtain the configuration script from the specified path. • If Export File Path and File Name are not set in the EXP CFGMML command, the configuration script will be saved in \bam\version_X\ftp\export_cfgmml\ on the OMU by default. The default file name is CFGMML-RNCX-YYYYMMDDHHMMSS.txt, where X is the RNC ID. • If Export File Path and File Name are specified in the EXP CFGMML command, the configuration script will be saved in the specified path. (The specified Export File Path must exist on the OMU and the File Name must be unique on the OMU.)
Historical alarms	1.Run the COL LOG command with Log File Type set to HISTORY_ALARM to obtain historical alarms. 2.Run the LST LOGRSTINFO command to query the path where the historical alarm file (the default file name is ALARM_INFO.zip) is saved. 3.Obtain the historical alarm file. The default save path is \bam\version_X\ftp\COLLOGINFVALM-LOG\.
Operation log	1.Run the COL LOG command with Log File Type set to OPT_LOG to obtain the operation log. 2.Run the LST LOGRSTINFO command to query the path where the operation log

Information to Be Collected	Collection Method
	(the default file name is OperateLog.zip) is saved. 3. Obtain the operation log. The default save path is \bam\version_X\ftp\COLLOGININFO\OPT-LOG\.
Performance measurement result file	Obtain the performance measurement result file. Save the file in \bam\common\MeasResult. The file name is YYYYMMDD.Start Time-End Time_EMS-*.mrf.bz2, where * is the measurement period. - The normal measurement period is 30 or 60 minutes by default, which can be set on the M2000. - The short measurement period is 5 or 15 minutes by default, which can be set on the M2000. - The long measurement period is 24 hours by default. For example, A20101203.0900+0800-0935+0800_EMS-SHORTPERIOD.mrf.bz2 indicates that the performance measurement result file contains the measurement result from 09:00 Eastern Time (UTC+8) to 09:35 Eastern Time (UTC+8) on December 3 in 2010. SHORTPERIOD indicates that the short measurement period is used.
OMU data	1. Run the BKP DB command with Path of Backup File and File Name set to appropriate values to back up the data to the specified directory on the OMU hard disk. 2. Obtain the backed up data file from the specified path. For the method of backing up system data, see the information about OMU service processes in the <i>BSC69000 UMTS OMU Administration Guide</i>.
OMU logs	1. Run the COL LOG command with Log File Type set to OMU_LOG to obtain the OMU logs. 2. Run the LST LOGRSTINFO command to query the path where the OMU logs are saved. 3. Obtain the running logs. The logs are saved in \bam\version_X\log by default, including the running log for each OMU service process. For details about the OMU service processes, see the <i>BSC69000 UMTS OMU Administration Guide</i>.
Running logs of the host	1. Run the COL LOG command with Log File Type set to HOST_LOG to obtain the running logs. 2. Run the LST LOGRSTINFO command to query the path where running logs of the host are saved. The file name is BSC0000_XXLog Start Time_End Time.log.zip, where XX indicates the subrack number. For example, BSC0000_00Log20101203102457_20101203113504.log.zip indicates that the log records the running information of the host from 10:24:57 to 11:35:04 on December 3, 2010. 3. Obtain the running logs. The default save path is \bam\common\fam\famlog\.
Common debugging logs	1. Run the COL LOG command with Log File Type set to DEBUG_LOG to obtain the common debugging logs. 2. Run the LST LOGRSTINFO command to query the path where the debugging logs are saved. The file name is BSC0000_[DEBUG]XXLog Start Time_End Time.log.zip, where XX

Information to Be Collected	Collection Method
	indicates the subrack number. For example, BSC0000_[DEBUG]00Log20101203102457_20101203113504.log.zip indicates that the log records the debugging information of subrack 0 from 10:24:57 to 11:35:04 on December 3, 2010. 3. Obtain the debugging logs. The default save path is \bam\common\fam\famlogfmt\.
CALLFAULT logs	1. Run the COL LOG command with 3G_CHR_LOG and CALLFAULT_LOG selected in the Log File Type drop-down list box to obtain the CHR and CALLFAULT logs. 2. Run the LST LOGRSTINFO command to query the path where the CHR and CALLFAULT logs are saved. - The CHR file name is BSC0000_[CHR]XXLog Start Time_End Time.log.zip, where XX indicates the subrack number. For example, BSC0000_[CHR]00Log20101203102457_20101203113504.log.zip indicates that the log records the call information of subrack 0 from 10:24:57 to 11:35:04 on December 3, 2010. - The CALLFAULT file name is BSC0000_[CALLFAULT]XXLog Start Time_End Time.log.zip, where XX indicates the subrack number. For example, BSC0000_[CALLFAULT]00Log20101203102457_20101203113504.log.zip indicates that the log records the call faults of subrack 0 from 10:24:57 to 11:35:04 on December 3, 2010. 3. Obtain the CHR and CALLFAULT logs. The default save path is \bam\common\fam\famlogfmt\.
PCHR logs	1. Run the COL LOG command with Log File Type set to PCHR_LOG to obtain the PCHR logs. 2. Run the LST LOGRSTINFO command to query the path where the PCHR logs are saved. The file name is BSC0000_[PCHR]XXLog Start Time_End Time.log.zip, where XX indicates the subrack number. For example, BSC0000_[PCHR]00Log20101203102457_20101203113504.log.zip indicates that the log records the PCHR information of subrack 0 from 10:24:57 to 11:35:04 on December 3, 2010. 3. Obtain the PCHR logs. The default save path is \bam\common\fam\famlogfmt\pchr\.
UE tracing result	1. Click Trace on the LMT main page. The Trace tab page is displayed. 2. In the Trace Navigation Tree, unfold Trace > UMTS Services and double-click UE Trace to trace various types of messages. For details, see the <i>BSC69000 UMTS LMT User Guide</i>.
Cell tracing result	1. Click Trace on the LMT main page. The Trace tab page is displayed. 2. In the Trace Navigation Tree, unfold Trace > UMTS Services and double-click Cell Trace to trace various types of messages. For details, see the <i>BSC69000 UMTS LMT User Guide</i>.
IOS tracing result	1. Click Trace on the LMT main page. The Trace tab page is displayed. 2. In the Trace Navigation Tree, unfold Trace > UMTS Services and double-click IOS Trace to trace various types of messages. For details, see the <i>BSC69000 UMTS</i>

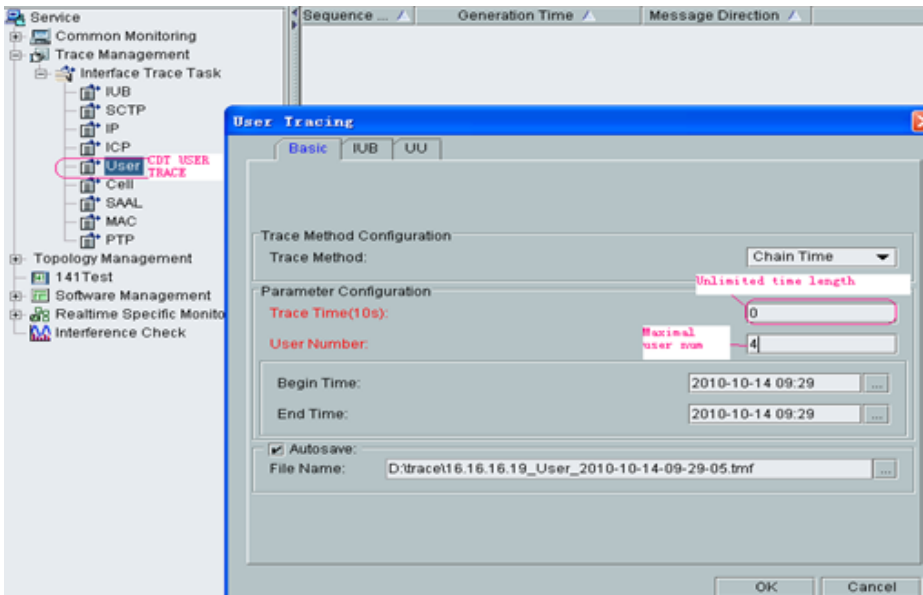
Information to Be Collected	Collection Method
	<i>LMT User Guide.</i>
Interface tracing result	1.Click Trace on the LMT main page. The Trace tab page is displayed. 2.In the Trace Navigation Tree , unfold Trace > UMTS Services , double-click the navigation node corresponding to tracing of an interface, and set related parameters. For details, see the <i>BSC69000 UMTS LMT User Guide</i> .
Cell status	Run the DSP UCELLCHK command to perform a health check on the cell.
Link performance monitoring result	1.Click Monitor on the LMT main page. The Monitor tab page is displayed. 2.In the Monitor Navigation Tree, unfold Monitor > Common Monitoring, and double-click Link Performance Monitoring. The Link Performance Monitoring dialog box is displayed. 3.In the Link Performance Monitoring dialog box, select the link to be monitored in the Monitor Item drop-down list box and set other parameters. Then click Submit to start monitoring. For details, see the <i>BSC69000 UMTS LMT User Guide</i> .

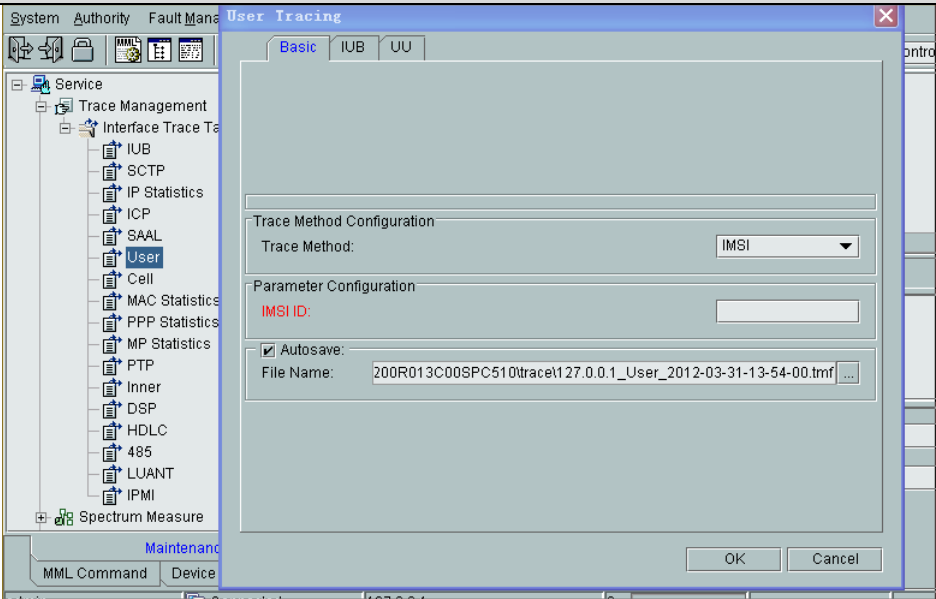
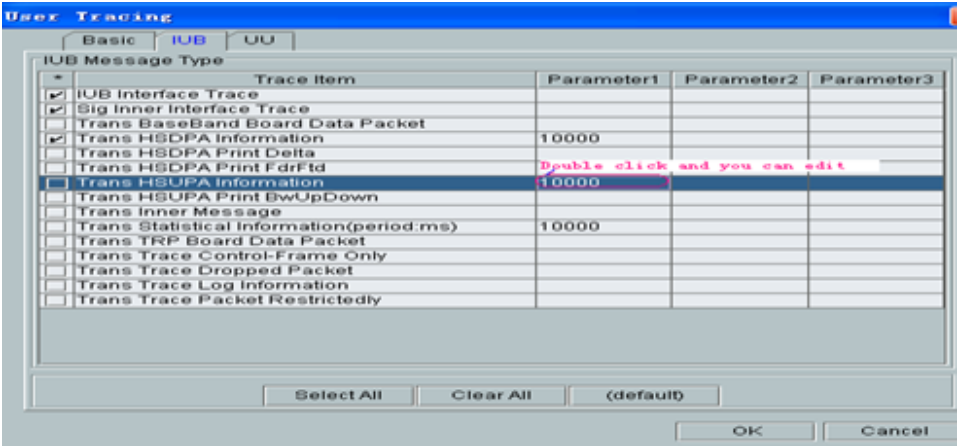
**NOTE**

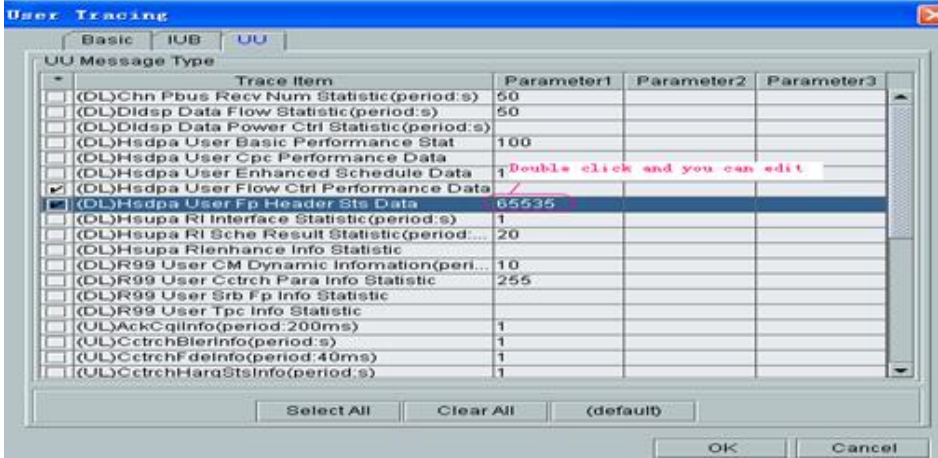
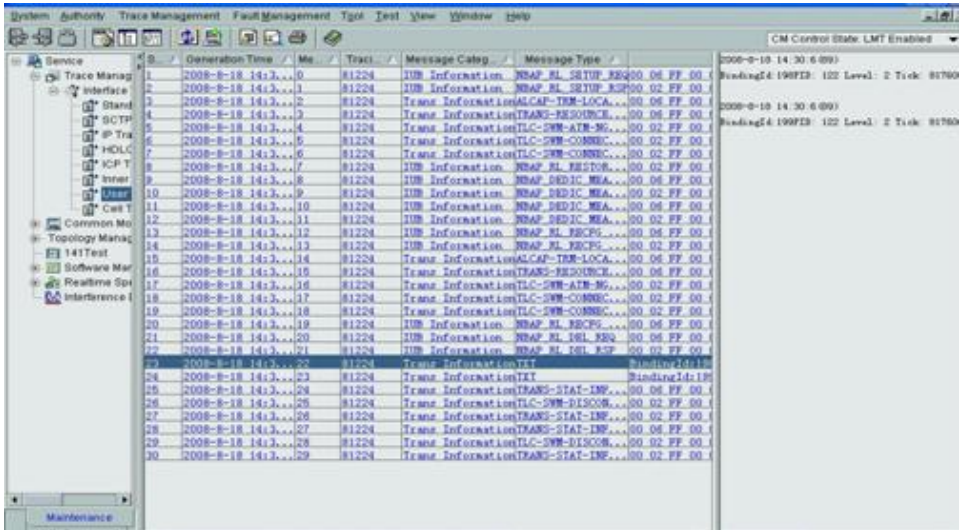
The version_X field indicates the directory where the active OMU workspace is installed. It can be queried by the **LST OMUAREA** command.

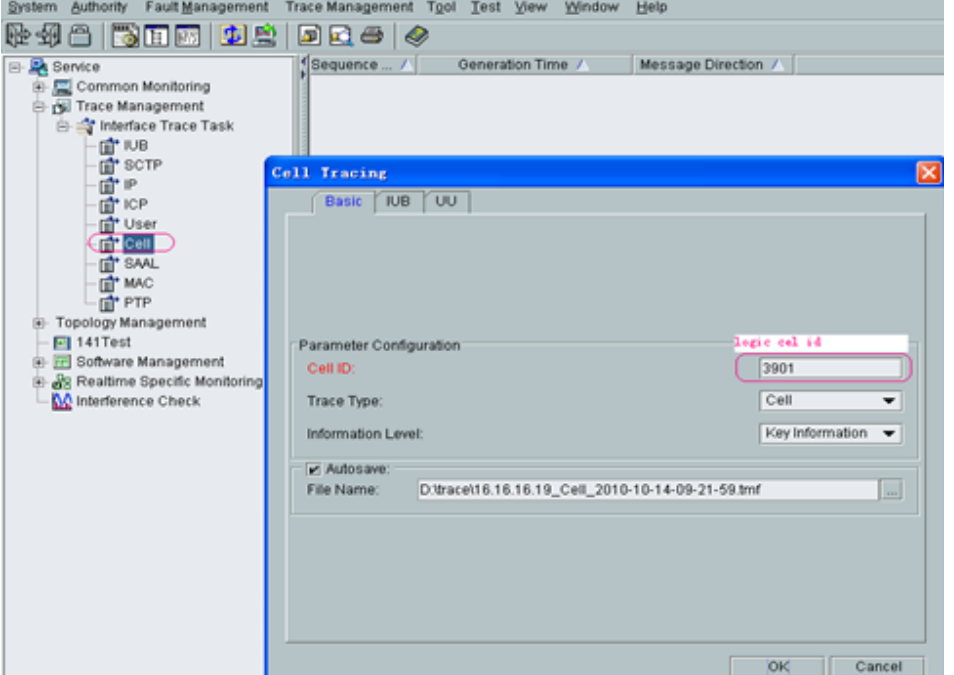
Table 15-2 Common methods of collecting fault information on the NodeB

Information to Be Collected	Collection Method
Version information of the faulty NE	Run the LST VER command on the NodeB LMT to query the NodeB software version.
Configuration script	1.Click Maintenance on the LMT main page. The Maintenance tab page is displayed. Unfold Service > Software Management and double-click Data Config File Transfer . The Data Config File Transfer dialog box is displayed. 2.In the Data Config File Transfer dialog box, set Transfer Type to Upload(NodeB to FTP Server). Then click Start to start monitoring. For detailed operations, see the information about backing up the configuration file in the NodeB LMT User Guide.
NodeB log	1.Click Maintenance on the LMT main page. The Maintenance tab page is displayed. 2.Unfold Service > Software Management and double-click Other File Transfer. The Other File Transfer dialog box is displayed. 3.In the Other File Transfer dialog box, set File Description to the corresponding types and other parameters to appropriate values. Then click Start to start the upload. For detailed operations, see the information about uploading NodeB logs in the NodeB LMT User Guide. NOTE - Log types of V100: maintenance log, main control log, board log, security log, baseband IQ data, and RTWP routine test log - Log types of V200: one-click log, security log, running log, operation log, abnormal configuration file, exception log, normal configuration file, Canbus log, alarm log, central

Information to Be Collected	Collection Method
	fault log, local fault log, test result log, transmission quality report log, debugging log, BSP report log, DSP memory log, DSP log, RTWP test log, BSP log, serial port redirection log, board replacement log, and board temperature log.
User information	- Click Maintenance on the LMT main page. The Maintenance tab page is displayed. Unfold Service > Trace Management > Interface Trace Task and double-click User. - Select the tracing mode. When no UEs are available for the drive test, select Chain Time, and the system will randomly trace a maximum of four UEs. When UEs are available for the drive test, select IMSI and specify the UEs to be traced. The two tracing modes can be selected as follows: - Select the time-based tracing mode as follows.  NOTE The entered time is based on the NodeB time. Figure 15-1 Selecting Chain Time  - Select the IMSI-based tracing mode as follows: - On the BSC LMT, run the following command: MOD UNODEB: NodeBId = xxx, NodeBTraceSwitch=ON; where xxx is the NodeB ID. - Select IMSI in the Trace Method drop-down list box and enter the IMSI ID, as shown in the following figure. Figure 15-2 Selecting IMSI

Information to Be Collected	Collection Method
	 3. On the IUB and UU tab pages, select the items to be traced, as shown in the following figures.  NOTE Set the parameters based on the problems to be located. Figure 15-3 Selecting Iub tracing items  Figure 15-4 Selecting Uu tracing items

Information to Be Collected	Collection Method
	 4. On the Basic tab page, click Auto save, specify the directory for saving the tracing result, and click OK to start tracing. The traced information is reported as follows. Figure 15-5 Traced UE information  5. Obtain the tracing result from the specified directory.
Cell information	1. Click Maintenance on the LMT main page. The Maintenance tab page is displayed. 2. Unfold Service > Trace Management > Interface Trace Task and double-click Cell. 3. On the Basic tab page, set Cell ID to the logic ID of the cell to be traced. Select Auto save and specify a directory, as shown in the following figure. Figure 15-6 Setting the cell ID

Information to Be Collected	Collection Method
	 4. Select tracing items on the IUB and UU tab pages. 5. Click OK to start tracing. 6. Obtain the tracing result from the specified directory.
IP packet statistics	Run the DSP IPSTAT command to collect statistics on IP packets transmitted on all links of a board.
Board manufacturing information	Run the DSP BRDMFRINFO command to obtain the model and bar code of a board.
MTU detection of the network interconnected to the NodeB	Run the TRACERT command to conduct statistics on IP packets transmitted on all links of a board.
Power consumption of the NodeB	- VS.BTS.EnergyCons.Adding - VS.BTS.EnergyCons.Measuring
CANBUS redirection	For detailed operations, see the information about how to start CANBUS redirection in the <i>BSC6900 UMTS LMT User Guide</i> .
Frequency spectrum scanning	For detailed operations, see the information about how to manage NodeB frequency spectrum scanning in the <i>BSC6900 UMTS LMT User Guide</i> .
Offline intermodulation interference detection	Run the STR RFTEST command. Then the RTWP value is reported for the antenna ports configured with carriers once a second, because signals are transmitted and received through the antenna ports configured with carriers. The test ends and the test result are displayed when the test time expires.  CAUTION

Information to Be Collected	Collection Method
	Starting the test on a module interrupts all services of the module.
Board hardware test	Run the STR HWTST command to check for faults in the components and interfaces of a board.  WARNING • The hardware self-diagnosis can be started only on one board in a NodeB at a time. • The hardware self-diagnosis lasts between 5 to 10 minutes, during which no operations can be performed on the board. • Ensure that no software or files are uploaded or downloaded during hardware self-diagnosis, because the operations may affect the effect of hardware self-diagnosis • Ensure that the power modules support a large power consumption before performing the hardware self-diagnosis, because the hardware self-diagnosis of TRXs triggers a single tone test, which causes excessive power consumption instantaneously. If the power modules do not meet the requirements, the RRU will be powered off and restarted repeatedly, and therefore the hardware self-diagnosis and single tone test will be started repeatedly after the hardware self-diagnosis is performed. • Ensure that the board to be tested has been configured before the hardware self-diagnosis. If the board to be tested is a traffic board, ensure that the board has been blocked before the hardware self-diagnosis.